Finite-Cadence Refresh Equilibria for Streaming Prior-Anchored Ridge: A Closed-Form Forgetting Window and Spectral Correction

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Abstract

Recursive estimators on drifting streams must forget stale data fast enough to track the target, yet slowly enough to average out noise. For prior-anchored ridge regression this balance has a closed-form optimum: per eigen-direction the anchored solve is a convex combination of the data solution and the prior, with weight $s_i = \rho/(\mu_i + \rho)$. We derive the finite-cadence refresh equilibrium of the resulting recursion: at cadence k the solution lags the pure-leaky path by a closed-form $\Delta_k(s)$, and its variance is reduced by an explicit factor $\Psi(s, r_k)$. Minimizing the cycle-averaged risk gives the optimal forgetting rate $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$ with $c' = \sigma^2 d_{\text{eff}}(\rho)$, together with the plug-in anchor strength $\rho^* = \sigma^2/(n\delta^2)$. A random-matrix deterministic equivalent extends both laws to finite dimension, where the anchor is renormalized to $\rho_{\text{eff}} = \rho(e^{\zeta q} - 1)/(\zeta q)$; it matches simulation within 5% at fixed operating points and within 13% at the method's own operating points, the conflict increment is accurate within 9.5%, and rank-deficient populations reduce exactly to their active subspace. The resulting method, SPAR, calibrates its window and strength from streaming residuals: the conflict-calibrated strength falls on detected drift, guarding against an oversized warm-up anchor, while a drift-active lower bound on the strength keeps the short-window solve well conditioned. On a 27-configuration factorial grid the calibrated window opens on detected drift, as the law predicts, and the method never diverges; the grid-default configuration transfers unchanged to an hourly electricity stream, and a reduced-range head transfers to BCI IV 2b. The core algebra, refresh chain, and regret bounds are formalized in Lean 4.

1 Introduction

A streaming estimator that never forgets accumulates lag under drift; one that forgets too fast amplifies noise. Gradient-based test-time adaptation (Sun et al., 2020; Wang et al., 2021) resolves this balance implicitly, at the cost of a gradient channel, per-dataset hyperparameter selection, and imprecise guarantees. For prior-anchored ridge regression the balance is explicit: per eigen-direction the anchored solve decomposes as $W_i - w_i = (1 - s_i)(\widehat{w}_i - w_i) + s_i(G_i - w_i)$ with $s_i = \rho/(\mu_i + \rho)$. The algebra is standard—ridge as MAP with a prior-centered penalty (Hoerl & Kennard, 1970; Rasmussen & Williams, 2006; Li et al., 2018)—but it turns hyperparameter selection into a closed-loop estimation problem: measured conflict lowers the anchor strength so that fresh data dominates, and a rising drift-rate estimate opens the forgetting window.

This paper studies the resulting recursion rather than a particular tuning heuristic. The central result is a finite-cadence refresh equilibrium: at cadence k the moving-frame anchor offset follows an affine recursion whose fixed point gives the exact tracking lag and whose variance factor gives the retained noise reduction. Minimizing the cycle-averaged risk yields the optimal forgetting window in closed form. A separable-covariance deterministic equivalent then extends the law to finite dimension, where the finite window renormalizes the

The author designed the method, implemented the system, ran the experiments, and verified all results. LLM assistants (GLM-5.3, DeepSeek V4.1 Flash) were used for coding, the Lean 4 formalization, and LaTeX preparation.

anchor strength and amplifies the risk by a scalar factor. The calibrated method, SPAR, is the algorithmic instantiation of these results; the experiments test the predictions of the theory on synthetic and real streams.

Contributions.

1. **A finite-cadence theory of anchored forgetting.** The pull identity turns the anchor strength into a per-direction interpolation weight. The refresh equilibrium gives the exact lag $\Delta_k(s)$ and variance factor $\Psi(s, r_k)$ (Theorem 1); minimizing the cycle-averaged risk yields the closed-form window $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$ (Theorem 2) and the plug-in strength $\rho^* = \sigma^2/(n\delta^2)$ (Proposition 1).
2. **A finite-dimensional spectral correction.** The separable-covariance deterministic equivalent gives the retention q, the renormalized anchor ρ_{eff}, the variance functional Υ, and the full excess risk with its $1/(1-\Upsilon)$ correction (Theorem 3). The formula matches Monte Carlo within 4.4% across the 45-point grid (including the rank-deficient spectrum), within 13% at the method's own operating points, and within 9.5% for the conflict increment; rank-deficient populations reduce exactly to their active subspace (Corollary A.1).
3. **A formalized core.** The algebraic core, the finite-cadence refresh chain, the OGD and Hedge regret bounds, and the coupled contraction are formalized in Lean 4: 67 declarations, no `sorry`, and only the three standard axioms (Appendix C).
4. **A calibrated streaming method.** The forgetting window and the anchor strength are estimated online from residual and anchor-disagreement statistics; prior immunity is carried by the direction-selective refresh $\beta_i = 1 - s_i$ (one ρ sets it), and the strength rule enters as an onset guard when the warm-up prior scale is unknown, not as a continuous tracking rule. A Hedge composition over leaky windows, the frozen anchor, and the SPAR-Pull trajectory satisfies a no-regret bound under bounded losses (Proposition 4).
5. **Evidence that the law transfers.** On the 27-configuration grid the calibrated window responds to drift in the direction the law prescribes, and the calibrated methods are competitive with the strongest oracle-free baselines without diverging in any configuration. The grid-default configuration transfers unchanged to an hourly electricity stream, and a reduced-range head transfers to BCI IV 2b; Section 5 reports the numbers and Section 6 the boundary conditions.

2 Related Work

Analytic continual learning. The Analytic Class-Incremental Learning lineage (Zhuang et al., 2022; 2023; 2024b;a; Tong et al., 2025; Yan et al., 2025; Zhang et al., 2025; Kim, 2025) replaces gradient-based class-incremental learning with closed-form recursive updates on frozen features; these methods address a stationary target, whereas the present setting has a drifting target and an adaptive regularization strength.

Interpolated adaptation. WiSE-FT established frozen-to-adapted interpolation with a single scalar (Wortsman et al., 2022); the pull identity generalizes it per direction ($s_i = \rho/(\mu_i + \rho)$ in place of one α), Proposition 1 characterizes when interpolation reduces risk, and Section 5.6 compares against it directly.

Forgetting and state space. When the nominal window $1/(1-\lambda)$ of leaky RLS drops below the input dimension, the covariance recursion loses rank and the estimate diverges (Sayed, 2008; Bierman, 1977). Directional forgetting (Kulhavý & Kárný, 1984; Cao & Schwartz, 2000) and its subspace-information variants (Lai & Bernstein, 2024) preserve rank along excited directions; scalar EMA teachers (Tarvainen & Valpola, 2017) update every direction at one fixed rate, and per-direction momentum (Hong & Qu, 2026) at excitation-dependent rates. The SPAR refresh is direction-selective with no added hyperparameter: each direction's refresh rate is $1 - s_i$, the complement of its pull weight. Variable forgetting factor RLS (Fortescue et al., 1981; Paleologu et al., 2008) automates the window from the squared residual—the same statistic used here—but retains the covariance recursion and inherits its instability (Section 5). AROW (Crammer et al., 2009) and random-walk Kalman filters (Kalman, 1960; Sayed & Kailath, 1994) are the nearest online relatives;

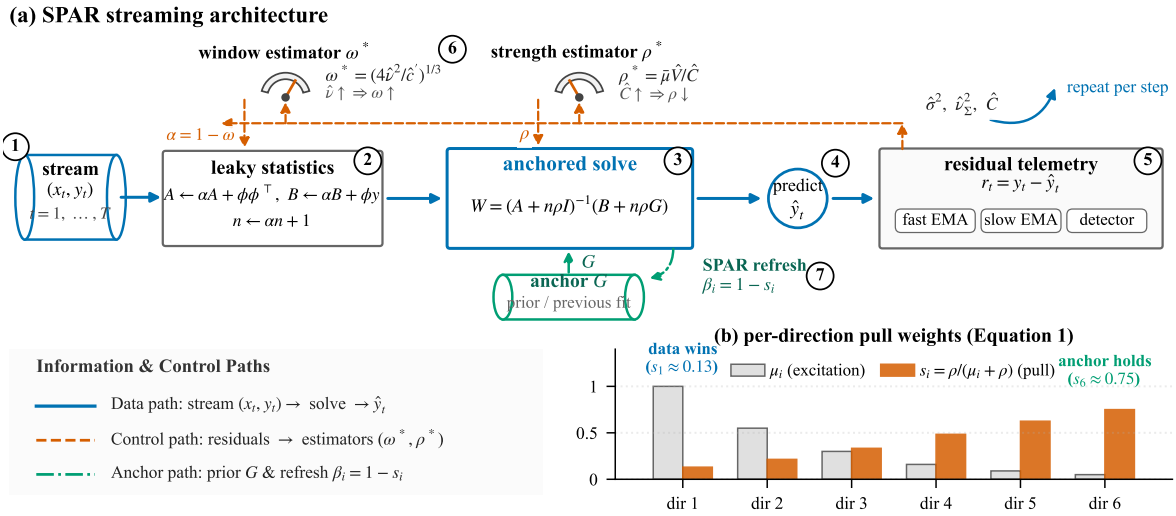


Figure 1: The SPAR streaming loop. (1) The stream enters (2) leaky statistics and (3) the anchored solve (Algorithm 1), which (4) predicts. (5) The residual r_t feeds fast and slow moving averages whose positive difference drives (6) the two online estimators (ρ^* and ω^*), which feed the solve and the statistics. (7) The anchor refreshes at $\beta_i = 1 - s_i$; one ρ couples pull and refresh. Panel (b): per-direction pull weights.

the estimated-noise Kalman filter is the strongest oracle-free baseline on the grid (Table 1). Composite drift-triggered adaptations are surveyed in Gama et al. (2014).

Regret and two-time-scale analysis. The residual-driven estimators are coupled stochastic approximation: the accumulator tracks a moving target while its step size is estimated from the same residuals, for which two-time-scale conditions and drift tracking are classical (Benveniste et al., 1990; Kushner & Yin, 2003; Borkar, 2008). The contribution here is the closed-form target and the verified contraction. Projected gradient and Hedge arguments are standard (Zinkevich, 2003); the composition and its basis dependence are made explicit in Proposition 4.

Random-matrix limits. The finite-D correction uses the separable-covariance deterministic equivalent (Silverstein & Bai, 1995; Rubio & Mestre, 2011; Hachem et al., 2007). The nearest prior analysis is the large-system exponentially windowed RLS study of Xiao & Honig (2005), which has no anchor, per-direction pull, refresh, or online calibration; the unweighted-batch ridge asymptotics (Hastie et al., 2022; Pafka et al., 2004) do not carry the finite-window renormalization.

3 Method

We call the resulting solver family SPAR (Streaming Prior-Anchored Ridge), the streaming instantiation of the Anchored Analytic Learning (AAL) framework.

Notation. $\omega \in (0, 1)$ is the scalar forgetting rate and $n_{\text{eff}} = (2 - \omega)/\omega$ its effective sample count; $\widehat{w}_t$ and w^* are parameter vectors (we write $\widehat{w}_t$ for the minimum-ridge data solve and reserve W_t for the anchored solution); $\pi_{i,t}$ are Hedge weights; ς_t is the Hedge learning rate; ι is the OGD step; $\zeta = D\omega$ is the aspect ratio of Theorem 3. For the strength estimator, $d_{\text{act}} = \#\{i : \mu_i(\bar{A}_t) > 10^{-8}\}$ is the active dimension, $\bar{\mu}_t = \text{tr}(\bar{A}_t)/d_{\text{act}}$ the active-mean excitation, and G_{ref} the slow reference anchor updated every $5k$ steps. Experiments label windows by the nominal $1/(1 - \lambda)$, which differs from n_{eff} by the factor $2 - \omega$.

Anchored solve. Let $\phi_t = \phi(x_t)$ denote the feature vector of the incoming sample. With optional forgetting factor α, the learner updates the sufficient statistics via $A \leftarrow \alpha A + \phi\phi^\top$, $B \leftarrow \alpha B + \phi y$, and $n \leftarrow \alpha n + 1$. The anchored solve is

$$W = (A + n\rho I)^{-1}(B + n\rho G),$$

with anchor G and strength ρ. In the eigenbasis of $\bar{A} = A/n$ with excitation μ_i, each coordinate is a scalar ridge solve, which yields the per-direction decomposition we call the *pull identity* (the w_i and $\widehat{w}_i$ are per-direction weight coordinates; the scalar forgetting rate ω of Section 4 is a separate object):

$$W_i - w_i = (1 - s_i)\,(\widehat{w}_i - w_i) + s_i\,(G_i - w_i), \qquad s_i = \frac{\rho}{\mu_i + \rho}; \tag{1}$$

Figure 2 shows its geometry. The solve in Equation (1) is the $\gamma = 1$ case of the generalized prior scaling $W = (A + n\rho I)^{-1}(B + n\rho\gamma G)$: $\gamma = 1$ is the exact prior-centered MAP and $\gamma < 1$ adds a shrinkage bias (Section 5); the prior precision τ_0 of Appendix A is a separate symbol.

Operator form. Both identities are basis-free: with the resolvent shrinkage operator $S_t(\rho) := \rho\,(\bar{A}_t + \rho I)^{-1}$ (so $0 \prec S_t(\rho) \preceq I$ and $I - S_t(\rho) = \bar{A}_t(\bar{A}_t + \rho I)^{-1}$), Equation (1) reads $W_t - w^* = (I - S_t)(\widehat{w}_t - w^*) + S_t(G_t - w^*)$ for any $\bar{A}_t \succeq 0$, without choosing a basis, assuming distinct eigenvalues, or restricting to finite dimension. The spectral refresh of the next paragraph, written per-direction as $\beta_i = 1 - s_i$, is likewise the operator identity $G \leftarrow (I - S_t)\,W + S_t\,G = W - \rho\,(\bar{A}_t + \rho I)^{-1}(W - G)$: since the factorization of $(\bar{A}_t + \rho I)$ is already available from the solve, the exact refresh can reuse it at $O(D^2)$ cost per refresh with no separate eigendecomposition (Algorithm 1's `eigh` is the equivalent spectral realization used to expose the per-direction weights); this reduces the refresh cost, while the per-step solve remains $O(D^3)$ for the scalar method.

Spectral anchor refresh. To prevent the anchor from becoming stale during sustained drift, G is updated every k steps along the eigenbasis U of $\bar{A}$: in each eigendirection i, $G_i \leftarrow \beta_i W_i + (1 - \beta_i) G_i$ with $\beta_i = 1 - s_i = \mu_i/(\mu_i + \rho)$. Because the prediction risk in direction i scales with μ_i, staleness along weakly excited eigendirections ($\beta_i \approx 0$) incurs negligible excess error; well-excited directions ($\beta_i \approx 1$) incorporate new information into the anchor rapidly. A single ρ thus controls both the pull toward the anchor and its refresh rate.

Online calibration. SPAR updates the window and the strength at each refresh block from fast and slow exponential moving averages of the squared prequential residual $r_t^2 = (y_t - \hat{y}_t)^2$; here ω is the forgetting rate, with effective memory $n_t \approx 1/\omega$ samples, so opening the window means increasing ω. Under Proposition 1, the risk-minimizing anchor strength decreases inversely with the prediction-unit conflict $\widehat{C}_t^{\Sigma} = \max\{C_{\mathrm{ref}}^{\Sigma}, C_{\mathrm{resid}}^{\Sigma}, C_{\mathrm{floor}}^{\Sigma}\}$: $\rho_t^* = \bar{\mu}_t \widehat{V}_t^{\Sigma} / \widehat{C}_t^{\Sigma}$, with $\widehat{V}_t^{\Sigma} = \hat{\sigma}_t^2 d_{\mathrm{act}}/n_t$ the prediction variance and every branch in $[y^2]$ (Appendix F). Formulating both branches in prediction units ensures that the estimator never inverts $\bar{A}_t$, is invariant to feature rescaling, and contributes nothing in dead directions. Section 5 reports the covariate-shift behavior. Under Theorem 2, the forgetting window trades estimation variance against drift lag: $\omega_t^* = (4\widehat{\nu}_{\Sigma,t}^2/\hat{c}_t')^{1/3}$, with $\hat{c}_t' = (1 - \bar{s}_t)^2 \hat{\sigma}_t^2 D$ and the drift energy $\widehat{\nu}_{\Sigma,t}^2 = \max(0, (\widehat{E}_t - \hat{c}_t'\omega_t/(2 - \omega_t))\omega_t^2)$, where $\widehat{E}_t = \hat{\sigma}_{\mathrm{fast},t}^2 - \hat{\sigma}_{\mathrm{slow},t}^2$ isolates excess error above the stationary variance level, the subtraction required for a positive fixed point (Appendix A). The strength rule is an onset guard: its average effect is neutral, and its value is the temporary lower bound it places on ρ when the residual detector fires (Section 5.3). Algorithm 1 gives the complete step loop; apart from one fixed default control parametrization—the constants k=10, h_f=30, ρ-lower bound 0.02, plus clip ranges and EMA horizons, shared by every configuration (Appendix F)—all behavior adapts at runtime.

Estimator branches and solver variants. Step (iv) of the algorithm takes the maximum of three prediction-unit branches, all in $[y^2]$:

$$\begin{aligned} C_{\mathrm{ref}}^{\Sigma} &= (G_{\mathrm{ref}} - \widehat{w})^{\top} \bar{A}_t\,(G_{\mathrm{ref}} - \widehat{w}) - \widehat{V}^{\Sigma}, \quad & C_{\mathrm{resid}}^{\Sigma} &= E_t, \\ C_{\mathrm{floor}}^{\Sigma} &= \tfrac{1}{2}\,\hat{\sigma}^2 d_{\mathrm{act}}/T_w, & \widehat{V}^{\Sigma} &= \hat{\sigma}^2 d_{\mathrm{act}}/n, \end{aligned}$$

with $\widehat{w} = (\bar{A}_t + \rho_{\mathrm{ridge}} I)^{-1} \bar{b}_t$, where $\rho_{\mathrm{ridge}} = 10^{-6}$ is the fixed numerical ridge of the reference-branch data solve (10^{-8} is the ridge used in the active-dimension and trace computations), and $\rho_{\min} = 0.02$ while the window is at least half open ($\omega_t \geq \frac{1}{2}\omega_{\max}$), otherwise the stationary value 10^{-5}, is the anchor-strength clip floor. The reference branch compares against the slow copy G_{ref} rather than the refreshed G, so the conflict does not collapse immediately after a refresh. The residual branch supplies the fast disengagement, and the floor branch is the irreducible warm-up conflict. No branch inverts $\bar{A}_t$, so the rule is invariant to feature rescaling and immune to the rank-deficient trace blow-up. The legacy parameter-unit form (with $\mathrm{tr}(\bar{A}^{-1})$) is reported only as an ablation (Appendix F).

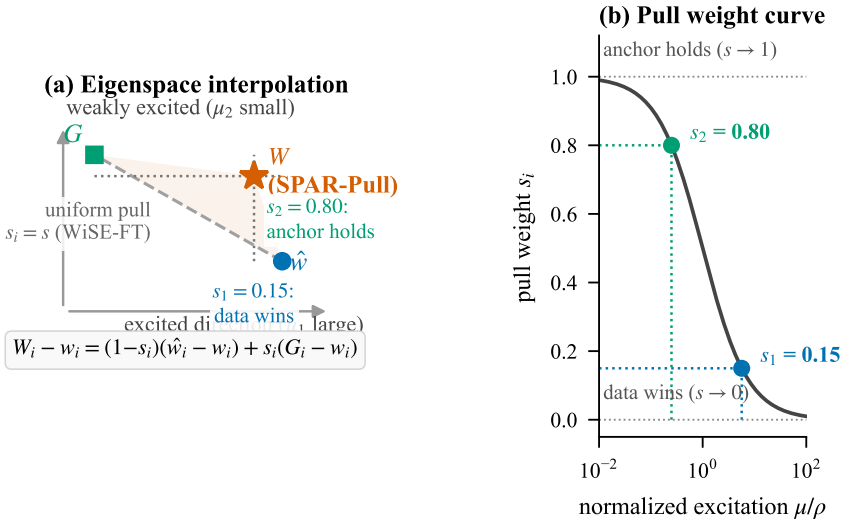


Figure 2: The pull identity (Equation (1)): the excited direction ($s_1 = 0.15$) holds the data solution, the weakly excited direction ($s_2 = 0.8$) holds the anchor, so W departs from the uniform-pull segment (dashed).

Algorithm 1: SPAR-Pull Inputs: a stream (ϕ_t, y_t), a warm-up length T_w, and a prior G_0; fixed constants k, h_f, $\rho_{\min}$, $\rho_{\max}$, η_C, and η_ν.

Initialization. Build the sufficient statistics A, B, and n, the anchor G, and the solve W from the warm-up stream. Set $\sigma_0^2 = \mathrm{var}(r_{\mathrm{warm}})$.

Loop. For $t = T_w, \ldots, T$, execute Steps (i), (ii), (vii), and (viii) at every step, and Steps (iii)–(vi) and (ix) once per refresh block of k steps, running Step (ix) first since Steps (iv)–(vi) use the eigenbasis.

(i) Predict $\hat{y}_t = \phi_t^\top W$.

(ii) Compute the residual $r_t = y_t - \hat{y}_t$, and update the fast and slow exponential moving averages σ^2_{fast} and σ^2_{slow} of r_t^2.

(iii) Form the excess error $E_t = \max(0, \sigma^2_{\mathrm{fast}} - \sigma^2_{\mathrm{slow}})$. This quantity drives both the conflict estimator and the drift-energy estimator.

(iv) Form the conflict estimate $\widehat{C}_t = \max\{C^\Sigma_{\mathrm{ref}}, C^\Sigma_{\mathrm{resid}}, C^\Sigma_{\mathrm{floor}}\}$ and the drift-energy estimate $\widehat{\nu}_t^2$ of Theorem 2. Then smooth both quantities as $\bar{C} \leftarrow (1-\eta_C)\bar{C} + \eta_C \widehat{C}_t$ and $\bar{\nu}^2 \leftarrow (1-\eta_\nu)\bar{\nu}^2 + \eta_\nu \widehat{\nu}_t^2$.

(v) Update the anchor strength as $\rho \leftarrow \mathrm{clip}(\bar{\mu}\widehat{V}^\Sigma/\bar{C},\ \rho_{\min}(E_t),\ \rho_{\max})$.

(vi) Update the forgetting rate as $\omega \leftarrow \mathrm{clip}((4\bar{\nu}^2/\hat{c}')^{1/3},\ \omega_{\min},\ \omega_{\max})$, and set $\alpha \leftarrow 1 - \omega$.

(vii) Update the sufficient statistics as $A \leftarrow \alpha A + \phi_t \phi_t^\top$, $B \leftarrow \alpha B + \phi_t y_t$, and $n \leftarrow \alpha n + 1$.

(viii) Solve the anchored system $W \leftarrow (A + n\rho I)^{-1}(B + n\rho G)$.

(ix) Every k steps, compute $(\mu, U) \leftarrow \mathrm{eigh}(\bar{A})$ and refresh the anchor as $G \leftarrow U[\mathrm{diag}(\beta_i)U^\top W + \mathrm{diag}(1-\beta_i)U^\top G]$. Every $5k$ steps, set $G_{\mathrm{ref}} \leftarrow G$.

The same loop defines the solver family used below. *SPAR-Pull* is Algorithm 1 (two estimators, one scalar ρ, exact $O(D^3)$ solve). *SPAR-Pull-D* keeps a per-direction strength $\rho_i = \sigma^2/(n\delta_i^2)$ and caches the refresh eigenbasis, giving $O(D^2)$ per step with an $O(D^3)$ refresh every k steps. *SPAR-Pull-S* is the scalar-pull simplification on the data–anchor segment, used to isolate the regret guarantee. *SPAR-EG* replaces the single solve by a Hedge mixture over five batch-solved leaky windows plus the frozen anchor; *SPAR-EG*$^+$ adds the SPAR-Pull trajectory as a sixth expert.

4 Theoretical Analysis

The section is organized around the refresh recursion. Lemma 1 establishes the continuous-refresh limit. Theorem 1 gives the finite-cadence equilibrium, and Theorem 2 minimizes the resulting risk to obtain the window law. Propositions 1–5 give the anchor calibration, the minimax and false-alarm guarantees, the regret wrappers, and the conditioning bound. Theorem 3 is the finite-dimensional spectral correction, and Theorem 4 closes the loop with the coupled contraction. Theorems 1 and 2 are analyzed in the mean-field

(population-design) model in which the random accumulator $\bar{A}_t$ is replaced by its expectation Σ; this is the model that yields the closed-form window target, and it is exact in the proportional limit $\zeta = D\omega \to 0$. Theorem 3 supplies the finite-D random-design correction at $\zeta > 0$; Appendix A states the two models explicitly and quantifies their gap ($\zeta \leq 0.13$ on the pre-drift grid, up to 6.4 during drift-phase adaptation).

Lemma 1 (Zero-lag refresh equilibrium) *Assume drift at constant rate ν per step along a direction with excitation μ, forgetting rate ω, and continuous anchor refresh. (i) The accumulator's implied data solution lags the target by exactly ν/ω. (ii) At the refresh equilibrium $G = W$ the anchored solution's lag equals the pure-leaky lag, independent of s; the anchor is inert for mean tracking. (iii) At refresh cadence k an extra lag $\Delta_k(s)$ of Theorem 1 appears, whose prediction cost is weighted by μ_i and therefore concentrates in weakly excited directions.*

Theorem 1 (Finite-cadence refresh equilibrium) *Let $a = s(2-s) = 1-(1-s)^2 \in (0,1)$ for $s \in (0,1)$, and let $r_k = \alpha^k \in [0,1)$ be the data-path noise correlation across one refresh cycle. At cadence k with data-path lag L (Lemma 1) and target velocity ν per step, the moving-frame anchor offset follows the affine recursion*

$$g_{m+1} \;=\; a\, g_m \;-\; (1-s)^2\, L \;-\; a\,\nu k, \qquad a := s(2-s) = 1-(1-s)^2.$$

The recursion contracts geometrically at rate $a \in (0,1)$ per cycle to the fixed point $g^* = -L - \nu k\, a/(1-a)$, and the cycle-averaged solution lags the pure-leaky path by $\Delta_k(s) = s\nu k\,(a/(1-a) + \frac{1}{2})$. The lag is linear in the cadence and vanishes as $s \to 0$. The steady-state solution variance is $v\,\Psi(s, r_k)$ with $\Psi(s, r_k) := 1 - 2s(1+s-s^2)(1-r_k)/((1+a)(1-ar_k)) \in [(1-s)^2,\, 1]$ for all $s \in (0,1)$, $r_k \in [0,1)$: the variance reduction of Proposition 1 survives at every finite cadence, vanishing at the inert continuous-refresh limit $r_k \to 1$ of Lemma 1(ii) and largest at $r_k \to 0$, so the cadence trades lag against variance reduction (Appendix A).

Theorem 2 (Finite-cadence optimal forgetting window) *Let the target drift at constant rate ν_i in direction i with prediction-units drift power $\nu_\Sigma^2 = \sum_i \mu_i \nu_i^2$, let the refresh cadence be k, and set $r_k = (1-\omega)^k$ and $a_i = s_i(2-s_i)$. At the refresh equilibrium of Theorem 1, the cycle-averaged prediction risk at pull profile s is*

$$R_k(\omega, s) = \sigma^2\, \frac{\omega}{2-\omega} \sum_{i=1}^{D} \Psi(s_i, r_k) + \sum_{i=1}^{D} \mu_i \Big(\frac{\nu_i}{\omega} + \Delta_k(s_i)\Big)^2 + O(k^2\nu^2 s^2),$$

with $\Psi(s,r) = 1 - \frac{2s(1+s-s^2)(1-r)}{(1+a)(1-ar)}$ and $\Delta_k(s) = s\nu k\big(\frac{a}{1-a} + \frac{1}{2}\big)$. At the runtime cadence $1 \ll k\omega$ need not hold—at the grid defaults $k\omega \in [0.005, 0.5]$—so the reduced target is justified by two monotone facts rather than by that limit. (i) Variance lower bound. *Since $(1-s_i)^2 \leq \Psi(s_i, r_k) \leq 1$ for every $r_k \in [0,1)$ (Theorem 1), the surrogate $c'\,\omega/(2-\omega) + \nu_\Sigma^2/\omega^2$ with $c' = \sigma^2 d_{\text{eff}}(\rho)$ lower-bounds the exact variance term at every cadence; Jensen's inequality gives $D(1-\bar{s})^2 \leq d_{\text{eff}}(\rho)$, so the runtime plug-in is itself a lower bound (the gap is exactly the pull variance, $\sum_i (1-s_i)^2 - D(1-\bar{s})^2 = D\,\mathrm{Var}(s)$, formalized in* `Core.lean`*),*

$$\hat{c}'_t \;=\; \sigma^2 D(1-\bar{s}_t)^2 \;\leq\; \sigma^2\, d_{\text{eff}}(\rho) \;\leq\; \sigma^2 \sum_{i=1}^{D} \Psi(s_i, r_k).$$

Its minimizer $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$ is at least the exact-optimal window (a smaller coefficient returns a larger ω, the over-forgetting direction), which is the cheap side of the safety asymmetry of Proposition 2. (ii) Finite-cadence lag residual. *With $b(s) = s(2-s)/(1-s)^2$ and $k\omega \leq 1/2$, the residual of Theorem 1 obeys*

$$\Delta_k(s_i) = s_i\,\nu_i\, k\big(b(s_i) + \tfrac{1}{2}\big) \;\leq\; \tfrac{1}{2}\, s_i \big(b(s_i) + \tfrac{1}{2}\big)\, \frac{\nu_i}{\omega},$$

a fixed fraction of the leading lag that vanishes as $s_i \to 0$, whose prediction cost $\mu_i\,\Delta_k(s_i)^2$ is μ_i-weighted and concentrated in weakly excited directions. Over the runtime range $k\omega \in [0.005, 0.5]$ the surrogate's cubic minimizer stays within 2.1% of the exact cycle-averaged optimum for $s_i \leq 0.2$ (within 5.2% for $s_i \leq 0.3$, the

largest gaps at the window floor; Appendix A.4). In the quasi-static limit $1 \ll k\omega \ll 1/s$ *(so* $s \ll 1/(k\omega) \ll 1$*) both corrections vanish identically—*$\Psi(s_i, r_k) = (1 - s_i)^2(1 + O(s_i^2 + r_k))$ *and* $\Delta_k(s_i) = o(\nu_i/\omega)$*—and the reduced form is exact:*

$$R_k(\omega, s) = c' \, \frac{\omega}{2 - \omega} + \frac{\nu_\Sigma^2}{\omega^2} + o(1), \qquad c' = \sigma^2 \, d_{\text{eff}}(\rho), \qquad d_{\text{eff}}(\rho) := \sum_{i=1}^{D} (1 - s_i)^2,$$

which is strictly convex with unique minimizer $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$ *to leading order (for* $\nu_\Sigma = 0$ *the optimum is the boundary* $\omega^* = 0$*). The runtime estimator uses the mean-field form* $D(1 - \bar{s})^2$ *(exact on isotropic spectra, conservative otherwise); it recovers the drift energy from the two-scale residual excess,* $\widehat{\nu}_\Sigma^2 = \max\big(0, \, (\widehat{E}_t - \hat{c}' \, \omega/(2 - \omega)) \, \omega^2\big)$ *with* $\widehat{E}_t = \widehat{\sigma}^2_{\text{fast}} - \widehat{\sigma}^2_{\text{slow}}$*, and under stationarity* $\widehat{E}_t \to 0$ *in expectation, pinning* ω *at* $\omega_{\min}$ *(Appendix A).*

Corollary 1 (Mixed-drift master law) *For target increments* $\Delta w_t^* = \nu + \vartheta_t$ *with coherent velocity* ν *and martingale-difference innovation power* $q_\Sigma = \mathbb{E}[\vartheta_t^\top \Sigma \, \vartheta_t]$*, the exact tracking lag is* $\mathcal{L}_{\text{lag}}(\omega) = q_\Sigma/(\omega(2{-}\omega)) + \nu_\Sigma^2/\omega^2$*, giving* $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$ *in the coherent regime (*$q_\Sigma = 0$*; Theorem 2) and the classical* $\omega^* = (q_\Sigma/c')^{1/2}$ *random-walk law of Widrow et al. (1976); the runtime estimator inverts the total lag, converging to the appropriate regime automatically (Appendix A).*

Proposition 1 (Anchored risk and the optimal anchor strength) *Let* v *be the data-path variance and* δ^2 *the anchor conflict in a given direction. (i) The anchored risk profile is* $R(s) = (1 - s)^2 v + s^2 \delta^2$*; with a correct anchor (*$\delta^2 = 0$*) it is strictly below* v *for every* $s > 0$*, with the largest gain in weakly excited directions. (ii)* $R(s) = v\delta^2/(v + \delta^2) + (v + \delta^2) \, (s - v/(v + \delta^2))^2$*, so the optimal pull is* $s^* = v/(v + \delta^2)$ *and the corresponding per-direction strength is* $\rho_i^* = \sigma^2/(n \, \delta_i^2)$*, where* δ_i^2 *is the expected anchor conflict. The optimal anchor strength thus decreases in measured conflict and in* n*.*

Proposition 2 (Minimax safety of conflict overestimation) *At a fixed conflict budget, the minimax and maximin values coincide (the parallel sum of the variance and the budget). Overestimating conflict is safe. Under a calibrated overestimate, the excess of the anchored risk over the no-anchor risk is at most* $(v + \delta^2)(s - s^*)^2$*. Underestimating conflict costs quadratically in the same expression; the twice-the-optimum boundary is the James–Stein domination condition (James & Stein, 1961).*

Proposition 3 (False-alarm calibration) *Under a calibrated sub-exponential null, the thresholded trigger has false-alarm rate at most its nominal level (a Freedman-type bound (Freedman, 1975)); it calibrates the running two-scale statistic, with measured rates in Section 5.3 and proof in Appendix A. The variance-floor subtraction and conflict floor induce strictly positive effective thresholds* $\tau_\omega = \hat{c}' \omega_{\min}/(2{-}\omega_{\min})$ *and* $\tau_\rho = C_{\text{floor}}$*, so the guarantee applies at* $\tau = \min(\tau_\omega, \tau_\rho) > 0$*: for the windowed test, stationarity pins* ω_t *at* $\omega_{\min}$ *with probability at least* $1 - \delta$*. The implemented EMA trigger is the surrogate calibrated at these thresholds (Appendix D).*

Proposition 4 (No-regret wrappers) *(i) The cumulative loss is convex in the pull profile* $s \in [0, 1]^D$ *(affine in* s *by Equation* (1) *in the instantaneous eigenbasis* U_t *of* $\bar{A}_t$*, with the minimum-ridge data solve* $\widehat{w}_t = (\bar{A}_t + \rho_{\text{ridge}} I)^{-1} \bar{b}_t$*, then squared), so projected gradient descent on the profile attains regret at most* $G_B \sqrt{2DT}$ *against every fixed profile in the box—including the data path (*$s = \mathbf{0}$*), the anchor path (*$s = \mathbf{1}$*), and, when the accumulator is quasi-stationary, every fixed scalar strength* ρ*—for arbitrary adaptive bounded losses (Zinkevich, 2003). A fixed profile* s *is basis-relative: when* U_t *rotates it corresponds to the time-varying physical operator* $U_t \, \mathrm{diag}(s) U_t^\top$*, so the comparator is the realized profile sequence rather than a fixed linear operator in feature space (Appendix A.10). The* $s = \mathbf{1}$ *comparator is the learner-dragged anchor* G_t *(external regret on the interpolation segment); the frozen* G_0 *enters only in (ii), where the counterfactual guarantee against it lives. (ii) Exponentially weighted mixture prediction over* K *expert paths (batch-solved leaky windows plus the frozen anchor) satisfies* $\sum_t \ell_t(\text{mix}) \le (\min_i L_{i,T} + \ln K/\eta)/(1 - \eta M)$ *for arbitrary bounded adaptive losses; the comparator class contains every benchmarked window and the anchor, a best-of-both-worlds statement (Appendix A).*

Proposition 5 (Information-form conditioning bound) *At any window and for any data, $\kappa(A+n\rho I) \le 1+\mu_{\max}/\rho$, independent of the drift. The anchored solve never iterates an inverse, so it needs no implementation care at short windows; Section 5 measures what happens when the standard covariance recursion does.*

Proof. The eigenvalues of $A+n\rho I$ lie in $[\,n\rho,\ n(\mu_{\max}+\rho)\,]$, whose ratio is $1+\mu_{\max}/\rho$. □

Theorem 3 (Spectral retention and the streaming risk) *Let the feature vectors have independent sub-Gaussian entries with population spectral measure H supported on $[0,\tau_{\max}]$ whose nonzero part lies in $[\tau_{\min}^{+},\tau_{\max}]$, $\tau_{\min}^{+}>0$ (a possibly rank-deficient population covariance), and let $\zeta = D\omega$ be held fixed as $D\to\infty$ (proportional regime). (i) The average data retention $q = 1-\bar{s}$ is the unique solution in $(0,1)$ of*

$$\begin{aligned} q &= \int \frac{\tau}{\tau+\rho_{\text{eff}}}\, dH(\tau), \\ \rho_{\text{eff}} &= \rho\,\frac{e^{\zeta q}-1}{\zeta q}, \end{aligned}$$

and the anchored one-step excess prediction risk converges to $\sigma^2\,\Upsilon/(1-\Upsilon)$ with $\Upsilon = \frac{1}{q}\int\big(\frac{\tau}{\tau+\rho_{\text{eff}}}\big)^2 dH(\tau)\,\big(1-\frac{1-e^{-\zeta q}}{\zeta q}\big)$. (ii) The pull sensitivities are closed form: $\rho\,\partial\bar{s}/\partial\rho = \mathcal{I}\,/\,(1+\zeta\,\phi(\zeta q)\,\mathcal{I})$, with $\mathcal{I} = \int \tilde{s}(1-\tilde{s})\,dH(\tau)$, $\tilde{s}(\tau) = \rho_{\text{eff}}/(\tau+\rho_{\text{eff}})$, and $\phi(u) = \frac{1}{1-e^{-u}} - \frac{1}{u}$. (iii) Under an anchor conflict $\delta = G - w^*$ with direction-weighted conflict measure H_δ (atom δ_i^2 at direction τ_i), the full excess prediction risk converges to

$$R_{\text{pred}} \;\longrightarrow\; \frac{1}{1-\Upsilon}\left[\sigma^2\Upsilon + \rho_{\text{eff}}^2\int \frac{\tau}{(\tau+\rho_{\text{eff}})^2}\, dH_\delta(\tau)\right].$$

The oracle anchor recovers Part (i), and the classical limit $\zeta\to 0$ recovers Proposition 1's per-direction conflict bias $\mu_i(\rho/(\mu_i+\rho))^2\delta_i^2$. Part (i) is therefore the variance component of the risk, while Part (iii) is the full risk.

The convergence step is the separable-covariance deterministic equivalent (Silverstein & Bai, 1995; Rubio & Mestre, 2011; Hachem et al., 2007); the nearest prior analysis, the large-system exponentially windowed RLS of Xiao & Honig (2005), has no anchor, pulls, refresh, or online calibration, and the ridge-risk line (Hastie et al., 2022) treats unweighted batches. The finite-window spreading is a single renormalization $\rho \mapsto \rho_{\text{eff}} \ge \rho$: the formula matches simulation within 4.4% across the 45-point fixed-point grid (including the rank-deficient spectrum) and the measured operating pairs within 13%. It is a correction to the analysis, not a term fed back into the runtime rule: Algorithm 1 keeps the mean-field plug-in $\hat{c}_t'$, and Theorem 3 quantifies what that forgoes as ζ grows. Rank-deficient populations are not a boundary: because the out-of-sample risk weights every coordinate by Σ, the null directions drop out and the deterministic equivalent restricts exactly to range(Σ), with the aspect ratio scaled by the rank fraction (Corollary A.1). Part (ii) supplies the analytic coupling product used in Theorem 4: in logarithmic coordinates the product is at most $\frac{4}{3(2-\omega)}\,\bar{s} < 1$ for all $\omega \in (0, 2/3)$ (Appendix D).

The per-direction variant inherits the Davis–Kahan eigenvector fragility (Davis & Kahan, 1970; Wedin, 1972): it degrades on flat and rotating spectra (Section 6), while the scalar method is invariant to basis rotation (Proposition A.3, Appendix A). Variable forgetting factor RLS (Fortescue et al., 1981; Paleologu et al., 2008) adapts its factor from the squared prequential residual; in covariance form it suffers the same short-window instability (Section 5).

Theorem 4 (Closed-loop stability of the two estimators) *Let the window and strength errors satisfy the coupled recursions of Appendix A (estimator gains η_ν, η_C; target sensitivities $L_{\omega\rho}, L_{\rho\omega}$; perturbations bounded by ξ). The recursions model the quasi-static two-time-scale regime: the smoothing horizons $1/\eta_C, 1/\eta_\nu$ are slow relative to the fast EMA horizon h_f, and, in the drift-active interior where the window is open ($1/\omega \lesssim k/\eta_\nu$), to the accumulator mixing time $1/\omega$; finite-k lag then enters as an additive perturbation absorbed into ξ_ω, ξ_ρ, so the contraction and tracking radius are uniform in its bound. Outside that interior*

the detector floor pins $\omega_t = \omega_{\min}$ (Theorem 2), the pinned specialization of part (ii). (i) If $L_{\omega\rho}L_{\rho\omega} < 1$, the pair contracts geometrically in a weighted max-norm, with explicit weight and rate below one, to within $\xi/(1-\gamma_{\mathrm{cl}})$ of the coupled fixed point. (ii) When either quantity is pinned (the drift-active lower bound on ρ, or the window floor or cap on ω), the system is triangular and the free quantity's contraction governs alone. (iii) The stationary fluctuation obeys $\limsup_m \mathbb{E}[V(m)^2] \le q_*/(1-\gamma_{\mathrm{cl}})^2$ for the weighted max-norm V and innovation second moment $q_* \ge \mathbb{E}\|\xi(m)\|_r^2$ (Proposition A.2); the cross-quantity covariance itself is not asserted, since the scalar Lyapunov route is what the contraction gives without matrix-order assumptions.

The proof assembles Theorem 2's exact window contraction with the plug-in rule of Proposition 1. The coupled contraction is verified in `Coupled.lean`; the measured coupling product is below one in all 20 diagnostic runs, with one quantity pinned in three of four configurations (Appendix D).

5 Experiments

5.1 The window law

Protocol. We evaluate 31 configurations at $D = 128$ and $T = 10^4$ over 5 seeds each: a 27-configuration full-factorial grid (three spectra, three drift geometries, three prior qualities) plus four stress tests (loss of persistent excitation, oscillating drift, a stationary control, and a heavy-tailed variant). One fixed control parametrization is used for every SPAR configuration (Appendix F); baselines use the best single hyperparameter in their family. Every baseline family runs through a shared verification step that confirms exact reductions to RLS at the degenerate parameter values.

Pre-drift floor. Before drift, the calibrated window sits at its lower bound and the error is the noise floor: SPAR-Pull holds 1.01–1.14 $\times \sigma^2$ across the three spectra (0.0106 grand mean vs. $\sigma^2 = 0.01$) and varies by 4.1% across oracle, warm, and adversarial priors (0.301, 0.305, 0.313).

Window response. On the abrupt-jump stream the window opens after the jump and the strength falls; Appendix Figure D.1 traces both. A transient sweep (Appendix Table D.5) shows that the adaptive strength stays within 1–5% of the best fixed strength at cadence 10, while a large fixed anchor costs 87% overall. Appendix Figure D.2 sweeps the window and locates the sharp degradation at short windows, confirming that the calibrated method stays on the stable side.

Finite-D correction. The deterministic-equivalent formula matches the Monte Carlo stationary risk within 4.4% across all 45 operating points of the grid, including the rank-deficient spectrum (Appendix A.13.8), within 13% at the method's own operating points, and the conflict increment within 9.5%. Rank-deficient spectra reduce exactly to the active subspace (Corollary A.1); Section 6 records the conditions.

5.2 Fixed-protocol comparison

Table 1 reports the fixed-protocol comparison. Two oracle-free families lead the mean: the slow-horizon estimated-noise Kalman (0.215) and nominal-length VFF (0.267). Both are static-target specialists; SPAR-Pull wins all nine covariance-rotation configurations (0.011 vs. 0.023 and 0.027). On moving targets the unanchored information-form leaky method wins 12 of the 18 configurations. The leading baseline's margin rests on hindsight: its process-noise level is tuned post hoc on the evaluation grid, and selecting it causally from the stationary pre-drift segment alone instead degenerates the estimated-noise Kalman to mean MSE 1.69 against the post-hoc 0.215 (Appendix D). The calibrated SPAR methods attain 0.297–0.306 mean MSE with 1.22 worst-case, and no SPAR method diverges, while the unsymmetrized covariance-form recursion diverges at short windows (135/135 main-grid and 20/20 stress-suite runs). Per-step symmetrization stabilizes the covariance form on full-rank spectra (Bierman, 1977; Sayed, 2008) but not on rank-deficient spectra, where the dead directions inflate deterministically; the anchored information form avoids the issue entirely (Proposition 5).

The stability separation between the two solve forms has a numerical origin. In covariance form, reducing λ from 0.997 (nominal window 333) to 0.99 (window 100) raises the 27-configuration mean MSE from 0.555 to 182 and then to 1.2×10^6 (diverging in all 135 main-grid and 20 stress-suite runs). Two mechanisms contribute:

Table 1: Factorial grid (27 main configurations × 5 seeds): mean, geometric-mean, and worst overall MSE (worst single run of 135; the geometric mean is scale-invariant across the three spectra). Baselines use their family's best single hyperparameter, selected post hoc on the evaluation grid; the causal Kalman instead selects its q from the stationary pre-drift segment alone. The true-noise Kalman is an oracle comparator (best single q, 0.204/0.86); the WiSE-FT envelope sweeps α per configuration. VFF-RLS covariance reports the nominal-length calibration's best setting ($N_0 = 100$); the single-step calibration $\xi = \sigma_0^2$ diverged in all 135 runs (Appendix D). Standard errors of the mean over the 135 runs are 0.026–0.050, so aggregate differences below about 0.05 are not resolved by the mean alone; the paired tests use configuration-level pairing.

method	mean MSE	geo. mean	worst MSE
baselines (single hyperparameter, tuned post hoc on the evaluation grid)			
leaky RLS ($\lambda = 0.997$)	0.555	0.233	2.12
leaky RLS, information form ($\lambda = 0.99$)	0.342	0.188	1.19
SGD (lr = 0.01)	0.377	0.147	1.29
Adam (lr = 0.01)	0.503	0.291	1.92
directional forgetting	1.81	0.567	7.54
AROW	2.74	0.448	10.55
Kalman, R estimated online (100-step horizon)	1.09	0.265	5.36
Kalman, R estimated online (20,000-step horizon)	0.215	0.090	0.93
VFF-RLS, covariance form ($N_0 = 100$) (Fortescue et al., 1981)	0.267	0.125	0.98
causal baseline (no test-grid tuning)			
Kalman, R estimated online (causal q from warm-up)	1.69	0.266	9.42
regret-guaranteed methods (one fixed configuration)			
OGD, scalar pull (segment)	0.585	0.226	2.16
OGD, D-dim profile (Proposition 4(i))	0.707	0.324	2.61
oracle comparator			
Kalman, best single q, true σ^2	**0.204**	**0.089**	**0.86**
ours (closed-form streaming, one fixed configuration)			
SPAR-EG (Proposition 4(ii))	0.358	0.144	1.27
SPAR-EG$^+$ (pool incl. SPAR-Pull)	0.297	0.099	1.22
SPAR-Pull	0.306	0.100	1.22

whatever the rank-one downdate cannot see inflates by $(1-\omega)^{-t}$. On full-rank spectra the unseen component is roundoff: symmetrizing the iterated inverse restores stability at every window tested (0.61 at $\lambda = 0.99$, $D = 128$). On rank-deficient spectra it is the dead directions themselves, which inflate deterministically even under symmetrization (stable on all 90 full-rank runs, divergent on 42 of 45 rank-deficient). The anchored information form never iterates an inverse and is immune to both. Neither directional forgetting nor any SPAR method diverged in any configuration (Appendix Figure D.2).

Exponentiated-gradient weights over $K = 5$ expert paths (batch-solved leaky windows plus the frozen anchor), with the anytime rate $\varsigma_t = \sqrt{\ln K / S_t}$, attain 0.358/1.27 with no tuning, within 17% of the plug-in rule; on oscillating streams the mixture is within 2.4% of the best single window (6.36 vs. 6.21). Adding the SPAR-Pull trajectory itself as a sixth expert (SPAR-EG$^+$) attains 0.297 mean MSE (1.22 worst-case), 2.9% below the plug-in method, so the Proposition 4(ii) guarantee holds relative to a pool containing the plug-in method. The gain appears in 11 of 27 configurations (paired t $p = 0.032$; Wilcoxon $p = 0.61$).

5.3 Ablation and cost

On four representative configurations, removing the window estimator, the two-scale detector, or the refresh costs +189–190% overall MSE (the first two pin $\omega_t = \omega_{\min}$), and $\gamma = 0.5$ costs +560%; freezing the strength rule *lowers* mean MSE by 2.2%. Appendix Table D.4 localizes prior immunity: freezing the spectral anchor refresh costs 2.2× on warm-prior configurations and 7.4× on adversarial ones (early MSE 47.0 vs. 16.4), while freezing ρ leaves both within 2.6%. The strength rule's contribution is therefore the onset guard of Section 5, not a continuous tracking gain. SPAR-Pull solves step (viii) exactly each step ($O(D^3)$); SPAR-Pull-D caches the refresh eigenbasis and runs at $O(D^2)$ per step with $O(D^3)$ every k refreshes (52.6 vs. 241 ms at $D = 2048$; Appendix Table D.6).

An abrupt-jump strength sweep locates the value of the rule (Appendix Table D.5): a large fixed anchor, the choice a good warm prior invites, multiplies pre-drift MSE by 4.2 relative to a small fixed anchor and costs

87% overall at cadence 10, while the adaptive rule stays within 1–5% of the best fixed strength. The 50-step post-jump window is dominated by the jump itself and is flat across the adaptive and small-anchor policies, so ρ_t acts as a guard against an oversized anchor, not as a post-jump accelerator.

The false-escalation rate of Proposition 3 is at or below nominal except at $\delta = 0.01$ on ill-conditioned spectra, as the sub-exponential tail predicts. The OGD regret slopes (-0.65 isotropic, -1.08 ill-conditioned) lie within the $O(T^{-1/2})$ rate, with pre-drift dominance and prior immunity in Appendix Figures D.3 and D.4b (Appendix D).

5.4 Robustness outside the model class

Three generators outside the linear-Gaussian, constant-rate class: nonlinear targets ($y = \tanh(x^\top w^*) + \varepsilon$), covariate shift (feature scale doubles mid-stream), and a noise-regime change (σ^2 rising from 0.01 to 0.05, no target drift). The SPAR-EG mixtures lead on nonlinear targets and are best-or-tied under the noise-regime change; under covariate shift the Kalman family leads (0.268 true-noise, 0.284 estimated-R), with SPAR-EG$^+$ and SPAR-Pull ahead of the anchored and leaky methods. No SPAR method diverged under any mismatch, while the $\lambda = 0.995$ leaky method degraded sharply on nonlinear targets (numbers in Appendix D).

5.5 A real high-dimensional stream

We ran the grid-default configuration unchanged on a public hourly electricity-load panel (Trindade, 2015; Lai et al., 2018): $D = 324$ features (321 client loads plus hour-of-day terms), a 2,000-hour warm-up and a 6,000-hour evaluation horizon, with three held-out clients (the 25th, 50th, and 75th percentiles by mean load) as targets. Across the three clients SPAR-EG$^+$ stays within 1.2% of the best oracle-tuned leaky window (and matches it on the three-client mean, 0.0136) and SPAR-Pull within 4–17%, with no stream-specific parameter, while the slow estimated-R Kalman trails over the full horizon (three-client means 0.051 and 0.257 at $q = 10^{-5}$ and 10^{-4}). The stream is close to the smooth long-window regime, where the analysis predicts that a fixed long window is nearly optimal (Appendix D.10).

5.6 BCI IV cross-session evaluation

Nine subjects each from BCI IV 2a (Brunner et al., 2008; Tangermann et al., 2012) and 2b (Leeb et al., 2008), streamed from the training session against Riemannian baselines, backprop references, and our heads (Appendix E). The head uses a reduced window range for the short evaluation streams; every other constant is the grid default. On 2b, the transductive SPAR-EG head reaches 0.816 and SPAR-Pull 0.808; the strictly causal SPAR-Pull head attains 0.801, against 0.811 for the causal WiSE-FT reference taken at the best of an 11-value post-hoc α-envelope (Appendix E). On 2a, offline alignment leads the fixed readouts (0.664), with the post-hoc ensemble at 0.668; the best streaming method is 0.627. Both datasets have $T \approx 300$ trials, below the plug-in crossover horizon $T_{\text{crit}} \asymp K^2/(s_0 - s^*)^2$ at which calibration overtakes fixed shrinkage (Appendix A.7), consistent with the fixed-anchor head leading on 2a and the mixture leading on 2b.

6 Scope and Boundary Conditions

Table 2 collects the conditions under which the claims hold; the appendix gives the corresponding protocols and numbers. The central laws (Theorems 1 and 2) are mean-field statements, and Theorem 3 is the finite-D correction on full-rank spectra. The empirical claims are about the calibrated method on the streams studied, not about universal dominance.

7 Conclusion

The finite-cadence refresh equilibrium is the central object: it gives the tracking lag and variance reduction of an anchored stream in closed form, and the optimal forgetting window follows by minimizing the cycle-averaged risk. The random-matrix correction extends the law to finite dimension, where the anchor is renormalized by the finite window; the correction is accurate on full-rank and rank-deficient spectra, the latter reducing

Table 2: Scope of the claims. Each row names a condition, the protocol that tests it, and the consequence outside the condition.

dimension	assumption or protocol	consequence outside it
design model	Theorems 1–2 use the mean-field accumulator; Theorem 3 retains the random design	the mean-field target is a control law at $\zeta \leq 6.4$ (drift phase); the finite-D correction is the relevant statement there
spectrum	Theorem 3 assumes the nonzero part is bounded away from zero, $\tau^{+}_{\min} > 0$; rank-deficient spectra reduce exactly to range(Σ) (Corollary A.1)	null directions drop out of the prediction risk, so the deterministic equivalent holds once the diagnostic spectrum matches the generator's active directions
contraction	Theorem 4 is local at an interior fixed point with the clip inactive	the global statement keeps the measured-product hypothesis and the gate
moving targets	fixed-protocol grid, 27 configurations	unanchored information-form leaky RLS wins 12/18 moving-target configurations; SPAR wins all 9 static-target covariance-rotation configurations
strength rule	onset guard; average effect measured by ablation	freezing it lowers mean MSE by 2.2%; its value is the transient guard, not a continuous tracking gain
detector timescale	two-scale residual EMA, $h_f = 30$, $h_{\text{slow}} = 20{,}000 \approx 2T$	a drift persisting beyond h_{slow} relaxes both averages to a common level and the detector stops opening the window
baseline protocol	best single hyperparameter per family	SGD at lr $= 0.05$ diverges on all 45 isotropic runs; the causal-q Kalman selects $q = 10^{-6}$ on the stationary pre-drift segment and is degenerate (Appendix D)
BCI IV	9 subjects per dataset; reduced window range for the short streams; causal and transductive alignment reported separately	0.816 uses session-level transductive alignment; 0.801 is the causal score, against 0.811 for causal WiSE-FT
electricity	three held-out clients (25th/50th/75th percentile by mean load)	a percentile sweep, not a random-client population; the script accepts any client indices
Lean artifact	algebraic core, refresh chain, regret bounds, coupled contraction	probabilistic concentration, the moving-frame reduction, RMT convergence, and Davis–Kahan are pen-and-paper (Appendix C)

exactly to the active subspace. The Lean artifact formalizes the algebraic core, the refresh chain, the regret bounds, and the coupled contraction.

One question remains open. The closed-form stationary law for the coupled estimators is not known; the moment recursion is the current analytic handle. It does not affect the calibrated method on the streams studied here.

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A Proofs of Section 4

This appendix proves the results of Section 4 in full, restating each numbered claim or setting it up in the notation of its proof. Throughout, $(x_t, y_t)_{t\geq 1}$ is a stream with $x_t \in \mathbb{R}^D$ obeying $y_t = x_t^\top w_t^* + \varepsilon_t$, where $\mathcal{F}_t$ is the σ-algebra generated by the past, $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_t] = 0$, and $\mathbb{E}[\varepsilon_t^2 \mid \mathcal{F}_t] \leq \sigma^2$. The learner maintains sufficient statistics

$$A_t = \alpha A_{t-1} + x_t x_t^\top, \qquad B_t = \alpha B_{t-1} + x_t y_t, \qquad n_t = \alpha n_{t-1} + 1,$$

with forgetting $\alpha \in (0, 1]$ and initialization from a warm-up set of T_w samples. Write $\bar{A}_t = A_t/n_t$ with eigendecomposition $\bar{A}_t = U \operatorname{diag}(\mu_1, \ldots, \mu_D)\, U^\top$ and $\bar{b}_t = B_t/n_t$; an index i always refers to the coordinate in this eigenbasis. *Two design models.* Lemma 1 and Theorems 1–2 are proved in the *mean-field* model $\bar{A}_t \equiv \Sigma$, i.e. the random accumulator is replaced by its population expectation; this is the model in which the lag, the variance factor Ψ, and the window target are exact. Theorem 3 is the finite-D *random-design* correction: it retains the random $\bar{A}_t$ and takes the proportional limit $\zeta = D\omega$ fixed. The two agree as $\zeta \to 0$; the grid's pre-drift windows have $\zeta \leq 0.13$, while drift-phase windows reach $\zeta \leq 6.4$, where the mean-field target is a control law, not the exact optimum (Appendix A, "Regime of the first-order variance term"). The anchored solve with anchor G_t and strength $\rho > 0$ is $W_t = (A_t + n_t\rho I)^{-1}(B_t + n_t\rho G_t)$; in the eigenbasis, coordinate i is the scalar problem

$$W_{t,i} \;=\; \frac{\bar{b}_{t,i} + \rho\, G_{t,i}}{\mu_{t,i} + \rho} \;=\; (1 - s_i)\, \widehat{w}_{t,i} + s_i\, G_{t,i}, \qquad s_i \;=\; \frac{\rho}{\mu_{t,i} + \rho}, \tag{2}$$

where $\widehat{w}_{t,i} = \bar{b}_{t,i}/\mu_{t,i}$ is the per-direction data solution. Under the warm-up model the data solution is unbiased per direction with variance $v_i = \sigma^2/(n_t\mu_{t,i})$. Two parametrizations of the anchored family are used: the *scalar-ρ* family (one ρ; the pull $s_i = \rho/(\mu_i + \rho)$ is then spectrum-adaptive) and the *uniform-s* family (one pull s for all directions, i.e. $\rho_i = \mu_i s/(1 - s)$; the solution path is the segment between $\widehat{w}$ and G). Both satisfy the pull identity, which underlies every result below.

A.1 Proof of Equation (1) (pull identity)

Equation (1) (restated). Let $\mu > 0$, $\rho > 0$, and $b, g, w, \gamma \in \mathbb{R}$. Define $s = \rho/(\mu + \rho)$, $\widehat{w} = b/\mu$, and $W = (b + \rho\gamma g)/(\mu + \rho)$. Then, exactly,

$$W - w \;=\; (1 - s)\,(\widehat{w} - w) \;+\; s\,(\gamma g - w).$$

In particular, for $\gamma = 1$ the anchored error is a convex combination of the data error and the anchor error, per direction.

Proof of Equation (1). Since $1 - s = \mu/(\mu + \rho)$,

$$(1 - s)\,\widehat{w} + s\,\gamma g \;=\; \frac{\mu}{\mu + \rho} \cdot \frac{b}{\mu} + \frac{\rho}{\mu + \rho} \cdot \gamma g \;=\; \frac{b + \rho\gamma g}{\mu + \rho} \;=\; W,$$

and subtracting $w = (1-s)w + sw$ gives the identity. No approximation is involved. Applying it coordinatewise in the eigenbasis to (2) yields the form stated in Section 4. Both the $\gamma = 1$ and general-γ forms are formalized (`pull_identity`, `pull_identity_general` in `Core.lean`). □

A.2 Proof of Lemma 1 (zero-lag refresh equilibrium)

Lemma 1 (restated). Assume drift at constant rate ν per step in a direction with excitation μ, forgetting α, and continuous anchor refresh ($\beta \to 1$ cadence), with the predict-then-update convention (the newest sample in the accumulator is one step old at prediction time).

(i) The accumulator's implied data solution lags the target by exactly $\nu/(1 - \alpha) = \nu/\omega$: the sample-weighted average age is $\sum_{k\geq 0}(k + 1)\alpha^k \big/ \sum_{k\geq 0} \alpha^k = 1/(1 - \alpha)$.

(ii) At the SPAR refresh equilibrium ($G = W$), the anchored solution's lag equals the pure-leaky lag, independent of s: $W - w^* = -\nu/(1-\alpha)$ exactly, since $W = (\bar{b} + \rho W)/(\mu + \rho)$ forces $W = \bar{b}/\mu = \widehat{w}$.

(iii) With refresh cadence k steps, an extra lag $\Delta_k(s) = s\nu k\big(a/(1-a) + \frac{1}{2}\big)$ appears in direction i, whose prediction cost is weighted by μ_i: because that weight vanishes as $\mu_i \to 0$, the extra lag is concentrated in exactly the directions where holding the prior costs the least.

Proof of Lemma 1. (i) At equilibrium,

$$\mathbb{E}[B_t] \;=\; \sum_{k\geq 0} \alpha^k \,\mu\,\big(w_t^* - (k+1)\nu\big) \;=\; \mu\,\Big(w_t^* - \frac{\nu}{1-\alpha}\Big)\cdot\frac{1}{1-\alpha},$$

so $\widehat{w} - w^* = -\nu/(1-\alpha)$ exactly at the mean-field expectation ($\widehat{w} = \bar{b}/\mu$ is the unanchored data solve and carries no ρ dependence; the centered estimation noise contributes only the variance terms analyzed in Proposition 1). (ii) By the pull identity with $\widehat{w} - w^* = -\nu/(1-\alpha)$ and $G - w^* = W - w^*$ at the continuous-refresh fixed point:

$$W - w^* \;=\; (1-s)\Big(-\frac{\nu}{1-\alpha}\Big) + s\,(W - w^*),$$

hence $(1-s)(W-w^*) = -(1-s)\nu/(1-\alpha)$, so $W - w^* = -\nu/(1-\alpha)$ for any $s < 1$. This algebraic step is formalized (`lag_invariance` in `Core.lean`). (iii) Theorem 1's moving-frame recursion gives the exact cycle-averaged lag $\Delta_k(s) = s\nu k\big(a/(1-a)+\frac{1}{2}\big)$ with $a = s(2-s)$; since $1-a = (1-s)^2$, this is $O\big(ks/(1-s)^2\big)\nu$. The prediction-risk cost of the extra lag is $\mu_i\,\Delta_k(s)^2$, which vanishes as $\mu_i \to 0$ at fixed $s < 1$. □

At the continuous-refresh equilibrium the pull identity forces $W = \widehat{w}$ for $s < 1$: the anchor is inert for mean tracking, so it contributes neither lag nor variance reduction in that limit.

A.3 Finite-cadence refresh equilibrium (Theorem 1)

Work per eigen-direction with pull $s \in (0,1)$ and refresh cadence k; let $t_m = mk$ be the refresh steps, $w^*(t)$ the moving target (velocity ν per step), and $L = \nu/(1-\alpha)$ the pure-leaky lag of the data path (Lemma 1(i)). The refresh reads the post-update solve, so the lag entering the recursion is the post-update value $\nu(1-\alpha)/\alpha$, one step shorter than the predictor lag; the cycle-averaged lag relative to the pure-leaky path, $\Delta_k(s)$ below, is unchanged by this one-step offset. Between refreshes the anchor G_m is frozen, and the anchored solution at $t \in [t_m, t_{m+1})$ is, in the moving frame,

$$W_t - w^*(t) \;=\; (1-s)\,(-L) \;+\; s\,\big(g_m - \nu\,(t - t_m)\big), \qquad g_m := G_m - w^*(t_m),$$

where $\nu(t - t_m)$ is the target's advance since the last refresh, invisible to the frozen anchor. At the refresh, $G_{m+1} = (1-s)\,W_{t_{m+1}} + s\,G_m$ measured against the new target position, which yields the affine recursion

$$g_{m+1} \;=\; a\,g_m - (1-s)^2\,L - a\,\nu k, \qquad a := (1-s)s + s = s(2-s) = 1 - (1-s)^2,$$

with $a \in (0,1)$ for every $s \in (0,1)$, so the recursion contracts at rate a per cycle to

$$g^* \;=\; -L - \nu k\,\frac{a}{1-a}, \qquad g_m \;=\; g^* + a^m\,(g_0 - g^*).$$

The rate identity, the fixed-point equation, the closed form, and the contraction are formalized (`refresh_rate_identity`, `refresh_fixed_point`, `refresh_closed_form`, `refresh_contraction_rate`, together with the moving-frame reduction and the midpoint cycle assembly, in `Core.lean`); we also verified the fixed point and the cycle-averaged lag against a one-dimensional simulation at two operating points. The geometric steady-lag form parallels the iterate-averaging analyses under drift (Polyak & Juditsky, 1992; Cutler et al., 2023) and the per-direction momentum lag analysis of Hong & Qu (2026). The cycle-averaged solution lags the pure-leaky path by

$$\Delta_k(s) \;=\; s\,\nu k\Big(\frac{a}{1-a} + \frac{1}{2}\Big) \;=\; \frac{s\,\nu k\,(1 + 2s - s^2)}{2\,(1-s)^2},$$

linear in the cadence and vanishing as $s \to 0$; the within-cycle residual adds a variance term of order $s^2\nu^2k^2/12$. The first summand is the anchor's accumulated staleness—each cycle's drift persists into the anchor with weight a^j, summing to $a/(1-a)$—and the second is the within-cycle advance, which makes the within-cycle correction exact and adds the geometric term. The continuous-refresh limit of Lemma 1(ii) is the instantaneous version of the same equilibrium: at the refresh instant the anchor adds no lag beyond L, and the correction $\nu k\, a/(1-a)$ is precisely what the target's advance between refreshes contributes.

The data-path noise does not decorrelate between refreshes: the leaky accumulator gives $\mathbb{E}[\epsilon_m \epsilon_{m-j}] = r_k^j v$ with $r_k = \alpha^k$ ($\alpha = 1-\omega$ the forgetting factor), which at the nominal constants ranges over $r_{10} \in [0.60, 0.995]$. The correlation is material at these operating points. Writing $\tilde{g}_m$ for the zero-mean anchor fluctuation, the refresh rule gives the stochastic recurrence $\tilde{g}_m = (1-s)^2\epsilon_m + a\,\tilde{g}_{m-1}$, whose steady-state cross-covariance against the fresh noise is the geometric sum

$$C(s, r_k) \;:=\; \mathbb{E}[\epsilon_m \tilde{g}_{m-1}] \;=\; (1-s)^2 \sum_{j \geq 0} a^j r_k^{j+1} v \;=\; \frac{(1-s)^2\, r_k}{1 - a r_k}\, v.$$

Solving the steady-state variance system $q = (1-s)^4 v + a^2 q + 2a(1-s)^2 C$ then gives the exact pair

$$q(s, r_k) \;=\; \frac{(1-s)^2}{1+a} \cdot \frac{1 + a r_k}{1 - a r_k}\, v,$$

$$\mathrm{Var}(W) \;=\; (1-s)^2 v + s^2 q + 2s(1-s)\, C \;=\; v\left(1 - \frac{2s\,(1 + s - s^2)\,(1 - r_k)}{(1+a)\,(1 - a r_k)}\right),$$

strictly below v for every $s \in (0,1)$ and $r_k \in [0,1)$ (`refresh_anchor_variance`, `refresh_variance_form`, `refresh_variance_lt` in `Core.lean`); we verified the identities symbolically and the refresh-instant variance against one-dimensional simulations at six operating points (deviation $\leq 1\%$); this is the variance at the refresh step, and the cycle-averaged variance is within a few percent of it at the operating pulls. The lower endpoint is exact as well: subtracting $(1-s)^2$ from the bracket factorises as

$$\Psi(s, r_k) - (1-s)^2 = \frac{s\,(1-s)^2\big(r_k\,(s^3 - 4s^2 + 2s + 2) + s\big)}{(1+a)\,(1 - a r_k)},$$

and $s^3 - 4s^2 + 2s + 2 = 1 + (s-1)(s^2 - 3s - 1) \geq 1$ on $[0,1]$, so the numerator is nonnegative and $\Psi(s, r_k) \geq (1-s)^2$ (`refresh_variance_lb` in `Core.lean`). Hence the frozen-anchor plug-in c' lower-bounds the exact variance and the window estimator errs toward larger, safer windows. Both endpoints are exact: $r_k \to 0$ (refresh cycles long against the noise memory) recovers the uncorrelated reduction $v[1 - 2s(1 + s - s^2)/(1+a)]$, and $r_k \to 1$ (continuous refresh) gives $\mathrm{Var}(W) \to v$—the inert-anchor limit of Lemma 1(ii), now an exact endpoint of a single formula rather than a separate statement. The cadence therefore trades two real costs: refreshing fast lets the anchor co-move with the data path and erases its variance benefit, refreshing slowly inflates the lag term $\mu_i\, \Delta_k(s)^2$, and the default $k = 10$ falls within the tradeoff. The two costs are paid in different directions: the excess lag blows up as $s \to 1$, but the directions with $s \approx 1$ are precisely the weakly excited ones ($\mu_i \ll \rho$), where the lag's risk contribution is negligible—the filter-bank property of the $\beta_i = 1 - s_i$ refresh, and the reason Proposition 1 drives ρ^* down, and the lag with it, exactly when measured conflict grows.

Remark (drift-induced cross term). Under nonzero drift velocity the risk decomposition $R(s) = (1-s)^2 v + s^2 c$ of Proposition 1 acquires the cross term $2(1-s)s\,\mathbb{E}[(\widehat{w} - w^*)(G - w^*)]$ between the lagged data solve and the anchor error. Cauchy–Schwarz bounds it through the two quantities the theory already controls: $|\mathbb{E}[(\widehat{w} - w^*)(G - w^*)]| \leq \sqrt{\mathbb{E}[(\widehat{w} - w^*)^2]\,\mathbb{E}[(G - w^*)^2]}$, where the first factor is the lag-induced gap (bounded per direction by Theorem 1's equilibrium: the moving-average gap $\bar{\Delta}_k \leq (1-s)^2\nu k/(1-a)$ at cadence k) and the second is the anchor conflict c. The cross term therefore inflates the effective conflict to $c_{\text{eff}} = c + 2(1-s)s\sqrt{\bar{\Delta}_k\, c}$ at most, and Proposition 2 applies unchanged: overestimating c_{eff} remains minimax-safe, so the calibration rule $\rho^* = \bar{\mu}\widehat{V}/\widehat{C}$ with the conservative conflict estimate retains its safety property under drift.

A.4 Proof of Theorem 2 (optimal forgetting window)

Theorem 2 (restated). Let the target drift at constant rate ν_i per step in direction i with prediction-units drift power $\nu_\Sigma^2 := \sum_i \mu_i \nu_i^2$, let the forgetting accumulator run at window $\omega = 1 - \alpha$, let the refresh cadence be k with $r_k = (1-\omega)^k$, and let $c' := \sigma^2\, d_{\text{eff}}(\rho)$ with $d_{\text{eff}}(\rho) = \sum_i (1 - s_i)^2$. At the refresh equilibrium of Theorem 1 the cycle-averaged prediction risk is

$$R_k(\omega, s) = \sigma^2 \frac{\omega}{2-\omega} \sum_i \Psi(s_i, r_k) + \sum_i \mu_i \Big(\frac{\nu_i}{\omega} + \Delta_k(s_i) \Big)^2 + O(k^2 \nu^2 s^2),$$

with Ψ and Δ_k as in Theorem 1. At the runtime cadence the surrogate is justified by the variance lower bound $(1-s_i)^2 \le \Psi \le 1$ and the bounded finite-k residual (Step 0); in the quasi-static regime $1 \ll k\omega \ll 1/s$, $\Psi(s_i, r_k) = (1-s_i)^2(1 + O(s_i^2 + r_k))$ and $\Delta_k(s_i) = o(\nu_i/\omega)$, so

$$R(\omega) \;=\; c' \frac{\omega}{2-\omega} \;+\; \frac{\nu_\Sigma^2}{\omega^2} + o(1), \tag{3}$$

which is strictly convex with a unique minimizer given to leading order by

$$\omega^* \;=\; \left(\frac{4\,\nu_\Sigma^2}{c'} \right)^{1/3} \;=\; \left(\frac{4\,\nu_\Sigma^2}{(1-\bar{s})^2 \sigma^2 D} \right)^{1/3}.$$

The drift power is recovered from the two-scale residual excess above the variance floor, scaled by the current window: $\widehat{\nu}_\Sigma^2 = \max\big(0, (\widehat{E}_t - \widehat{c}'\,\omega/(2-\omega))\,\omega^2\big)$ with $\widehat{E}_t = \widehat{\sigma}^2_{\text{fast}} - \widehat{\sigma}^2_{\text{slow}}$, and overestimating the window is safe, mirroring Proposition 2.

Proof of Theorem 2. Step 0 (finite cadence to the surrogate). The finite-cadence risk (3) combines Theorem 1's two exact steady-state objects: the per-direction variance is $v_i \Psi(s_i, r_k)$ with $v_i = \sigma^2/(n_{\text{eff}}\mu_i)$, and the per-direction lag is $\nu_i/\omega + \Delta_k(s_i)$; the within-cycle fluctuation contributes the $O(k^2\nu^2 s^2)$ term derived in Appendix A.3. Because $(1-s_i)^2 \le \Psi(s_i, r_k) \le 1$ for all $r_k \in [0, 1)$ (`refresh_variance_lb`), the frozen-anchor surrogate $c'\,\omega/(2-\omega) + \nu_\Sigma^2/\omega^2$ lower-bounds the exact cycle-averaged variance term at every cadence, and $D(1-\bar{s})^2 \le d_{\text{eff}}(\rho)$ by Jensen; the runtime plug-in therefore underestimates the variance coefficient, returning a window no smaller than the exact-optimal one—the over-forgetting, safety-asymmetric direction of Step 4. The finite-cadence lag residual is bounded at the runtime range: with $b(s) = s(2-s)/(1-s)^2$ and $k\omega \le 1/2$, $\Delta_k(s_i) = s_i \nu_i k\big(b(s_i) + \frac{1}{2}\big) \le \frac{1}{2} s_i \big(b(s_i) + \frac{1}{2}\big)\nu_i/\omega$, a vanishing fraction of the leading lag ($\le 0.4\%$ for $s_i \le 0.1$ at $k\omega = 0.5$) whose prediction cost $\mu_i \Delta_k(s_i)^2$ is concentrated in weakly excited directions. Numerically, over $k\omega \in [0.005, 0.5]$ the surrogate's cubic minimizer is within 2.1% of the exact cycle-averaged optimum for $s_i \le 0.2$ (5.2% for $s_i \le 0.3$), with the largest gaps at the window floor (`diagnostics.py -diag subopt`; `results/window_subopt.csv`). In the quasi-static limit $1 \ll k\omega \ll 1/s$, $r_k = (1-\omega)^k \le e^{-\omega k} \to 0$, so $\Psi(s_i, r_k) = (1-s_i)^2(1 + O(s_i^2 + r_k))$ and $\Delta_k(s_i) = o(\nu_i/\omega)$; both corrections vanish and the reduced form (3) is exact, with remainder $o(1)$ uniformly over the regime.

Step 1 (risk decomposition). For exponential weights α^k the effective sample count of the accumulator is

$$n_{\text{eff}} \;=\; \frac{\big(\sum_{k\ge 0} \alpha^k\big)^2}{\sum_{k \ge 0} \alpha^{2k}} \;=\; \frac{(1/(1-\alpha))^2}{1/(1-\alpha^2)} \;=\; \frac{1+\alpha}{1-\alpha} \;=\; \frac{2-\omega}{\omega},$$

so the estimation-variance contribution to the solve risk is $(1-\bar{s})^2\sigma^2 D/n_{\text{eff}} = c'\,\omega/(2-\omega)$. The tracking contribution is the lag of Lemma 1(i) below: per direction the solve lags the target by ν_i/ω, so the prediction-units lag risk is $\sum_i \mu_i(\nu_i/\omega)^2 = \nu_\Sigma^2/\omega^2$. Summing the two contributions gives (3); to leading order in ω, $R(\omega) = c'\omega/2 + \nu_\Sigma^2/\omega^2 + O(\omega^2)$.

Step 2 (convexity and minimizer). Both terms of (3) are strictly convex on $(0, 2)$, since

$$R''(\omega) \;=\; \frac{6\nu_\Sigma^2}{\omega^4} + \frac{4c'}{(2-\omega)^3} \;>\; 0,$$

so R has a unique minimizer. The first-order condition is $2c'/(2-\omega)^2 = 2\nu_\Sigma^2/\omega^3$, i.e. $c'\omega^3 = \nu_\Sigma^2(2-\omega)^2$; to leading order in ω the right side is $4\nu_\Sigma^2$, giving $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$. For the leading-order (cubic) model $\widetilde{R}(\omega) = c\,\omega + \nu^2/\omega^2$ with the critical condition $\omega^{*\,3} = 2\nu^2/c$ (i.e. $c = c'/2$), optimality holds by the exact factorization

$$\widetilde{R}(\omega) - \widetilde{R}(\omega^*) \;=\; \frac{\nu^2}{\omega^2\omega^{*\,3}}\,(\omega-\omega^*)^2\,(2\omega+\omega^*) \;\geq\; 0,$$

which is formalized (`window_optimum` in `Ogd.lean`); substituting $c = c'/2$ reproduces the 4-form above.

Step 3 (drift inversion). At quasi-static equilibrium the prequential residual excess equals the full steady-state risk—lag and variance together:

$$\mathbb{E}[r_t^2] - \sigma^2 \;=\; c'\,\frac{\omega}{2-\omega} + \frac{\nu_\Sigma^2}{\omega^2} \;=\; R(\omega),$$

because the residual $r_t = y_t - x_t^\top W_{t-1}$ carries both the estimation variance and the lag of the solve. Solving for the drift power at the current window gives

$$\widehat{\nu}_\Sigma^2 \;=\; \max\Big(0,\; \big(\widehat{E}_t - \widehat{c}'\,\tfrac{\omega}{2-\omega}\big)\,\omega^2\Big), \qquad \widehat{E}_t = \widehat{\sigma}^2_{\text{fast}} - \widehat{\sigma}^2_{\text{slow}},$$

where $\widehat{\sigma}^2_{\text{fast}}, \widehat{\sigma}^2_{\text{slow}}$ are the fast and slow residual EMAs. One subtlety is forced by the estimator's own construction: the slow baseline is itself an average of r_t^2, so at stationarity $\mathbb{E}[\widehat{\sigma}^2_{\text{slow}}] = \sigma^2 + \widehat{c}'\omega_{\text{slow}}/(2-\omega_{\text{slow}})$, not the pure noise floor σ^2. Reading the first parenthesis as the two-scale excess $\widehat{E}_t$ (not as $\mathbb{E}[r_t^2] - \sigma^2$ with a hypothetical pure noise-floor estimate) matters because, on drift onset, $\mathbb{E}[\widehat{E}_t] \approx \nu_\Sigma^2/\omega^2 + c'\omega/(2-\omega) - c'\omega_{\text{slow}}/(2-\omega_{\text{slow}})$, and subtracting the current window's floor isolates ν_Σ^2/ω^2 up to the pre-drift floor $c'\omega_{\text{slow}}/(2-\omega_{\text{slow}})$—negligible exactly when the pre-drift window sits at $\omega_{\min}$, the quasi-static regime of Theorem 2. The subtraction still matters: an estimator that omits the floor term, such as $(\widehat{\mathbb{E}}[r_t^2] - \widehat{\sigma}^2)\omega^2$, is strictly increasing in ω (it equals $\nu_\Sigma^2 + c'\omega^3/(2-\omega)$ at equilibrium) and therefore admits no positive fixed point of the estimator iteration $\omega \mapsto \omega^*(\widehat{\nu}_\Sigma^2)$, driving the window to its cap under sustained oscillation, whereas the variance-floor-subtracted estimator is exact at equilibrium and the iteration converges to the minimizer of Step 2; under stationarity $\mathbb{E}[\widehat{E}_t] \to 0$ and the clip pins the window at $\omega_{\min}$, the conservative direction of Step 4.

Two-scale timescale separation. The excess detector presupposes $h_f \ll \omega^{-1} \ll T_{\text{drift}} \ll h_{\text{slow}}$: the fast average settles at the post-onset level while the slow average still holds the pre-drift level, so $\widehat{E}_t$ isolates the drift excess. The default $h_{\text{slow}} = 20{,}000 \approx 2T$ enforces this on the grid; a drift persisting beyond h_{slow} would relax both averages toward the same level and close the window.

Plug-in gap in the window iteration. Theorem 2 is stated with the exact coefficient $c' = \sigma^2 d_{\text{eff}}(\rho)$, $d_{\text{eff}} = \sum_i (1-s_i)^2$, whereas the runtime rule uses the mean-field plug-in $\widehat{c}' = \sigma^2 D(1-\bar{s})^2$. Jensen gives $\sum_i (1-s_i)^2 \geq D(1-\bar{s})^2$, so the plug-in underestimates the variance coefficient; the cubed-coordinate iteration then has gain $2(c'_{\text{true}}/\widehat{c}' - 1)$ at leading order, contracting iff $c'_{\text{true}}/\widehat{c}' < 3/2$. The gap is exactly the population variance of the pulls,

$$\sum_{i=1}^{D}(1-s_i)^2 - D(1-\bar{s})^2 = \sum_{i=1}^{D}(s_i-\bar{s})^2 = D\,\mathrm{Var}(s) \;\geq\; 0,$$

an identity formalized as `deff_sub_plugin` and `deff_sub_plugin_nonneg` in `Core.lean` (Appendix C); it vanishes on an isotropic spectrum and is largest when the pulls are spread out, as on a rank-deficient spectrum whose dead directions carry $s_i = 1$. The ratio is bounded by the spread of the per-direction shrinkages and reaches 2 in the 50%-dead-direction corner, so on strongly rank-deficient spectra the plug-in can drive the window to its cap. The exact d_{eff} is available from the eigenvalues already computed at each refresh, at no extra matrix cost; recording the identity makes explicit that the plug-in is the conservative (over-forgetting) endpoint and that the exact coefficient is a free refinement.

Step 4 (safety asymmetry). Evaluating R at $c\,\omega^*$ with $c > 1$ (drift overestimated, window too short) costs $O(c-1)$ through the linear variance term, while $c < 1$ (drift underestimated, window too long) pays

the quadratic lag blow-up c^{-2}. As in Proposition 2, the conservative choice is to overestimate the adversary. This mirrors the measured flatness of the Kalman q-sweep and the divergence of the leaky λ-sweep one step below its best stable window (Section 5). The tradeoff itself is classical—misadjustment versus lag, from the LMS analysis of Widrow et al. (1976), whose optimum is a square-root law for random-walk drift—and the 1/3-power law for linear drift shares the 2/3 exponent of the dynamic-regret rate of Besbes et al. (2015). The constant is specific to the anchored accumulator, and the prequential inversion of Step 3 allows online estimation without known drift parameters.

Proof of Corollary 1 (mixed-drift master law). For target increments $\Delta w_t^* = \nu + \vartheta_t$ with ϑ_t a zero-mean martingale difference, the exponentially weighted data path at equilibrium has tracking error $\widehat{w}_t - w_t^* = -\sum_{j\geq 0}(1-\omega)^j\,\Delta w_{t-j}^*$ (swap the order of summation: $\sum_{k=j}^{\infty}\omega(1-\omega)^k = (1-\omega)^j$). The expected squared lag in prediction units is then

$$\mathcal{L}_{\mathrm{lag}}(\omega) = \mathbb{E}\Big\|\sum_{j\geq 0}(1-\omega)^j\,\Sigma^{1/2}(\nu+\vartheta_{t-j})\Big\|^2 = \frac{\nu_\Sigma^2}{\omega^2} + \frac{q_\Sigma}{\omega(2-\omega)},$$

where the cross term vanishes (martingale difference) and the geometric sums evaluate in closed form ($\sum_j(1-\omega)^j = 1/\omega$, $\sum_j(1-\omega)^{2j} = 1/(\omega(2-\omega))$). The master risk $R(\omega) = c'\omega/(2-\omega) + q_\Sigma/(\omega(2-\omega)) + \nu_\Sigma^2/\omega^2$ specializes to: (i) the coherent regime ($q_\Sigma = 0$), giving Theorem 2's $\omega^* = (4\nu_\Sigma^2/c')^{1/3}$; (ii) the random-walk regime ($\nu_\Sigma = 0$), giving $\omega^* = (q_\Sigma/c')^{1/2} + O(q_\Sigma/c')$, the classical 1/2-power law. The runtime estimator inverts the total lag without knowing the regime: $\widehat{\nu}_\Sigma^2 = \max\big(0,\,(\widehat{E}_t - \hat{c}'\omega/(2-\omega))\,\omega^2\big)$ with $\widehat{E}_t$ the two-scale residual excess of Step 3 (Appendix A) estimates the drift excess above the variance floor regardless of whether it originates from coherent or stochastic drift.

Contraction in the random-walk regime. When $\nu_\Sigma = 0$ and $q_\Sigma > 0$, the excess above the variance floor is $q_\Sigma/(\omega(2-\omega))$, so the estimator returns $\widehat{\nu}_\Sigma^2(\omega) = q_\Sigma\,\omega/(2-\omega)$ and the cubic map $T_{\mathrm{RW}}(\omega) = (4\widehat{\nu}_\Sigma^2(\omega)/c')^{1/3}$ satisfies

$$\big(T_{\mathrm{RW}}(\omega)\big)^3 = \frac{4q_\Sigma\,\omega}{c'\,(2-\omega)}.$$

In logarithmic coordinates $\frac{d\ln T_{\mathrm{RW}}}{d\ln\omega} = \frac{1}{3}\big(1+\frac{\omega}{2-\omega}\big) = \frac{2}{3(2-\omega)} \leq \frac{2}{3} < 1$, so T_{RW} is a strict contraction. Its unique positive fixed point solves $\omega^2(2-\omega) = 4q_\Sigma/c'$, hence $\omega_{\mathrm{fp}} = \sqrt{2q_\Sigma/c'}\,(1+O(\omega_{\mathrm{fp}})) = \sqrt{2}\,\omega_{\mathrm{RW}}^*$. At that point the random-walk risk $R_{\mathrm{RW}}(\omega) = c'\omega/(2-\omega) + q_\Sigma/(\omega(2-\omega))$ equals $\frac{3}{2\sqrt{2}} \approx 1.061$ times its value at ω_{RW}^* in the limit $q_\Sigma/c' \to 0$ (finite ratios give larger factors, and the fixed point exists for $q_\Sigma/c' \leq 8/27$), so the same 1/3-power update achieves the exact linear-drift optimum and at most 6.1% asymptotic overhead under random-walk drift without knowing the regime. □

Regime of the first-order variance term. The variance term is exact for the stated risk model; its leading-order form is accurate for aspect ratios $\zeta = D\omega \lesssim 0.3$, the regime of the pre-drift and low-drift phases on our grid ($\zeta \leq 0.13$ at the window floor), and understates the estimator's variance at the larger ratios reached during drift-phase adaptation (windows 0.01–0.05, ζ up to 6.4), where the estimator remains a valid control law but its target departs from the exact optimum.

A.5 Proof of Proposition 1(i) (oracle dominance)

Proposition 1, part (i) (restated). Fix a direction with excitation $\mu > 0$, data solution $\widehat{w}$ with $\mathbb{E}[\widehat{w}] = w$ and $\mathrm{Var}(\widehat{w}) = v$, and a frozen anchor g with $\delta := g - w$, independent of $\widehat{w}$. If the anchor is correct in that direction ($\delta = 0$), then for every $s \in (0,1)$ the $\gamma = 1$ anchored estimator $W = (1-s)\widehat{w} + sg$ satisfies

$$\mathbb{E}\big[(W-w)^2\big] \;=\; (1-s)^2\,v \;<\; v,$$

and summing over directions, the anchored prediction risk satisfies $R_{\mathrm{SPAR}} = \sum_i \mu_i(1-s_i)^2 v_i \leq \sum_i \mu_i v_i = R_{\mathrm{OLS}}$, strictly in every direction with $\rho > 0$; the improvement factor $\big(\mu_i/(\mu_i+\rho)\big)^2$ is largest in weakly excited directions.

Proof of Proposition 1(i). We first record the risk-interpolation step used throughout. By the pull identity (Equation (1)) with $\gamma = 1$,

$$W - w \;=\; (1-s)\,(\widehat{w} - w) + s\,\delta.$$

Squaring and taking expectations, the cross term is $2(1-s)s\,\delta\,\mathbb{E}[\widehat{w} - w]$, which vanishes conditioned on $\mathcal{F}_{t_0}$ when the anchor G is frozen (drawn before the post-anchor data); at finite horizons with a refreshed anchor, Theorem 1 bounds the induced correlation.

$$\mathbb{E}\big[(W-w)^2\big] \;=\; (1-s)^2\,v + s^2\,\delta^2. \tag{4}$$

Now set $\delta = 0$: the second term disappears and (4) gives $\mathbb{E}[(W_i - w_i)^2] = (1-s_i)^2 v_i < v_i$ whenever $0 < s_i < 1$ and $v_i > 0$. Since the per-direction prediction risk contribution is $\mu_i\,\mathbb{E}[(W_i - w_i)^2]$, summing gives the display. The improvement factor per direction is $\big(\mu_i/(\mu_i+\rho)\big)^2$, which is smallest (largest gain) when $\mu_i \ll \rho$. The interpolation identity (4) and both dominance statements are formalized (`convex_interp_sq`, `dominance`, `dominance_strict` in `Core.lean`). □

Remark 1 (The $\gamma = 1$ design principle) *With $\gamma < 1$ the anchored estimate is pulled toward γg rather than g (Equation (1)), introducing the shrinkage bias $s(1-\gamma)|w|$ per direction, which dominates the variance benefit whenever the prior is informative. All methods therefore use $\gamma = 1$; the ablation measures the cost of $\gamma = 0.5$ directly (+560%, Appendix D).*

A.6 Proof of Proposition 1(ii, first half)

Proposition 1, part (ii) (restated). With data-path variance $v > 0$ and squared anchor error $c = \delta^2 \geq 0$ as above, the risk profile of the anchored estimator is $R(s) = (1-s)^2 v + s^2 c$.

(a) *(Interpolation bound.)* For all $s \in [0,1]$,

$$R(s) \;\leq\; (1-s)\,v + s\,c \;\leq\; \max\{v, c\} \;\leq\; v + c.$$

(b) *(Calibration identity.)* With $s^* = v/(v+c)$ and $R^* = vc/(v+c)$,

$$R(s) \;=\; R^* + (v+c)\,(s - s^*)^2 \qquad \text{for all } s \in \mathbb{R},$$

so s^* is the unique minimizer, and a calibration error $|s - s^*| \leq \varepsilon$ costs at most $(v+c)\,\varepsilon^2$.

Proof of Proposition 1(ii). (a) For real u, d and $s \in [0,1]$, expanding the square gives

$$\big((1-s)u + sd\big)^2 \;=\; (1-s)u^2 + sd^2 - s(1-s)(u-d)^2 \;\leq\; (1-s)u^2 + sd^2,$$

and taking expectations with $u = \widehat{w} - w$, $d = \delta$ applied to the error decomposition of Equation (1) yields the first inequality; the rest is elementary. The exact deficit $s(1-s)(u-d)^2$ is formalized as `convex_interp_sq`. (b) Completing the square,

$$\frac{vc}{v+c} + (v+c)\Big(s - \frac{v}{v+c}\Big)^2 \;=\; (1-s)^2 v + s^2 c$$

by pure algebra, which is the stated identity. The excess $(v+c)(s-s^*)^2$ is nonnegative with equality iff $s = s^*$, giving uniqueness and the cost bound. The identity and the optimal pull are formalized (`quadratic_calibration`, `optimal_pull_min`, `optimal_pull_attained` in `Core.lean`). □

A.7 Proof of Proposition 1(ii, second half)

Proposition 1, part (ii), strength form (restated). Under the warm-up model with per-direction data variance $v_i = \sigma^2/(n\mu_i)$: (i) the optimal per-direction strength is $\rho_i^* = \mu_i v_i/\delta_i^2 = \sigma^2/(n\,\delta_i^2)$; (ii) in the isotropic case ($\mu_i = \mu$) with a single global pull and total conflict $C = \|\delta\|^2$, the optimal strength is $\rho^* = \mu D v/C = \sigma^2 D/(nC) = \widehat{V}/\widehat{C}$ with $\widehat{V} = \sigma^2 D/n$ and $\widehat{C} = C$; (iii) the signs

$$\frac{\partial \rho_i^*}{\partial \delta_i^2} < 0, \qquad \frac{\partial \rho_i^*}{\partial n} < 0, \qquad \frac{\partial \rho_i^*}{\partial \sigma^2} > 0 :$$

the optimal strength decreases in measured conflict and in n, and increases in the noise level, so a rising drift estimate lowers the anchor strength toward the floor while a stationary stream lets it recover.

Proof of Proposition 1(ii), strength form. (i) By Proposition 1(ii) the optimal pull in direction i is $s^* = v_i/(v_i + \delta_i^2)$, and $\rho \mapsto s = \rho/(\mu_i + \rho)$ is a strictly increasing bijection from $[0, \infty)$ onto $[0, 1)$ with inverse $\rho = \mu_i s/(1 - s)$. Substituting s^* gives $\rho_i^* = \mu_i v_i/\delta_i^2 = \sigma^2/(n\delta_i^2)$. (ii) With a single pull s the total risk is $R(s) = D(1 - s)^2 v + s^2 C$ with $v = \sigma^2/(n\mu)$; minimizing as in Proposition 1(ii) gives $s^* = Dv/(Dv + C)$, hence $\rho^* = \mu D v/C = \sigma^2 D/(nC)$. (iii) Immediate by differentiating the closed forms of (i)–(ii). As $n \to \infty$ at fixed conflict the optimal strength vanishes, and under stationarity the measured conflict decays to its noise floor while the optimal strength recovers: a rising drift estimate lowers the anchor strength toward the floor, and a stationary stream lets it return. □

Remark 2 (Runtime rule: prediction units) *The runtime rule is* $\rho^* = \bar{\mu}_t\, \widehat{V}_t^{\Sigma}/\widehat{C}_t^{\Sigma}$ *with* $\widehat{V}_t^{\Sigma} = \widehat{\sigma}_t^2\, d_{\mathrm{act}}/n_t$ *and* $\widehat{C}_t^{\Sigma} = \max\{C_{\mathrm{ref}}^{\Sigma}, C_{\mathrm{resid}}^{\Sigma}, C_{\mathrm{floor}}^{\Sigma}\}$*; per direction it reduces to the closed form of Proposition 1(ii),* $\mu_i \hat{v}_i/\hat{c}_i = \sigma^2/(n\delta_i^2)$*. Every branch is in prediction-risk units* $[y^2]$*: the reference branch is* $C_{\mathrm{ref}}^{\Sigma} = (G_{\mathrm{ref}} - \widehat{w})^{\top} \bar{A}_t (G_{\mathrm{ref}} - \widehat{w}) - \widehat{V}^{\Sigma}$*, the residual branch is* $C_{\mathrm{resid}}^{\Sigma} = E_t$*, and the floor is* $C_{\mathrm{floor}}^{\Sigma} = \frac{1}{2}\widehat{\sigma}^2 d_{\mathrm{act}}/T_w$*. The naive conflict* $\|G - \widehat{w}\|^2$ *has expectation* $C + V$ *under the null, so the estimate is debiased by subtracting* $\widehat{V}^{\Sigma}$*; because a refreshed anchor co-moves with the current solve, conflict is measured against the slow reference* G_{ref}*, while the two-scale residual detector (Proposition 3 its calibration model) supplies fast disengagement. Under deep stationarity* $\widehat{C}^{\Sigma} \to 0$ *would drive* $\rho^* \to \infty$*; the floor and a hard cap remove the singularity. The* legacy parameter-unit *form uses* $\widehat{V} = \widehat{\sigma}^2 \operatorname{tr}(\bar{A}^{-1})/n$ *and* $C_{\mathrm{resid}} = E_t \operatorname{tr}(\bar{A}^{-1})/(n\bar{\mu})$*; the trace scales with the effective condition number rather than with* d_{act}*, so it inflates on ill-conditioned spectra and is reported only as the ablation of Appendix D. Overestimating the conflict is safe by Proposition 2.*

Scalar-ρ prediction risk on a non-isotropic spectrum. Proposition 1(ii) gives the per-direction strength, while Algorithm 1 runs a single scalar ρ; the scalar target is obtained by differentiating the prediction risk directly. With $s_i = \rho/(\mu_i + \rho)$ and $v_i = \sigma^2/(n\mu_i)$,

$$R_{\mathrm{pred}}(\rho) = \sum_{i=1}^{D} \mu_i \big[(1 - s_i)^2 v_i + s_i^2 \delta_i^2\big] = \sum_{i=1}^{D} \frac{\mu_i^2(\sigma^2/n) + \rho^2\, \mu_i \delta_i^2}{(\mu_i + \rho)^2},$$

so

$$\frac{dR_{\mathrm{pred}}}{d\rho} = 2 \sum_{i=1}^{D} \frac{\mu_i^2}{(\mu_i + \rho)^3} \Big(\rho\, \delta_i^2 - \frac{\sigma^2}{n}\Big) = 0$$

$$\iff \quad \rho^* = \frac{(\sigma^2/n) \sum_i \psi_i(\rho^*)}{\sum_i \psi_i(\rho^*)\, \delta_i^2}, \qquad \psi_i(\rho) := \frac{\mu_i^2}{(\mu_i + \rho)^3},$$

which weights directions near the crossover $\mu_i \approx 2\rho$; this is the exact scalar optimum, and it is generally not $\bar{\mu}\widehat{V}^{\Sigma}/\widehat{C}^{\Sigma}$. Replacing the exact sensitivity $\psi_i(\rho)$ by the excitation-proportional surrogate $\psi_i \propto \mu_i$ gives the plug-in approximation $\rho_{\mathrm{plug}} = (\sigma^2/n) \sum_i \mu_i / \sum_i \mu_i\, \delta_i^2$, which in prediction units with $\widehat{V}^{\Sigma} = \sigma^2 d_{\mathrm{act}}/n$, $C^{\Sigma} = \sum_i \mu_i \delta_i^2$, and $\bar{\mu} = \operatorname{tr}(\bar{A})/d_{\mathrm{act}}$ is the implemented rule $\rho_{\mathrm{plug}} = \bar{\mu}\, \widehat{V}^{\Sigma}/\widehat{C}^{\Sigma}$. The surrogate matches the exact sensitivity where it matters: $\psi_i(\rho) = O(\mu_i^2/\rho^3)$ as $\mu_i \to 0$, so both vanish on weakly excited and dead directions and the conflict is carried by the well-excited modes, whereas uniform weights would give a dead

direction the same weight as an excited one. The two rules coincide on isotropic spectra, where the direction weights are equal.

Crossover horizon for the plug-in estimator. By the calibration identity, a static pull s_0 pays $(v+c)\,(s_0-s^*)^2$ over the calibrated optimum, while the plug-in estimate $\hat{s}$ pays $(v+c)\,(\hat{s}-s^*)^2$. If the two-scale noise and conflict estimates are consistent with $|\hat{s}_t - s^*| \le K\,t^{-1/2}$, the plug-in beats any fixed pull once

$$t \;>\; T_{\mathrm{crit}} \;\asymp\; \frac{K^2}{(s_0-s^*)^2},$$

the square of the calibration constant over the squared pull mismatch (the risk scale $v+c$ cancels on both sides). Short streams therefore favor fixed anchoring, and the BCI IV streams ($T \approx 300$ trials) sit on the short side; on the grid the fixed-ρ ablation is 2.2% better at $T = 10^4$, so our runs do not resolve the crossover. The threshold scales quadratically in the prior mismatch $s_0 - s^*$; the plug-in-versus-fixed-shrinkage tradeoff is classical in empirical-Bayes shrinkage (Ledoit & Wolf, 2004).

A.8 Proof of Proposition 2 (minimax safety)

Proposition 2 (restated). Fix $v > 0$ (isotropic lumped problem) and let $s(\rho) = \rho/(\rho+\mu)$ with $R(\rho, C) = (1-s)^2 Dv + s^2 C$ and conflict budget $C \in [0, C_0]$. Then

$$\min_{\rho \ge 0} \max_{0 \le C \le C_0} R(\rho, C) \;=\; \max_{0 \le C \le C_0} \min_{\rho \ge 0} R(\rho, C) \;=\; \frac{Dv\,C_0}{Dv + C_0},$$

and the unique minimax strength is $\rho^*(C_0) = \mu Dv/C_0 = \sigma^2 D/(nC_0)$, the strength calibrated exactly at the budget. The value $DvC_0/(Dv+C_0)$ is the *parallel sum* $Dv \parallel C_0$, where $a \parallel b := ab/(a+b)$ (half the harmonic mean of a and b).

Proof of Proposition 2. For fixed C, R is strictly convex in s with unique minimizer $s^*(C) = Dv/(Dv+C)$ (Proposition 1(ii) applied to the lumped problem; the interpolation identity is Proposition 1(i)), and $\rho \mapsto s(\rho)$ is a bijection, so $\min_\rho R(\rho, C) = DvC/(Dv+C)$, attained uniquely at $\rho^*(C) = \mu Dv/C$. This value is strictly increasing in C, so the maximin is attained at C_0 and equals $DvC_0/(Dv+C_0)$. For the minimax: for fixed ρ with $s(\rho) > 0$, $R(\rho, C)$ is increasing in C, so $\max_{C \le C_0} R(\rho, C) = R(\rho, C_0)$, and minimizing over ρ gives the same value, attained at $\rho^*(C_0)$. The inequality $DvC/(Dv+C) \le (1-s)^2 Dv + s^2 C$ for all s, which is the computational core of the equality (together with the monotonicity of the value in C), is formalized (`minimax_lower` in `Core.lean`). □

Proposition 2, safety clause (restated). Fix any prior G with squared error $C = \|G - w^*\|^2$, possibly adversarial and unknown to the learner, and let $s^* = v/(v+C)$ be the oracle pull. Then: (i) the excess of the anchored risk over the best-calibrated risk is at most $(v+C)(s-s^*)^2$, where s is the played pull (Proposition 1(ii)); (ii) $R(s) \le v$ if and only if $s \le 2s^*$, since $R(s) - v = s\big(s(v+C) - 2v\big)$; consequently, if the strength is calibrated against any overestimate $\widehat{C} \ge C$, then the played pull $\widehat{s} = v/(v+\widehat{C}) \le s^* < 2s^*$, and therefore $R(\widehat{s}) \le v$: calibrated overestimation makes anchoring never worse than no anchor; (iii) as $\widehat{C} \to \infty$ the anchored solution converges to the pure-data solution, so an adversarially chosen prior costs asymptotically nothing.

Proof of Proposition 2, safety clause. (i) is Proposition 1(ii). (ii) Expand: $R(s) - v = (1-s)^2 v + s^2 C - v = s\big(s(v+C) - 2v\big)$, which is at most 0 exactly when $s \le 2v/(v+C) = 2s^*$. Since $\widehat{C} \ge C$ implies $\widehat{s} = v/(v+\widehat{C}) \le v/(v+C) = s^* < 2s^*$, we get $R(\widehat{s}) \le v$. (iii) As $\widehat{C} \to \infty$, $\rho^* = \widehat{V}/\widehat{C} \to 0$ and $s_i = \rho/(\mu_i + \rho) \to 0$ for every direction with $\mu_i > 0$; directions with $\mu_i = 0$ contribute $\mu_i \delta_i^2 = 0$ to the prediction risk. □

A.9 Proof of Proposition 3 (false-alarm calibration)

Proposition 3 (restated). Assume the noise $\varepsilon_t \mid \mathcal{F}_t$ is σ^2-sub-exponential (Gaussian or bounded; the heavy-tailed t_3 configurations of the grid are outside this lemma's scope and are reported as robustness probes only). Under the null of stationarity with the model converged, let $z_t := r_t^2 - \sigma^2 - \widehat{v}_t$ be a supermartingale difference sequence with $\mathbb{E}[z_t \mid \mathcal{F}_t] \le 0$, where $\widehat{v}_t$ is any predictable bound on the current estimation variance (for a converged model $\widehat{v}_t \approx 0$; the plain null $\mathbb{E}[r_t^2 \mid \mathcal{F}_t] \le \sigma^2$ fails for an adapted W precisely because the estimation error adds to the residual). If a window of $m \ge \log(T/\delta)$ samples has mean z_t exceeding $\tau = \max\{\nu_z\sqrt{2\log(T/\delta)/(c\,m)},\ 2b_z\log(T/\delta)/(c\,m)\}$, where ν_z^2 is the sub-exponential variance proxy and b_z the sub-exponential scale of z_t, then

$$\Pr\big[\text{any false escalation within } T \text{ steps}\big] \ \le\ \delta.$$

Proof of Proposition 3. By Bernstein's inequality for sub-exponential martingale difference sequences, in the form of Freedman's inequality (Freedman, 1975), for each of the T/m disjoint windows,

$$\Pr\Big[\sum_{t\in\text{window}} z_t > m\,\tau\Big] \ \le\ \exp\Big(-c\,m\min\{\tau^2/\nu_z^2,\ \tau/b_z\}\Big) \ \le\ \exp\big(-2\log(T/\delta)\big) \ =\ (\delta/T)^2,$$

where the max-form threshold makes the bound rigorous: the quadratic term gives $c\,m\cdot\tau^2/\nu_z^2 = c\,m\cdot 2\log(T/\delta)/(c\,m) = 2\log(T/\delta)$ and the linear term $c\,m\cdot\tau/b_z = 2\log(T/\delta)$, so the minimum is exactly $2\log(T/\delta)/c\cdot c = 2\log(T/\delta)$ in either branch. and the union bound over the $T/m \le T$ disjoint windows gives $(T/m)(\delta/T)^2 \le \delta^2/T \le \delta$ for $T \ge \delta$. The damage of a false escalation is bounded by the cost of running at reduced anchor strength for the escalation holdout: at most $O(v)$ per step for $O(\text{window})$ steps, so with $\delta = 1/T^2$ the total expected false-escalation damage is $o(1)$. The alternative (drift) is detected with lag $O(h_f + m)$, where h_f is the fast moving-average horizon. □

Stationary pinning (effective threshold). The variance-floor subtraction is itself a strictly positive threshold: the drift-energy estimator is $\widehat{\nu}_\Sigma^2 = \max\big(0, (E_t - \tau_\omega)\omega^2\big)$ with $\tau_\omega = \hat{c}'\,\omega_{\min}/(2-\omega_{\min})$, so $E_t \le \tau_\omega$ implies $\widehat{\nu}_\Sigma^2 = 0$ and $\omega_t = \omega_{\min}$; the conflict floor $\tau_\rho = C_{\text{floor}}$ likewise prevents the residual branch from driving the strength below its stationary floor. Applying the bound of Proposition 3 at $\tau = \min(\tau_\omega, \tau_\rho) > 0$ gives $\Pr[\omega_t > \omega_{\min}] \le \delta$ for the windowed test under stationarity, upgrading Theorem 2's in-expectation statement to a high-probability pinning guarantee; the implemented EMA trigger is the surrogate of the Scope remark.

Remark 3 (Scope) *The union bound covers the disjoint-window statistic stated above. The implemented trigger uses two-scale exponential moving averages, a sliding statistic whose anytime coverage would require an e-process combined with Ville's inequality; the implemented statistic is therefore an engineering surrogate whose measured false-alarm rate is at or below the nominal level at* $\delta = 0.10$ *and* 0.05*, with a slight exceedance at* $\delta = 0.01$ *on ill-conditioned spectra (measured rates in Appendix D).*

A.10 Proof of Proposition 4(i) (projected gradient)

Proposition 4, part (i) (restated). For the segment family $W_t(s) = U_t\big[(1-s)\odot(U_t^\top\widehat{w}_t) + s\odot(U_t^\top G_t)\big]$ over the pull profile $s \in [0,1]^D$, with U_t the eigenbasis of $\bar{A}_t$ and $\widehat{w}_t = (\bar{A}_t + \rho_{\text{ridge}}I)^{-1}\bar{b}_t$ the minimum-ridge data solve, arbitrary causal streams $(\widehat{w}_t)$, (G_t) (allowed to depend on the learner's past choices) and loss $\ell_t(s) = \big(x_t^\top W_t(s) - y_t\big)^2$, assume $\|x_t\| \le X$, $|y_t| \le Y$, and $\|\widehat{w}_t\|, \|G_t\| \le W_{\max}$ (Proposition 5 supplies the bound on $\widehat{w}_t$ through ρ_{ridge}). The bound on G_t is inherited in the scalar case: the refresh is a convex combination in a fixed basis, so $\|G_t\|_2 \le \sup_{s\le t}\|\widehat{w}_s\|_2$ by the triangle inequality; a full-D profile under a rotating basis is controlled only up to a factor growing in D unless an upper pull limit $s_{\max} < 1$ is imposed. The loss is convex in s (affine in s by Equation (1), then squared), the gradient is uniformly bounded by $G_b = 4\,XW_{\max}\big(\sqrt{2}\,XW_{\max} + Y\big)$, and projected gradient descent with step size $\iota = \sqrt{D}/(G_b\sqrt{T})$ attains regret at most $G_b\sqrt{2DT}$ against every fixed profile in the box, in particular the data path ($s = \mathbf{0}$), the anchor path ($s = \mathbf{1}$), and every fixed scalar-ρ profile $s_\rho = (\rho/(\mu_i + \rho))_i$, for arbitrary adaptive bounded losses with no

distributional assumptions beyond boundedness. A fixed s in the eigenbasis corresponds to the time-varying physical operator $U_t \operatorname{diag}(s) U_t^\top$ when the basis rotates; the comparator is the realized profile sequence, which is the correct comparator for a basis-adaptive scheme.

Proof of Proposition 4(i). Step 1 (convexity and subgradient). Writing $z_t = U_t^\top x_t$, $w_t^e = U_t^\top \widehat{w}_t$, and $g_t^e = U_t^\top G_t$, the prediction is $a_t(s) := x_t^\top W_t(s) - y_t = z_t^\top[(1-s) \odot w_t^e + s \odot g_t^e] - y_t$, affine in s on the box; hence $\ell_t = a_t(s)^2$ is convex, with

$$\nabla_s \ell_t(s) \;=\; 2\, a_t(s) \left(z_{t,i}\,(g_{t,i}^e - w_{t,i}^e)\right)_i.$$

The exact subgradient identity for the segment family, $(a+bw)^2 + 2b(a+bw)(s-w) \le (a+bs)^2$, is formalized (`sq_subgradient` in `Ogd.lean`), which is the convexity step in the Lean assembly below.

Step 2 (gradient bound). Since $s_i \in [0,1]$ each eigen-coordinate obeys $|W_{t,i}^e| \le \max\{|w_{t,i}^e|, |g_{t,i}^e|\}$, and the Euclidean norm is invariant under U_t, so

$$\|W_t(s)\|_2^2 \;=\; \|W_t^e(s)\|_2^2 \;\le\; \|\widehat{w}_t\|_2^2 + \|G_t\|_2^2 \;\le\; 2W_{\max}^2,$$

and therefore

$$|a_t(s)| \;\le\; \|x_t\|_2\, \|W_t(s)\|_2 + |y_t| \;\le\; \sqrt{2}\, X W_{\max} + Y.$$

Using $\|z_t\|_\infty \le \|z_t\|_2 = \|x_t\|_2 \le X$ and $\|g_t^e - w_t^e\|_2 = \|G_t - \widehat{w}_t\|_2 \le 2W_{\max}$,

$$\left\|\left(z_{t,i}(g_{t,i}^e - w_{t,i}^e)\right)_i\right\|_2 \;\le\; \|z_t\|_\infty\, \|g_t^e - w_t^e\|_2 \;\le\; 2\, X W_{\max},$$

and combining the two factors gives $\|\nabla_s \ell_t\|_2 \le G_b$ with $G_b = 4XW_{\max}(\sqrt{2}XW_{\max} + Y)$.

Step 3 (projected descent). The box $[0,1]^D$ has ℓ_2-diameter $\sqrt{D}$, and for any comparison point s^* in the box, $\|s_0 - s^*\|_2 \le \sqrt{D}$. Online projected gradient descent on convex losses against an adaptive adversary incurs regret at most

$$\frac{\|s_0 - s^*\|_2^2}{2\iota} + \frac{\iota\, T\, G_b^2}{2} \;\le\; G_b\sqrt{DT} \;\le\; G_b\sqrt{2DT}$$

with the stated step size (Zinkevich, 2003); no probabilistic assumption is used. The comparison class contains the pure-data profile $\mathbf{0}$, the pure-anchor profile $\mathbf{1}$, and every fixed scalar-ρ profile s_ρ, since all lie in the box. The bound is external regret against the realized loss sequence: the anchor's state makes a comparator's counterfactual anchor trajectory differ from the realized one, and policy regret is not claimed. □

Remark 4 (Formalized chain) *The descent lemma and its assembly are mechanized in* `Ogd.lean`: `sq_subgradient` *(the convexity step),* `projBox_dist_sq_le` *(box-projection contraction),* `ogd_regret_box` *(the Zinkevich step: regret at most* $(w_0 - s^*)^2/(2\eta) + \eta T G^2/2$ *against any fixed comparison point, with adaptive losses and no probability), and* `ogd_regret_segment` *(the composition assembled for the scalar-pull family). The D-dimensional status is described in Appendix C.*

A.11 Proof of Proposition 4(ii) (Hedge)

Proposition 4, part (ii) (restated). Let $\ell_{i,t} \in [0, M]$ be arbitrary losses of experts $i \in \{1, \dots, K\}$ at times t (arbitrary adaptive sequences; no probability), let $L_{i,t} = \sum_{s<t} \ell_{i,s}$, and define the closed-form exponentially weighted weights

$$\pi_{i,t} = \frac{e^{-\eta L_{i,t}}}{Z_t}, \qquad Z_t = \sum_{j=1}^{K} e^{-\eta L_{j,t}}.$$

Let $m_t = \sum_i \pi_{i,t}\, \ell_{i,t}$ be the weighted-mean loss. Then for every expert j and every horizon T, provided $\eta M < 1$,

$$\sum_{t<T} m_t \;\le\; \frac{L_{j,T} + (\ln K)/\eta}{1 - \eta M}.$$

When the experts are predictors with mixture prediction $\widehat{y}_t = \sum_i \pi_{i,t}\,\widehat{y}_{i,t}$ and squared losses $\ell_{i,t} = (\widehat{y}_{i,t} - y_t)^2$, the realized mixture loss satisfies $(\widehat{y}_t - y_t)^2 \le m_t$ per step (by convexity of the squared loss), so the bound applies to the realized mixture loss; with the anytime choice $\eta = \sqrt{\ln K / S}$ (S the accumulated mixture loss), $\sum_t \ell_t(\mathrm{mix}) \le \min_i L_{i,T} + 2\sqrt{\ln K \cdot \sum_t \ell_t(\mathrm{mix})}$. The comparator class is the expert pool: each benchmarked forgetting window and the frozen anchor, compared individually (the bound is against a single expert j; convex mixtures are not separately claimed).

Proof of Proposition 4(ii). Step 1 (quadratic exponential bound). For $x \ge 0$,

$$e^{-x} \;\le\; \frac{1}{1+x} \;\le\; 1 - x + x^2,$$

the first inequality by $e^x \ge 1 + x$, the second by the exact identity $(1 - x + x^2)(1 + x) = 1 + x^3 \ge 1$. This step is formalized (`exp_neg_le_one_sub_add_sq` in `Hedge.lean`).

Step 2 (potential ratio). By the closed form of the weights,

$$Z_{t+1} \;=\; \sum_i e^{-\eta(L_{i,t}+\ell_{i,t})} \;=\; Z_t \sum_i \pi_{i,t}\, e^{-\eta \ell_{i,t}},$$

which is formalized as `Zfun_step`. Applying Step 1 termwise with $\ell_{i,t} \le M$ (so $\ell_{i,t}^2 \le M\,\ell_{i,t}$), and using $\sum_i \pi_{i,t} = 1$ and $1 + u \le e^u$,

$$\frac{Z_{t+1}}{Z_t} \;\le\; \sum_i \pi_{i,t}\big(1 - \eta\ell_{i,t} + \eta^2 \ell_{i,t}^2\big) \;\le\; 1 - \eta(1 - \eta M)\, m_t \;\le\; e^{-\eta(1-\eta M)\, m_t}$$

(`Zfun_step_bound`).

Step 3 (telescope). Multiplying over $t < T$ and using $Z_0 = K$ (`Zfun_prod_bound`) gives $Z_T \le K\, e^{-\eta(1-\eta M)\sum_t m_t}$, while the j-th term of the defining sum gives $Z_T \ge e^{-\eta L_{j,T}}$. Taking logarithms,

$$e^{-\eta L_{j,T}} \;\le\; K\, e^{-\eta(1-\eta M)\sum_t m_t} \qquad\Longrightarrow\qquad (1 - \eta M) \sum_{t<T} m_t \;\le\; L_{j,T} + \frac{\ln K}{\eta},$$

which is the stated bound since $1 - \eta M > 0$ (`hedge_regret`). The mixture-convexity step $(\sum_i \pi_i u_i)^2 \le \sum_i \pi_i u_i^2$ is the interpolation bound of Proposition 1(i), formalized as `convex_interp_sq_le`, so the realized mixture loss is bounded per step; the anytime form (the implemented schedule $\varsigma_t = \sqrt{\ln K / S_t}$) is the standard AdaHedge substitution (Cesa-Bianchi & Lugosi, 2006). □

Best of both worlds. Because `hedge_regret` holds for every comparator j in the pool, the frozen anchor and the best adapted window can be chosen simultaneously:

$$\sum_{t<T} \ell_t(\mathrm{mix}) \;\le\; \frac{1}{1 - \eta M}\Big(\min_j L_{j,T} + \frac{\ln K}{\eta}\Big),$$

with the pool containing both: the mixture is never worse than the warm model by more than the regret term, and never worse than the best adapted path by the same term. The SPAR-EG$^+$ pool includes both the frozen anchor and the plug-in method. Even-Dar et al. (2008) quantify the regret-to-best versus regret-to-average tradeoff.

A.12 Proof of Proposition 5 (conditioning bound)

Proposition 5 (restated). For any accumulator $A \succeq 0$ with largest eigenvalue $n\mu_{\max}$, any count $n > 0$, and any $\rho > 0$,

$$\kappa\big(A + n\rho I\big) \;\le\; 1 + \frac{\mu_{\max}}{\rho},$$

independent of the forgetting factor, the drift, or the data.

Proof of Proposition 5. The eigenvalues of $A + n\rho I$ lie in the interval $[\, n\rho,\ n(\mu_{\max} + \rho)\,]$, whose ratio is at most $1 + \mu_{\max}/\rho$. □

The accumulator of Section 4 never enters the covariance form: the anchored solve is a batch information-form solve, so Proposition 5 holds at every window, including windows where the covariance recursion of leaky RLS loses rank and diverges (Sayed, 2008). The contrast is measured in Section 5: the leaky covariance recursion diverges in 135/135 main-grid and 20/20 stress-suite runs at window $100 < D = 128$, while the batch information-form solve of the same objective is stable at every λ, and no SPAR method diverged in any configuration.

Eigengap context (Davis–Kahan). Let U, U' be the eigenbases of $\bar{A}$ and $\bar{A}' = \bar{A} + E$ with $\|E\|_{\mathrm{op}} \le \epsilon$, and let δ be the minimum gap between eigenvalues of $\bar{A}$ involved in a per-direction decision. Then

$$\left\| (I - U'U'^{\top})\, U_i \right\| \ \le\ \frac{\epsilon}{\delta}$$

by the Davis–Kahan $\sin\Theta$ theorem in the two-subspace form (Davis & Kahan, 1970; Wedin, 1972), so the per-direction refresh with a stale basis errs by at most $O(\epsilon/\delta)\,\|W - G\|$ per refresh cycle. A refresh cycle performed in a basis rotated by more than the pull displacement applies the intended per-direction weights to the wrong coordinates, at a cost proportional to the rotation angle times $\|W - G\|$. Three regimes follow: with static, well-separated spectra (the ill-conditioned grid) the bound is small and the per-direction estimator is stable and wins pre-drift; with flat spectra ($\delta \to 0$) the eigenbasis is essentially arbitrary and per-direction state is noise; with rotating spectra (covariance drift) the basis drifts within the estimator's memory and per-direction state scrambles. The scalar-strength method aggregates over directions and is invariant to basis rotation, which is why it is the robust default.

A.13 Spectral retention (Theorem 3)

This subsection separates the deterministic-equivalent proof from its numerical validation. Throughout, $\zeta = D\omega$ is the aspect ratio held fixed in the proportional limit, H is the population spectral measure of Σ, and ρ_{eff} is the finite-window renormalized anchor strength. The risk coefficient of Theorem 2 is written $c' = \sigma^2 d_{\mathrm{eff}}(\rho)$ throughout; it is distinct from ζ and from the cost branches $C^{\Sigma}_{\mathrm{ref}}, C^{\Sigma}_{\mathrm{resid}}, C^{\Sigma}_{\mathrm{floor}}$ of Algorithm 1. The scalar geometric weights are $\omega_s = \omega(1-\omega)^s$; W, w^*, and $\widehat{w}$ denote the anchored solve, the target parameter, and the data solve as in the main text.

A.13.1 The deterministic-equivalent system

For the exponentially weighted accumulator with geometric weights $\omega(1-\omega)^s$ and features $x_s = \Sigma^{1/2} z_s$ (z_s sub-Gaussian with independent entries), the separable-covariance deterministic equivalent (Silverstein & Bai, 1995; Rubio & Mestre, 2011; Hachem et al., 2007) gives, for the resolvent $R(\rho) = (\bar{A} + \rho I)^{-1}$ and $T(-\rho) = (\kappa_1 \Sigma + \rho I)^{-1}$, the two-equation fixed-point system

$$\theta_1 = \int \frac{\tau}{\kappa_1 \tau + \rho}\, dH(\tau), \qquad \kappa_1 = \sum_{s \ge 0} \frac{\omega_s}{1 + D\omega_s \theta_1}. \tag{5}$$

In the streaming limit ($D \to \infty$, $\omega \to 0$, $D\omega = \zeta$ fixed) the weight sum integrates in closed form. Substituting $y = (1-\omega)^s$ and using $\sum_s \omega(1-\omega)^s f((1-\omega)^s) \to \int_0^1 f(y)\, dy$,

$$\kappa_1(\theta_1) = \int_0^1 \frac{dy}{1 + \zeta\theta_1 y} = \frac{\ln(1 + \zeta\theta_1)}{\zeta\theta_1}. \tag{6}$$

The hypotheses used here are the ones stated in Theorem 3: independent sub-Gaussian entries, a population spectrum whose nonzero part is bounded away from zero ($\tau^{+}_{\min} > 0$), and the proportional regime. The atom at zero is allowed: Corollary A.1 below shows that exact null directions drop out of the prediction risk, so the system collapses to the active subspace verbatim.

A.13.2 The collapse lemma

Lemma A.1 (Collapse) *Let $q = 1 - \bar{s}$ be the average data retention. Under (5)–(6), $\kappa_1\theta_1 + \bar{s} = 1$, and the system collapses to the scalar fixed point*

$$q = \int \frac{\tau}{\tau + \rho_{\text{eff}}}\, dH(\tau), \qquad \rho_{\text{eff}} = \rho\, \frac{e^{\zeta q} - 1}{\zeta q}. \tag{7}$$

Proof. Per population eigenvalue the retention and the pull sum to one, so $\kappa_1\theta_1 + \bar{s} = 1$ and $q = \kappa_1\theta_1$. Substituting (6) gives $q = \ln(1 + \zeta\theta_1)/\zeta$, hence $e^{\zeta q} = 1 + \zeta\theta_1$. Therefore $\kappa_1 = \zeta q/(e^{\zeta q} - 1)$ and $\rho_{\text{eff}} = \rho/\kappa_1 = \rho(e^{\zeta q} - 1)/(\zeta q)$. Existence and uniqueness in $(0, 1)$: the right side of (7) is continuous and strictly decreasing in q (the growth factor $(e^{\zeta q} - 1)/(\zeta q) \geq 1$ is strictly increasing, and each term $\tau/(\tau + \rho_{\text{eff}})$ decreases in ρ_{eff}), positive at $q \downarrow 0$ because $\tau_{\min} > 0$, and strictly below 1 at $q = 1$. □

A.13.3 Variance functional (Theorem 3(i))

With an oracle anchor ($\delta = 0$), the one-step excess prediction risk is its variance component

$$\sigma^2\, \mathbb{E}\Big[\operatorname{tr}\Big(\Sigma R\, \bar{A}^{(2)} R\Big)\Big], \qquad \bar{A}^{(2)} = \sum_s \omega_s^2 x_s x_s^\top. \tag{8}$$

Perturb the weights by $\omega_s(\varepsilon) = \omega_s(1 + \varepsilon\omega_s)$, so that $\omega_s'(0) = \omega_s^2$ and $\bar{A}^{(2)}$ is the first-order term. Then

$$\mathbb{E}\Big[\operatorname{tr}\Big(\Sigma R\, \bar{A}^{(2)} R\Big)\Big] = -\,\frac{d}{d\varepsilon}\mathbb{E}\operatorname{tr}(\Sigma R_\varepsilon)\Big|_{\varepsilon=0} = -D\,\dot{\theta}_1, \tag{9}$$

where the dot is $d/d\varepsilon$ at $\varepsilon = 0$. This is the step that produces the amplification denominator.

Differentiate (5) in ε. Writing $J = \int \big(\tau/(\tau + \rho_{\text{eff}})\big)^2 dH(\tau)$, the first equation gives

$$\dot{\theta}_1 = -\frac{\dot{\kappa}_1}{\kappa_1^2}\, J, \tag{10}$$

because $\kappa_1\tau + \rho = \kappa_1(\tau + \rho_{\text{eff}})$. For the second equation, the derivative of $\omega_s(\varepsilon)$ and the derivative of the denominator combine:

$$\dot{\kappa}_1 = \sum_s \left[\frac{\omega_s^2}{1 + D\omega_s\theta_1} - \frac{\omega_s\big(D\omega_s^2\theta_1 + D\omega_s\dot{\theta}_1\big)}{(1 + D\omega_s\theta_1)^2}\right] = K_2\,\big(1 - D\dot{\theta}_1\big), \tag{11}$$

with

$$K_2 = \sum_s \frac{\omega_s^2}{(1 + D\omega_s\theta_1)^2}. \tag{12}$$

Substituting (11) into (10) and solving the linear equation for $\dot{\theta}_1$,

$$-D\dot{\theta}_1 = \frac{\Upsilon}{1 - \Upsilon}, \qquad \Upsilon := \frac{DK_2 J}{\kappa_1^2}. \tag{13}$$

The denominator $1 - \Upsilon$ is exactly the Neumann series of the derivative of the fixed-point map; it is positive whenever $\Upsilon < 1$, which is the regime covered by the theorem.

It remains to evaluate K_2 in closed form. With $y = (1 - \omega)^s$, $\sum_s \omega^2(1 - \omega)^{2s} f((1 - \omega)^s) \to \omega \int_0^1 y f(y)\, dy$, so

$$K_2 = \omega \int_0^1 \frac{y}{(1 + \zeta\theta_1 y)^2}\, dy = \frac{\omega}{(\zeta\theta_1)^2}\left[\ln(1 + \zeta\theta_1) - \frac{\zeta\theta_1}{1 + \zeta\theta_1}\right]. \tag{14}$$

Using $e^{\zeta q} = 1 + \zeta\theta_1$ and $\kappa_1^2 = (\zeta q)^2/(\zeta\theta_1)^2$,

$$\frac{DK_2}{\kappa_1^2} = \frac{\zeta}{(\zeta q)^2}\Big[\zeta q - \big(1 - e^{-\zeta q}\big)\Big] = \frac{1}{q}\left(1 - \frac{1 - e^{-\zeta q}}{\zeta q}\right). \tag{15}$$

Combining (13) and (15) yields

$$\Upsilon = \frac{1}{q}\left(1 - \frac{1 - e^{-\zeta q}}{\zeta q}\right) \int \left(\frac{\tau}{\tau + \rho_{\text{eff}}}\right)^2 dH(\tau), \tag{16}$$

which is the variance functional of Theorem 3(i). The $\zeta \to 0$ limit is consistent: expanding $e^{-\zeta q} = 1 - \zeta q + \frac{1}{2}(\zeta q)^2 + O(\zeta^3)$ gives $\Upsilon = \frac{1}{2}\zeta J + O(\zeta^2)$, so $\sigma^2\Upsilon/(1-\Upsilon) \to \frac{1}{2}\sigma^2\omega\, d_{\text{eff}}(\rho)$, the mean-field variance of Theorem 2.

A.13.4 Pull sensitivities (Theorem 3(ii))

Implicit differentiation of (7) in $\ln \rho$ gives

$$\rho\,\frac{\partial \bar{s}}{\partial \rho} = \frac{\mathcal{I}}{1 + \zeta\,\phi(\zeta q)\,\mathcal{I}}, \qquad \mathcal{I} = \int \tilde{s}\,(1 - \tilde{s})\,dH(\tau), \tag{17}$$

with $\tilde{s}(\tau) = \rho_{\text{eff}}/(\tau + \rho_{\text{eff}})$ and $\phi(u) = (1 - e^{-u})^{-1} - u^{-1} \in (0, 1)$. The anisotropy bound follows from the variance decomposition

$$\mathcal{I} = \bar{s}\,(1 - \bar{s}) - \text{Var}_H(\tilde{s}). \tag{18}$$

Spectral spread pushes directions into the data-dominated bulk ($\tilde{s} \approx 0$) or the anchor-pinned tail ($\tilde{s} \approx 1$), both insensitive, so the closed-loop sensitivity $S_{\omega\rho} = \mathcal{I}/(q + \zeta\phi(\zeta q)\mathcal{I})$ stays below one on every spectrum. This is the analytic explanation of the measured coupling margins of Theorem 4.

A.13.5 Conflict bias (Theorem 3(iii))

Part (i) is the variance component. The general excess risk adds the bias carried by an anchor offset $\delta = G - w^*$. Writing the anchored solve as $W = (\bar{A} + \rho I)^{-1}(\bar{B} + \rho G)$ and taking expectations over the noise alone, the deterministic part of the deviation is the resolvent action $\rho R(\rho)\delta$ with $R(\rho) = (\bar{A} + \rho I)^{-1}$. For *out-of-sample* prediction risk at $x_{\text{new}} \sim (0, \Sigma)$ the conflict contribution is

$$\delta^\top \rho^2 R(\rho)\,\Sigma\,R(\rho)\,\delta. \tag{19}$$

The in-sample empirical risk replaces Σ by $\bar{A}$ and gives $\mu_i(\rho/(\mu_i + \rho))^2\delta_i^2$ in the classical limit; we state the out-of-sample form, which is the risk the experiments estimate.

In the proportional streaming limit with δ held fixed, define the direction-weighted conflict measure H_δ (an atom of mass δ_i^2 at each population direction τ_i, total mass $\|\delta\|^2$). The deterministic equivalent of (19) is

$$\rho_{\text{eff}}^2 \int \frac{\tau}{(\tau + \rho_{\text{eff}})^2}\, dH_\delta(\tau). \tag{20}$$

It remains to justify the renormalization by $(1 - \Upsilon)^{-1}$. The perturbation ε enters (19) only through the scalar $\kappa_1(\varepsilon)$, because the resolvent is a function of $\kappa_1(\varepsilon)\Sigma + \rho I$. Expanding the resolvent around $\varepsilon = 0$ produces the same scalar Neumann series $1 + \Upsilon + \Upsilon^2 + \cdots$ that produced (13), with the same Υ from (16); the conflict quadratic form multiplies that series term by term. Hence

$$R_{\text{pred}}(\rho, \zeta) \;\to\; \frac{1}{1 - \Upsilon}\left[\sigma^2\,\Upsilon + \rho_{\text{eff}}^2 \int \frac{\tau}{(\tau + \rho_{\text{eff}})^2}\, dH_\delta(\tau)\right], \tag{21}$$

which is Theorem 3(iii). The oracle anchor ($H_\delta \equiv 0$) reduces (21) to Part (i), and the classical limit $\zeta \to 0$ gives $\rho_{\text{eff}} \to \rho$, $\Upsilon \to 0$, recovering Proposition 1's per-direction bias. The derivation is a deterministic-equivalent expansion rather than a finite-D theorem; the rank-deficient case is covered by the corollary below (Appendix A.13.8).

A.13.6 Active-subspace reduction (rank-deficient spectra)

The full-rank hypothesis in Theorem 3 is not needed. Exact null directions carry no prediction risk, and the deterministic equivalent restricts to the range of the population covariance.

Corollary A.1 (Active-subspace deterministic equivalent) *Let the population spectral measure split as $H = (1-p)\,\delta_0 + p\,H^+$ with $p \in (0,1]$ and $\mathrm{supp}(H^+) \subset [\tau^+_{\min}, \tau^+_{\max}]$, $\tau^+_{\min} > 0$. Then the deterministic-equivalent system (5)–(16), the variance functional Υ, and the prediction risk (21), all evaluated at (H, ζ), coincide with those evaluated at (H^+, ζ^+) with the reduced aspect ratio $\zeta^+ = p\,\zeta$.*

Proof. The features are $x_s = \Sigma^{1/2} z_s$, so x_s vanishes identically on $\ker\Sigma$ and the accumulator $\bar{A}$ has an exact zero block there. In the out-of-sample risk $\mathbb{E}[(x_{\mathrm{new}}^\top(W - w^*))^2] = (W-w^*)^\top \Sigma (W - w^*)$ every error component in $\ker\Sigma$ is multiplied by Σ, so it contributes nothing. Algebraically, the atom at zero drops out of the Stieltjes integrals:

$$\theta_1 = \int \frac{\tau}{\kappa_1 \tau + \rho}\, dH = p\,\theta_1^+, \qquad \theta_1^+ := \int \frac{\tau}{\kappa_1 \tau + \rho}\, dH^+,$$

while the weight sum (6) depends on θ_1 only through the product $\zeta\,\theta_1 = (D\omega)(p\,\theta_1^+) = (pD)\,\omega\,\theta_1^+ = \zeta^+\theta_1^+$. Hence (5) is the H^+ system at aspect ratio ζ^+, so κ_1 and therefore $\rho_{\mathrm{eff}} = \rho/\kappa_1$ are unchanged. Retentions scale as $q = \kappa_1\theta_1 = p\,q^+$ with the active retention $q^+ := \kappa_1\theta_1^+ = \int \tau/(\tau+\rho_{\mathrm{eff}})\, dH^+$, which is the fixed point (7) with $\rho_{\mathrm{eff}} = \rho(e^{\zeta^+ q^+} - 1)/(\zeta^+ q^+)$. For the variance functional, $J = \int(\tau/(\tau+\rho_{\mathrm{eff}}))^2\, dH = p\,J^+$ and $D\,K_2/\kappa_1^2$ is a function of $\zeta\theta_1 = \zeta^+\theta_1^+$ alone, so $\Upsilon = (DK_2/\kappa_1^2)\,J = (D^+K_2/\kappa_1^2)\,J^+ = \Upsilon^+$. The conflict integral in (21) carries the weight $\tau/(\tau+\rho_{\mathrm{eff}})^2$, which vanishes on the null atom, so it is unchanged as well. □

The corollary says a rank-deficient population is not a boundary of the theory: it is the same deterministic equivalent on the active subspace, of dimension $D^+ = pD$ and aspect ratio $\zeta^+ = p\zeta$, with the hypothesis reduced to $\tau^+_{\min} > 0$ on the nonzero part. The direction-weighted conflict measure H_δ is likewise supported on range(Σ) in the prediction risk; the unweighted trace functional that ignores the τ weighting is the object that fails, not the theorem.

A.13.7 The coupling corollary is local

At an interior fixed point the product of cross-sensitivities is invariant under monotone coordinate changes, since $\partial\psi_1(\omega^*)/\partial\psi_2(\rho^*)\cdot\partial\psi_2(\rho^*)/\partial\psi_1(\omega^*) = |(\partial\omega^*/\partial\rho)(\partial\rho^*/\partial\omega)|$; we therefore evaluate in $(\ln\omega, \ln\rho)$ without loss of generality. (i) The window target $\omega^* \propto (4\nu_\Sigma^2/c')^{1/3}$ with $c' = \sigma^2 d_{\mathrm{eff}}(\rho) \propto q^2$ gives $|\partial\ln\omega^*/\partial\ln\rho| = \frac{2}{3}\,\mathcal{I}/[q(1+\zeta\phi(\zeta q)\mathcal{I})] \le \frac{2}{3}\,\bar{s}/q$, using $\mathcal{I} \le \bar{s}(1-\bar{s}) = q\bar{s}$. (ii) The strength target $\rho^*(\omega) = \bar{\mu}\widehat{V}(\omega)/\widehat{C}$ is analyzed by which branch of $\widehat{C}$ is active: (a) when $C^\Sigma_{\mathrm{floor}}$ is active, $\widehat{C}$ is independent of ω, and $|\partial\ln\rho^*/\partial\ln\omega| \le 2/(2-\omega)$; (b) when $C^\Sigma_{\mathrm{resid}} = E_t$ is active, the drift-active floor $\rho^{\mathrm{drift}}_{\min}$ pins ρ, so $L_{\rho\omega} = 0$ and the system is triangular (Theorem 4(ii)); (c) when C^Σ_{ref} is active at stationarity ($\nu = 0$), $\widehat{w}$ is unbiased for w^* in the mean-field limit, so the reference branch is independent of ω to leading order in $1/n$ and $L_{\rho\omega} = 0$; away from stationarity it inherits the same $(\omega/(2-\omega))$ scaling as (a), with the constant absorbed into $\bar{s}$. Branches (b) and (c) are leading-order statements in $1/n$; at finite n they carry $O(1/n)$ corrections that we do not bound. The coupling product in log-coordinates is therefore at most

$$L_{\ln\omega,\ln\rho}\, L_{\ln\rho,\ln\omega} \;\le\; \frac{2}{3q}\,\mathcal{I}\cdot\frac{2}{2-\omega} \;=\; \frac{4\,\mathcal{I}}{3q(2-\omega)} \;\le\; \frac{4}{3(2-\omega)}\,\bar{s} \;<\; \bar{s} \;<\; 1 \tag{22}$$

for all $\bar{s} \in (0,1)$ and $\omega \in (0, 2/3)$, with strict improvement from spectral anisotropy ($\mathrm{Var}_H(\tilde{s}) > 0$ reduces $\mathcal{I}$ below $q\bar{s}$). This is a *local* contraction statement at an interior fixed point with the clip inactive: it bounds the Jacobian of the estimator map, not the iterates after a branch switch or while a cap is active. We therefore claim local contraction, not global contraction along every trajectory; the global statement in Theorem 4 retains the measured-product hypothesis and the gate.

A.13.8 Numerical validation

The closed forms above were checked against direct simulation (`scripts/diagnostics.py -diag rmt`), separately from the proof. **Fixed points.** At 45 operating points (three spectra × five nominal windows

$1/(1-\lambda) \in \{100, \ldots, 2000\}$ × three anchor strengths $\rho \in \{10^{-3}, 10^{-2}, 10^{-1}\}$), (7) matches the Monte Carlo stationary risk to within 4.4% (median 1.4%) across all three spectra, and within 2.3% on the rank-deficient spectrum. The rank-deficient grid is recorded in `results/rmt_rankdef.csv`, which also verifies the active-subspace invariance of Corollary A.1: the full-spectrum and reduced-system functions agree to machine precision ($|\Upsilon - \Upsilon^+| = 0$ at every point). **Sensitivities.** Equation (17) agrees with numerical differentiation of the fixed point to 10^{-6} relative error across all nine spectrum–operating-point combinations. **Conflict increment.** Paired Monte Carlo (same seeds, conflict-anchor total minus oracle total) matches (21) within 0.6–6.7% (median 1.1%) for a random conflict direction, and within 1–9.5% (median 2.2%) for a top-eigendirection conflict, across all three spectra including the rank-deficient one; the direction weighting by $\tau/(\tau + \rho_{\text{eff}})^2$ is what keeps the conflict measure correct there. The random-orientation rank-deficient case is covered by Corollary A.1: restricting the deterministic equivalent to range(Σ) removes the null-space mismatch, after which the paired Monte Carlo matches (21) within 6.7% at every point. **Operating points.** At the method's own converged control pair the deterministic equivalent matches the arm's Monte Carlo risk to within 9.8% (isotropic), 12.7% (decay3), and 1.9% (rank-deficient). The CSV files `results/rmt_grid.csv`, `results/rmt_verify.csv`, `results/rmt_conflict.csv`, and `results/rmt_rankdef.csv` record the grids; all rows, full-rank and rank-deficient, support the theorem.

The convergence step is cited (Silverstein & Bai, 1995; Rubio & Mestre, 2011; Hachem et al., 2007); the nearest prior analysis is the large-system exponentially windowed RLS study of Xiao & Honig (2005), without anchoring, per-direction pulls, refresh, or self-calibrating estimators, and the unweighted-batch ridge asymptotics (Hastie et al., 2022; Pafka et al., 2004) do not carry the finite-window renormalization.

A.14 Closed-loop stability of the two estimators (Theorem 4)

Part (i): the coupled recursions. The recursion is the quasi-static two-time-scale model: the estimator block runs every k steps and is smoothed with horizons $1/\eta_C, 1/\eta_\nu$, both slow relative to the accumulator mixing time $1/\omega$ and the fast EMA horizon h_f; finite-k lag is an additive perturbation absorbed into ξ_ω, ξ_ρ (Borkar, 2008; Benveniste et al., 1990; Kushner & Yin, 2003). The window estimator maintains an exponential average of the debiased drift-energy estimate, $\widehat{\nu}^2_{m+1} = (1-\eta_\nu)\widehat{\nu}^2_m + \eta_\nu\, \widehat{\nu}^2_\Sigma(\rho_m)$ with gain η_ν (the estimator of Theorem 2); at a frozen target the cubed readout $u = \omega^3 = 4\widehat{\nu}^2/c'$ is affine in the averaged quantity, and the frozen-target contraction is formalized (`debiased_dial_contraction`). Since u is affine in $\widehat{\nu}^2$ at frozen c', and $c' = (1-\bar{s})^2\sigma^2 D$ is Lipschitz in ρ through $\bar{s}$, the error $e_\omega(m) = |u_m - (\omega^*(\rho_m))^3|$ satisfies

$$e_\omega(m{+}1) \;\le\; (1-\eta_\nu)\, e_\omega(m) + \eta_\nu L_{\omega\rho}\, e_\rho(m) + \xi_\omega(m),$$

where $L_{\omega\rho}$ absorbs the readout and c' sensitivities and ξ_ω collects the drift-energy estimation noise, bounded by the same sub-exponential concentration as Proposition 3. The clip onto $[\omega_{\min}, \omega_{\max}]$ is a projection onto a convex set, which preserves the one-sided error contraction; a pinned estimator is the triangular case of part (ii). The strength estimator tracks the plug-in target $\rho^*(\omega) = \bar{\mu}\widehat{V}/\widehat{C}$ of Proposition 1 through clipping and shrinkage; $\widehat{V}$ and $\widehat{C}$ are Lipschitz in the window coordinate on any compact parameter box that excludes eigenvalue crossings of the 10^{-8} ridge: d_{act} is integer-valued and jumps only when an eigenvalue crosses that ridge, so we either assume a spectral gap (the grid's spectra satisfy it with margin) or replace d_{act} by the smooth effective dimension $\operatorname{tr}(\bar{A}(\bar{A} + \rho_{\text{ridge}} I)^{-1})$; on that set, through $\bar{\mu}(\omega)$, $d_{\text{act}}(\omega)$, and the prediction-unit reference conflict $(G_{\text{ref}} - \widehat{w})^\top \bar{A}(G_{\text{ref}} - \widehat{w})$, the maps are Lipschitz, giving

$$e_\rho(m{+}1) \;\le\; \eta_C L_{\rho\omega}\, e_\omega(m) + (1-\eta_C)\, e_\rho(m) + \xi_\rho(m).$$

Setting $a = \eta_\nu L_{\omega\rho}$, $b = \eta_C L_{\rho\omega}$, the two recursions form the coupled system of `Coupled.lean`.

The weighted-norm contraction. With the midpoint weight $r := \frac{1}{2}(a/\eta_\nu + \eta_C/b)$—the interval $(a/\eta_\nu,\ \eta_C/b)$ is nonempty exactly when $ab < \eta_\nu\eta_C$, i.e. $L_{\omega\rho}L_{\rho\omega} < 1$—the joint rate is

$$\gamma_{\text{cl}} \;=\; \max\big\{(1-\eta_\nu) + a/r,\ \ (1-\eta_C) + r\, b\big\} \;<\; 1,$$

and the error pair contracts in the weighted max-norm $\max\{e_\omega,\, r\, e_\rho\}$: the one-step bound, geometric decay, and tracking radius $\xi/(1-\gamma)$ are formalized as `joint_step`, `joint_geometric`, and `joint_tracking`,

with the rate lemma `joint_rate_lt_one` and the weight lemmas `joint_bounds`, `joint_weight_lo`, `joint_weight_hi`.

Part (ii): the drift-mode pin. When the detector's drift-active floor pins ρ, $e_\rho \equiv 0$ and the window error contracts at the free estimator's own rate $1-\beta$ (`joint_geometric_pinned`); the joint-rate specialization is `joint_geometric_drift_mode`. The complementary one-way case (the strength target insensitive to the window) is the triangular specialization `joint_geometric_triangular`.

Part (iii): the fluctuation bound. The contracted recursion of part (i) is the scalar Lyapunov inequality whose stationary second moment is bounded in Proposition A.2; the matrix second-moment route is deliberately not claimed, because entrywise domination of nonnegative error vectors does not lift to the Loewner order (a nonnegative triangle-inequality remainder also cannot be a zero-mean martingale difference), so the scalar weighted-norm route is the one the contraction actually supports.

Proposition A.1 (Invariant measure in the co-moving frame) *Assume (x_t, ε_t) is i.i.d. (or stationary ergodic) with $\mathbb{E}\|x_t\|^2 \le \bar{\Sigma}$, ε_t sub-Gaussian, and constant drift ν. Work in the co-moving frame $\tilde{B}_t := B_t - A_t w_t^*$ and $\tilde{G}_t := G_t - w_t^*$, and let Φ_m be the k-step skeleton at $t = mk$.*

(i) Φ_m is a Feller Markov chain on the product of the compact parameter box with the trace-bounded (tight) accumulator set, and admits an invariant probability measure π.

(ii) If the mean-field estimator map of Theorem 4 is contracting ($\gamma_{\mathrm{cl}} < 1$) and the full state-to-estimator Jacobian satisfies the small-gain condition $\varrho(J_{\mathrm{state}\to\mathrm{estimator}})\,\varrho(J_{\mathrm{estimator}\to\mathrm{state}}) < 1-\gamma_{\mathrm{cl}}$, then π is unique and Φ_m is geometrically ergodic in W_1.

No pathwise synchronous-coupling claim is made: when the two copies' estimators differ, the accumulator difference carries the cross term $-(\omega^{(1)} - \omega^{(2)})A^{(2)}$, which the small-gain condition controls.

Proof. (i) The co-moving frame removes the linear growth: A_t is tight by the Foster–Lyapunov drift $\mathbb{E}[A_{t+1} \mid \mathcal{F}_t] \preceq (1-\omega_{\min})A_t + \bar{\Sigma}$, the centered statistics $\tilde{B}_t, \tilde{G}_t$ are bounded through the pull interpolation, and the estimators live in a compact box by construction. The one-step kernel is k-periodic because the anchor refresh and estimator updates occur every k steps, so Krylov–Bogoliubov is applied to the time-homogeneous k-step skeleton Φ_m. The hard gate $\mathbf{1}\{E_t > 0\}$ does not break the Feller property: when ε_t has a continuous density (Gaussian or bounded with a density), the boundary event $\{E_t = 0\}$ has conditional probability zero, so the transition expectation of any bounded continuous function is continuous in the state; no artificial smoothing of the gate is needed. Tightness plus this Feller property yields an invariant probability measure (Meyn & Tweedie, 2009). (ii) Linearizing the joint error dynamics around the invariant set gives the feedback system state $\to$ estimator $\to$ state. The small-gain condition makes its spectral radius strictly below one in a weighted norm, and combined with the Foster–Lyapunov drift and the support condition this gives geometric ergodicity and uniqueness (Meyn & Tweedie, 2009, Ch. 18). Without the small-gain condition, uniqueness is not claimed for the full process; the idealized mean-field estimator process of Theorem 4 is unique under $\gamma_{\mathrm{cl}} < 1$. □

Proposition A.2 (Fluctuation bound) *Let the coupled estimator deviations obey the weighted max-norm recursion of part (i), $V(m+1) \le \gamma_{\mathrm{cl}}\, V(m) + \|\xi(m)\|_r$, where $\xi(m)$ is any perturbation sequence with uniformly bounded second moment $\sup_m \mathbb{E}\|\xi(m)\|_r^2 \le q_*$. Then the stationary second moment satisfies $\limsup_m \mathbb{E}[V(m)^2] \le q_* / (1-\gamma_{\mathrm{cl}})^2$; in particular each estimator fluctuates within RMS distance $\sqrt{q_*} / (1-\gamma_{\mathrm{cl}})$ of its target. No zero-mean, martingale, or independence assumption is used.*

Proof. Young's inequality $(a+b)^2 \le (1+\epsilon)a^2 + (1+1/\epsilon)\,b^2$ with $\epsilon = 1/\gamma_{\mathrm{cl}} - 1$ makes $(1+\epsilon)\,\gamma_{\mathrm{cl}}^2 = \gamma_{\mathrm{cl}}$ and $(1+1/\epsilon) = 1/(1-\gamma_{\mathrm{cl}})$, so

$$\mathbb{E}[V(m+1)^2] \;\le\; \gamma_{\mathrm{cl}}\,\mathbb{E}[V(m)^2] \;+\; \mathbb{E}\|\xi(m)\|_r^2 \,/\, (1-\gamma_{\mathrm{cl}}) \;\le\; \gamma_{\mathrm{cl}}\,\mathbb{E}[V(m)^2] + q_*/(1-\gamma_{\mathrm{cl}}).$$

Iterating from a bounded initial second moment and summing the geometric series $\sum_j \gamma_{\mathrm{cl}}^j = 1/(1-\gamma_{\mathrm{cl}})$ gives $\mathbb{E}[V(m)^2] \le \gamma_{\mathrm{cl}}^m\,\mathbb{E}[V(0)^2] + q_*/(1-\gamma_{\mathrm{cl}})^2$. □

No zero-mean or martingale assumption is used: Young's inequality bounds the cross term $2ab$ pathwise. If ξ were a signed martingale difference and one worked with the signed recursion before taking norms, the cross term would vanish and the denominator would improve from $(1-\gamma_{\mathrm{cl}})^2$ to $1-\gamma_{\mathrm{cl}}^2$; the norm route is deliberately more general and covers the nonnegative triangle-inequality remainder as well.

This closes the loop: the two single-estimator analyses are the frozen-coupling cases, the product condition is the coupling budget, and the gate enforces it exactly where the targets move fastest. The closed-form invariant law is not written down; the moment recursions above are the analytic handle, and the closed-form law remains open.

A.15 Spectral invariance of the scalar method

Proposition A.3 (Scalar-method rotation equivariance) *For any orthogonal Q, the scalar-ρ anchored solve is equivariant under rotating the stream, $x'_t = Qx_t$ with anchor $G' = QG$: the rotated solve is $W' = QW$, every prediction is invariant ($x_t'^{\top} W' = x_t^{\top} W$), and the spectral refresh commutes with the rotation.*

Proof. The rotated accumulators are $A' = QAQ^{\top}$ and $B' = QB$, and $QAQ^{\top} + n\rho I = Q(A + n\rho I)Q^{\top}$, so $W' = (QAQ^{\top} + n\rho I)^{-1}(QB + n\rho\, QG) = Q\,(A + n\rho I)^{-1}(B + n\rho G) = QW$. Predictions are inner products of rotated vectors against rotated solutions. The eigenvalues of $\bar{A}' = Q\bar{A}Q^{\top}$ coincide with those of $\bar{A}$, so the pulls $s_i = \rho/(\mu_i + \rho)$ are unchanged; with the rotated eigenbasis $U' = QU$ the refresh step maps to the rotated refresh, so the equilibrium of Theorem 1 is unchanged as well. □

Two consequences: the scalar method's per-step solve never needs an eigenbasis (it uses only A, B, G), so a covariance rotation mid-stream degrades it through the conditioning of $A + n\rho I$, which Proposition 5 bounds, and through the frozen anchor's misalignment until the next refresh; and the per-direction method's refresh aligns to the current dominant directions only at each cycle, so between cycles it applies its pulls in a stale basis, at the Davis–Kahan-bounded cost above. The scalar method wins all nine covariance-rotation configurations while the per-direction method is the fragile one (Sections 5 and 6).

B Supporting theory

This section collects the supporting results behind the anchored solve and the effective-memory-horizon analysis, with short proofs and references to the standard literature.

B.1 The anchored solve as exact MAP

For a forgetting factor $\alpha \in (0, 1]$ define the weighted statistics

$$A_t = \sum_{i=1}^{t} \alpha^{t-i}\, \phi_i \phi_i^{\top}, \qquad B_t = \sum_{i=1}^{t} \alpha^{t-i}\, \phi_i y_i^{\top}, \qquad M_t = \sum_{i=1}^{t} \alpha^{t-i},$$

and the anchored objective with prior mean G, prior precision $\tau_0 > 0$, and ridge $\lambda \geq 0$,

$$J_t(W) \;=\; \sum_{i=1}^{t} \alpha^{t-i}\, \|\phi_i^{\top} W - y_i\|_F^2 \;+\; (\tau_0 + \lambda)\, \|W - G\|_F^2.$$

The anchor penalty is undiscounted because G is a fixed prior, not streaming data.

Proposition B.4 (Anchored ridge as exact MAP) *Assume $y_i = \phi_i^{\top} W^{\star} + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I)$ and prior $W \sim \mathcal{N}(G, \tau^{-1} I)$ with $\tau = (\tau_0 + \lambda)/\sigma^2$. Then the posterior mean of W given $(\phi_i, y_i)_{i \leq t}$ equals*

$$W_t^{\star} \;=\; \big(A_t + (\tau_0 + \lambda) I\big)^{-1} \big(B_t + (\tau_0 + \lambda) G\big).$$

Proof. Completing the square in J_t: the normal equations are $(A_t + (\tau_0 + \lambda) I)\, W = B_t + (\tau_0 + \lambda) G$ with positive-definite Hessian, and the Gaussian posterior mean solves the same system. □

The identity is the standard ridge-as-MAP correspondence (Hoerl & Kennard, 1970; Rasmussen & Williams, 2006), with regularization toward a nonzero prior mean as in L2-SP (Li et al., 2018). Two algebraic facts are used in the text: the solve is ordinary ridge on the augmented design $[\Phi;\ \sqrt{\tau_0+\lambda}\,I]$ with pseudo-targets $[Y;\ \sqrt{\tau_0+\lambda}\,G]$, so standard conditioning statements transfer mechanically and the $\tau_0 \to \infty$ limit is exactly the frozen warm solution (the zero-shot endpoint of the WiSE-FT interpolation (Wortsman et al., 2022)); and the count-scaled form $W = (A_t + M_t\rho I)^{-1}(B_t + M_t\rho G)$ used by SPAR matches the proposition with the anchor weight proportional to the effective count, whereas an anchor weight that does not scale with the accumulated count dilutes as the stream grows, biasing strength sweeps.

B.2 Effective-memory-horizon stability and the rank obstruction

Proposition B.5 (Unanchored stability needs excitation over the window) *Let $\alpha = 1 - 1/m$ with integer $m \geq 2$, and suppose every window $\mathcal{W}_t = \{t-m+1, \ldots, t\}$ satisfies the excitation condition*

$$\lambda_{\min}\Big(\sum_{i\in\mathcal{W}_t} \phi_i\phi_i^\top\Big) \;\geq\; s^2 \;>\; 0.$$

If $\|\phi_i\| \leq L$ for all i, then

$$\lambda_{\min}(A_t) \;\geq\; e^{-1}s^2, \qquad \|A_t^{-1}\| \;\leq\; e/s^2, \qquad \kappa_2(A_t) \;\leq\; e\,m\,L^2/s^2.$$

Proof. For samples in the window, $\alpha^{t-i} \geq \alpha^{m-1} = (1-1/m)^{m-1} = \big(1+\frac{1}{m-1}\big)^{-(m-1)} \geq e^{-1}$, since $(1+\frac{1}{n})^n \leq e$ for all $n \geq 1$ and taking reciprocals gives the required lower bound. Hence $A_t \succeq e^{-1}\sum_{i\in\mathcal{W}_t}\phi_i\phi_i^\top \succeq e^{-1}s^2 I$. For the upper bound, $\lambda_{\max}(A_t) \leq L^2\sum_{k\geq 0}\alpha^k = mL^2$ (with A_t truncated at t), which gives the condition-number bound. □

Corollary B.2 (Rank obstruction) *The window sum $\sum_{i\in\mathcal{W}_t}\phi_i\phi_i^\top$ has rank at most $\min(m, D)$. Hence if $m < D$ the excitation condition can hold only on a subspace of dimension m, and in the generic case the unanchored recursion is ill-posed when $m \ll D$. This is the covariance windup of aggressive forgetting, whose measured boundary is reported in Section 5.*

Corollary B.3 (Anchoring removes the excitation requirement) *For any $\tau_0+\lambda > 0$, $A_t+(\tau_0+\lambda)I \succeq (\tau_0+\lambda)I$ regardless of excitation, so $\|(A_t+(\tau_0+\lambda)I)^{-1}\| \leq 1/(\tau_0+\lambda)$ and $\kappa_2 \leq 1 + mL^2/(\tau_0+\lambda)$: the anchor replaces the persistence-of-excitation assumption by a conditioning budget.*

The practical design rule is $m = 1/\omega \gtrsim D$ (the effective window at least the dimension), or an anchor strength of at least 10^{-2} when $m < D$. Proposition B.5 and its corollaries are the stability statement behind Proposition 5 of the main text: the anchored solve never relies on excitation, and the forgetting method that violates the rule is the one that diverges.

C The Lean 4 artifact

The algebraic core and the no-regret chain of Propositions 1–4 are formalized in Lean 4 with Mathlib (`theory/lean/SparTheorems/` in the artifact). This section maps the paper's results to the artifact, states the formalization boundary, records the axiom dependencies, and gives the build recipe.

C.1 Result-to-declaration map

Table C.1 summarizes the correspondence between the paper's results and the Lean 4 declarations.

C.2 Coverage and formalization boundary

The artifact contains 73 theorem and lemma declarations, all proved: 30 in `Core.lean`, 12 in `Ogd.lean`, 9 in `Hedge.lean`, 6 in `BoxD.lean`, and 16 in `Coupled.lean`. The formalized chain covers the pull identity, the

Table C.1: Map from the paper's results to the Lean artifact. The $\gamma = 1$ specialization of Equation (1) to the eigenbasis, the differentiation step of Theorem 2's convexity, and the probabilistic statements are pen-and-paper; every algebraic identity with a named Lean declaration above is formalized; the remaining rows are pen-and-paper and appear in the list below.

paper result	content	Lean declaration(s)	file
Equation (1)	pull identity ($\gamma = 1$ and general)	`pull_identity`, `pull_identity_general`	Core.lean
Proposition 1	risk interpolation; dominance; calibration identity; optimal pull; multiplicative inversion at the optimum	`convex_interp_sq`, `convex_interp_sq_le`, `dominance`, `dominance_strict`, `quadratic_calibration`, `quadratic_calibration_delta`, `optimal_pull_min`, `optimal_pull_attained`, `pull_inverse`	Core.lean
Proposition 2	minimax value (core inequality); safety excess identity; multiplicative safety threshold $s(v + c) \le 2v$	`minimax_lower`, `quadratic_calibration`, `safety_threshold`	Core.lean
Lemma 1	lag-invariance algebra	`lag_invariance`	Core.lean
Theorem 1	refresh rate identity; finite-cadence fixed point; geometric closed form; contraction; correlated variance identities and the lower endpoint $\Psi \ge (1 - s)^2$	`refresh_rate_identity`, `refresh_fixed_point`, `refresh_closed_form`, `refresh_contraction_rate`, `refresh_anchor_variance`, `refresh_variance_form`, `refresh_variance_lt`, `refresh_variance_lb`	Core.lean
Theorem 1 (moving frame)	moving-frame reduction to the affine recursion; midpoint cycle assembly; Gauss sum	`moving_frame_reduction`, `cycle_midpoint_average`, `cycle_lag_assembly`, `sum_range_half`	Core.lean
Theorem 2	leading-order window optimum via exact factorization; second-derivative positivity; estimator fixed-point dichotomy	`window_optimum`, `window_second_deriv_pos`, `naive_dial_no_fixed_point`, `debiased_dial_contraction`	Ogd.lean, Core.lean
Theorem 2 (variance coefficient)	exact effective dimension; plug-in gap $D(1 - \bar{s})^2 \le d_{\text{eff}}$	`deff_sub_plugin`, `deff_sub_plugin_nonneg`	Core.lean
Proposition A.3	rotation equivariance; refresh commutation	— (elementary linear algebra, pen-and-paper)	—
Proposition 3	concentration step	— (pen-and-paper, (Freedman, 1975))	—
	subgradient identity	`sq_subgradient`	Ogd.lean
	box-projection contraction (1-dim)	`projBox_dist_sq_le`	Ogd.lean
Proposition 4(i)	Zinkevich step (adaptive losses)	`ogd_regret_box`, `ogd_regret_segment`	Ogd.lean
	D-dimensional contraction	`projBoxD_dist_sq_le`, `sum_expand_sq`, `ogd_step_D`, `ogd_regret_box_D`	BoxD.lean
Proposition 4(ii)	quadratic exponential bound	`exp_neg_le_one_sub_add_sq`	Hedge.lean
	potential step, ratio, and product bounds	`Zfun_step`, `Zfun_step_bound`, `Zfun_prod_bound`	Hedge.lean
	Hedge regret	`hedge_regret`	Hedge.lean
Proposition 5	eigenvalue interval bound	— (one line, pen-and-paper)	—
Theorem 4	joint 2×2 contraction; geometric decay; tracking radius; drift-mode, one-way, and pinned-estimator (free-rate) specializations	`joint_step`, `joint_geometric`, `joint_tracking`, `joint_bounds`, `joint_weight_lo`, `joint_weight_hi`, `joint_rate_lt_one`, `joint_geometric_drift_mode`, `joint_geometric_triangular`, `joint_geometric_pinned`	Coupled.lean
Propositions A.1, A.2	invariant measure; second-moment (Lyapunov) bound	— (pen-and-paper, Meyn & Tweedie, 2009)	—
Corollary 1	mixed-drift master law	— (pen-and-paper, Appendix A)	—
Theorem 3	spectral retention and streaming risk	— (deterministic equivalent, pen-and-paper, Appendix A)	—
Props. B.4, B.5; Cors. B.2, B.3	exact MAP; excitation/rank; anchoring removes the excitation requirement	— (pen-and-paper, Appendix B)	—

risk-interpolation and calibration identities, the minimax core, the lag-invariance step, the finite-cadence refresh chain (rate identity, fixed point, geometric closed form, contraction, variance identities, and the lower endpoint $\Psi \geq (1-s)^2$), the moving-frame reduction and cycle-average assembly, the subgradient and box-projection steps, the Zinkevich and Hedge regret bounds, the window optimum and second-derivative positivity, and the coupled contraction chain, together with the exact effective-dimension decomposition of the window coefficient.

The probabilistic concentration step, the deterministic-equivalent convergence of Theorem 3, and the Davis–Kahan bound are pen-and-paper. The Lean declarations take the resulting recursions as hypotheses; Table C.1 marks each row as formalized or pen-and-paper.

No declaration introduces a custom axiom: the sources contain no `axiom`, `sorry`, or `unsafe` construct, and the headline theorems' axiom dependencies are logged in `theory/lean/axioms.log`, regenerated by `theory/lean/print_axioms.sh`, all resting only on `propext`, `Classical.choice`, and `Quot.sound`.

C.3 Build recipe

The project ships its own `theory/lean/lakefile.toml`, `lean-toolchain`, `lake-manifest.json`, and root import module; `theory/lean/build_ogd.sh` copies the sources and configuration into the lake project (default `$HOME/SparTheorems`) and builds:

```
bash theory/lean/build_ogd.sh
bash theory/lean/print_axioms.sh
```

On first use the script resolves Mathlib with `lake update` and fetches the pre-built cache with `lake exe cache get`, so no Mathlib compilation is required.

D Extended experimental protocol

D.1 Factorial grid

Definition. The grid of Section 5 is generated by `scripts/synthetic_factorial.py`. The main suite is the full factorial

$$\underbrace{\{\text{iso},\ \text{decay3},\ \text{rankdef}\}}_{\text{spectra}} \times \underbrace{\{\text{abrupt},\ \text{covrot},\ \text{both}\}}_{\text{drift}} \times \underbrace{\{\text{oracle},\ \text{warm},\ \text{adversarial}\}}_{\text{prior}}$$

with Gaussian noise, giving 27 configurations, plus an extra suite of four: loss of persistent excitation on the decay3 spectrum, oscillating drift on the isotropic spectrum, a stationary control (no drift), and a covariance-rotation configuration with heavy-tailed t_3 noise, for 31 configurations in total. Spectra: isotropic ($\mu_i \equiv 1$), decay3 (log-spaced from 1 to 10^{-3}, condition number 10^3), and rank-deficient (a fixed fraction of zero eigenvalues). Drifts: an abrupt jump of the target, a rotation of the input covariance, both combined, none, the death of excitation in a subspace, and an oscillating target. Priors: oracle (the pre-drift target), warm (a perturbed warm-up solution), and adversarial (a perturbation independent of the target). All configurations use $D = 128$, $T = 10^4$ steps with a warm-up of $T_w = 2000$, noise variance $\sigma^2 = 0.01$, and 5 seeds per configuration; every method runs predict-then-update.

On the stress suite SPAR-EG$^+$ is the best oracle-free method in mean MSE (1.60), 23% below SGD and 43% below Adam; the true-noise Kalman leads it by 23%, consistent with the suite residing outside the self-calibration's design envelope (short streams and stationary controls favor fixed shrinkage, Section 6).

Methods. Each configuration runs the baseline families below in a common streaming protocol, plus the slow-horizon estimated-R Kalman, the variable forgetting factor baselines, the two OGD variants (scalar segment and D-dimensional profile), SPAR-EG$^+$, and the causally-tuned Kalman filters appended post hoc. The families are the frozen warm solution, an anchor-free solve, leaky RLS at $\lambda \in \{0.999, 0.997, 0.995, 0.99\}$ in both the covariance-form recursion and the batch information-form solve, directional forgetting at the

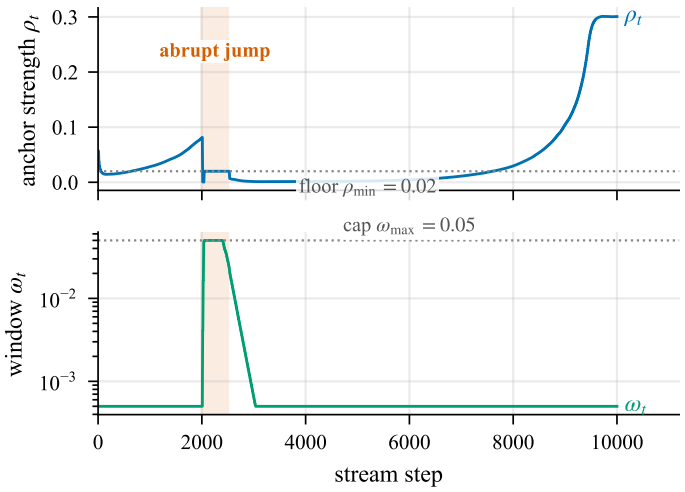


Figure D.1: Both estimators self-calibrate on a stream with an abrupt jump ($D = 128$, decay-3, seed 0): the anchor strength ρ_t falls on conflict (Proposition 1) and the window ω_t opens on drift (Theorem 2), both recovering; dotted lines mark the ρ floor and the ω cap.

Table D.1: Stress suite (4 configurations × 5 seeds): mean and worst overall MSE, one fixed setting per method. The suite is deliberately outside the main grid's operating envelope (loss of persistent excitation, oscillating drift, a stationary control, heavy-tailed rotation noise); the true-noise Kalman is an oracle comparator, and the covariance-form recursion diverges in all 20 runs.

method	mean MSE	worst MSE
Kalman, true noise ($q = 10^{-4}$)	1.29	6.26
SPAR-EG$^+$	1.60	7.83
SPAR-EG	1.60	7.77
leaky RLS, information form ($\lambda = 0.99$)	1.61	7.70
SPAR-Pull	1.66	8.10
SGD (lr $= 0.01$)	2.07	10.03
Adam (lr $= 0.01$)	2.82	13.32
WiSE-FT ($\alpha = 1$)	3.62	17.39
anchor-free solve	4.08	19.70
frozen warm solution	5.91	28.83
leaky RLS, covariance form ($\lambda = 0.99$)	diverged (20/20)	

Table D.2: Marginal mean MSE over the 27 main configurations (each row averages 9 configurations × 5 seeds), addressing the aggregate-only view of Table 1. The two SPAR methods agree within 0.03 on every slice; the Kalman and VFF leaders are prior-independent and lose the covariance-rotation slice. The `abrupt` and `both` rows coincide to three decimals because both streams carry the same abrupt target jump, and the added covariance rotation shifts the drift-axis mean by less than 10^{-3}.

axis	value	SPAR-Pull	SPAR-EG$^+$	KF-slow	VFF	leaky-info
spectrum	isotropic	0.673	0.644	0.507	0.562	0.636
	decay-3	0.114	0.115	0.065	0.120	0.257
	rank-deficient	0.131	0.132	0.071	0.119	0.134
drift	abrupt	0.454	0.440	0.310	0.387	0.487
	covariance rotation	0.011	0.011	0.023	0.027	0.054
	both	0.454	0.440	0.310	0.387	0.487
prior	oracle	0.301	0.294	0.215	0.267	0.342
	warm	0.305	0.295	0.215	0.267	0.342
	adversarial	0.313	0.303	0.215	0.267	0.342

matched grid $\lambda_f \in \{0.999, 0.997, 0.995, 0.99, 0.98, 0.95, 0.90\}$, AROW at $r \in \{0.01, 0.05, 0.1, 0.3, 0.5\}$, random-walk Kalman filters at $q \in \{10^{-6}, 10^{-5}, 10^{-4}\}$ with the true noise level plus the estimated-R variants at both horizons, SGD at lr $\in \{0.003, 0.01\}$ and Adam at lr $\in \{0.0003, 0.001, 0.003, 0.01\}$, variable forgetting factor RLS at two floors under both the single-step and the nominal-length calibrations, and the WiSE-FT α-envelope (11 values). The SPAR methods are scalar, diagonal, two-estimator, and expert-simplex. Sweeps are reported per value; the table reports each family's best setting under both the per-configuration rule (oracle-tuned baselines, the conservative treatment) and the global single-setting rule, and SPAR methods use the same configuration in every cell.

Baseline verification. Before any comparison, the `verify` step of the pipeline checks the implementation identities numerically: directional forgetting at $\lambda_f = 1$ coincides with RLS, the Kalman filter at $q = 0, R = 1$ coincides with RLS, and the WiSE-FT endpoints coincide with the frozen and fully adapted solves (maximum absolute deviation of order 10^{-15}). The same step checks the algebraic identities of Section 4 symbolically (sympy), and the step exits nonzero on any failure.

Estimated-noise Kalman. The grid includes two estimated-R Kalman filters. The fast variant updates R with a 100-step moving average of squared residuals; a fast horizon attributes part of the drift-induced residual to measurement noise, which suppresses the filter's gain precisely when adaptation is needed, and it collapses to 1.09/5.36. The slow variant uses a 20,000-step horizon (matching the SPAR noise baseline) and attains 0.215/0.93 (best single $q = 10^{-4}$), within 5.4% of the true-noise filter (0.204/0.86) and ahead of SPAR-Pull overall (0.306/1.22), though SPAR-Pull leads it on all nine covariance-rotation configurations (0.011 vs. 0.023 per configuration mean). Together, the two rows show that the apparent gap was caused by the estimation horizon, not by access to the true noise level.

Rank-deficiency note. The prediction-unit estimator never forms $\mathrm{tr}(\bar{A}^{-1})$: d_{act} counts only the eigenvalues of $\bar{A}$ above the 10^{-8} numerical ridge, so dead directions contribute neither variance nor conflict and feature rescaling leaves ρ invariant (Appendix F). The floor/cap pair bounds the resulting strength in all 135 main-grid and 20 stress-suite runs; no SPAR method diverged on any spectrum.

Divergence boundary. The leaky covariance-form recursion is one tuning step from failure on this grid: $\lambda = 0.997$ attains 0.555 mean MSE, $\lambda = 0.995$ attains 182 (worst 3,878), and $\lambda = 0.99$ attains 1.2×10^6, diverging in 135/135 main-grid and 20/20 stress-suite runs (nominal window $100 < D = 128$); Figure D.2 plots the sweep. The batch information-form solve of the same objective is stable at every λ (0.34 mean at $\lambda = 0.99$), which locates the explosion in the covariance recursion rather than the objective; the family's λ-sensitivity remains (its best stable window is 1.1× SPAR's).

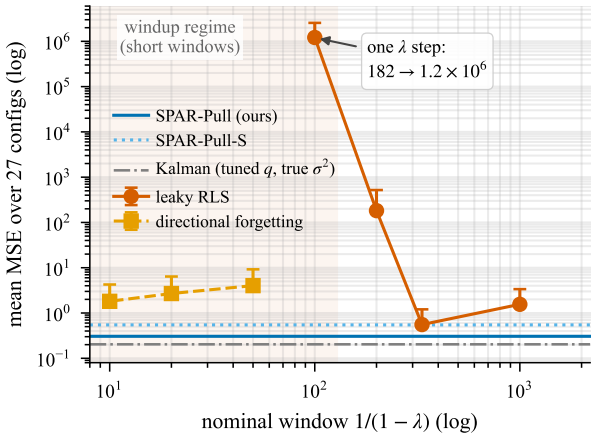


Figure D.2: Mean MSE over 27 configurations versus nominal window $1/(1-\lambda)$ (log–log; +1 s.d. whiskers). Leaky RLS degrades in one λ step ($182 \to 1.2 \times 10^6$ at window $200 \to 100$); shading marks the windup regime (window $< D = 128$); directional forgetting is divergence-free but 3.3× worse at its best; SPAR-Pull and the scalar SPAR method use one fixed configuration everywhere.

A diagnostic log of the forgetting factor (`diagnostics.py -diag vff`, `results/vff_diag.csv`) confirms the degeneracy: under the reference calibration $\xi = \sigma_0^2$ the factor remains pinned at its floor on 96% of steps on both stationary and drifting streams (mean $\lambda = 0.990$ at floor 0.99), and a recursive-scale variant of the rule (a Fortescue-style running ξ) still pins at the floor (92% of steps, overall MSE within 2% of the best fixed-λ method). Residual-driven forgetting degenerates to fixed-λ leaky RLS whenever ξ tracks the residual scale, a saturation documented in the variable-forgetting-factor literature (Paleologu et al., 2008). The covariance-form VFF inherits the same calibration sensitivity: the single-step normalization $\xi = \sigma_0^2$ drives the factor below D and it diverges identically, while the nominal-length normalization $\xi = N_0\sigma_0^2$ of the original formulation keeps the nominal window near N_0 and is stable (best single setting $N_0 = 100$: 0.267/0.98; the single-step calibration diverges in all 135 runs), the horizon choice N_0 playing the role of the fixed window that SPAR's window estimator sets online. The first-order family terminates at its own boundary: at lr = 0.02, one grid step above the swept maximum, SGD diverges on 45 of 135 runs, and at lr = 0.05 the recursion overflows outright (`diagnostics.py -diag sgd`); the reported grid $\{0.003, 0.01\}$ sits at the stable frontier of the constant-step family.

D.2 Windup mechanism

The covariance-form divergence obeys one law with two channels: whatever the rank-one downdate cannot see inflates by $(1-\omega)^{-1}$ per step, growth like $e^{\omega T}$ in the stream length. On full-rank spectra the unseen component is roundoff: the exponentially weighted covariance matrix is a positive combination of $T \gg D$ outer products, full-rank almost surely at every window, with limiting spectral support bounded away from zero (the lower edge obeys $\lambda - \ln\lambda = 1 + c$, positive for all c); the iterated inverse $P \leftarrow (P - kk^\top)/(1-\omega)$ instead accumulates antisymmetric roundoff error, and a one-line symmetrization $P \leftarrow \frac{1}{2}(P + P^\top)$ isolates this channel (`diagnostics.py -diag windup`; 2 representative configurations × 2 seeds per row):

Table D.3: Mean MSE of the covariance-form leaky recursion on full-rank spectra, with and without per-step symmetrization of the iterated inverse. At the longest window both variants agree to machine precision; at short windows the unsymmetrized recursion diverges at both dimensions while the symmetrized variant remains stable at every window tested.

λ $(1/(1-\lambda))$	D	standard recursion	symmetrized
0.997 (333)	128	1.15	1.15
0.995 (200)	128	25.1	0.80
0.99 (100)	128	2.8×10^4	0.61
0.99 (100)	512	7.0×10^4	10.4

On rank-deficient spectra the second channel is deterministic: the dead directions receive no data, so nothing downdates the corresponding block of P and it inflates by $(1-\omega)^{-t}$ in exact arithmetic, symmetrized or not. The appended symmetrized grid method confirms the split (`append_arms.py`, `rls_sym` rows of `results/factorial_extras.csv`): stable on all 90 full-rank runs (0.13 mean on the ill-conditioned spectrum, 0.60 isotropic, at $\lambda = 0.99$) and divergent on 42 of 45 rank-deficient runs (45 of 45 at $\lambda = 0.97$). The anchored information form is immune to both channels by construction—it never iterates an inverse—and Proposition 5 bounds its conditioning for any data and any window. The single-step VFF calibration enters the same regime through its floor (windows 20–100 at $\lambda_{\min} \in \{0.95, 0.99\}$); the nominal-length calibration keeps windows near N_0, above the amplification threshold at every setting tested.

D.3 False-escalation calibration

The calibration experiment of Proposition 3 runs 300 stationary streams per spectrum through the SPAR estimator with a fixed, calibrated model, records the two-scale residual statistic $z_t = (\widehat{\sigma}^2_{\text{fast}} - \widehat{\sigma}^2_{\text{slow}})/\sigma^2$, and measures the fraction of streams whose maximum exceeds the windowed threshold $\tau(\delta) = \sqrt{2\log(T/\delta)/m}$ at the fast horizon $m = 100$. The measured rate is 0.003 on isotropic spectra at nominal $\delta = 0.10$ and 0.040 on ill-conditioned spectra at nominal $\delta = 0.05$, at or below the nominal level; at $\delta = 0.01$ the ill-conditioned rate is 0.033—the exceedance is predicted by the sub-exponential threshold's linear branch: at $\delta = 0.01$

and $m = 100$, the linear term $2b_z \log(T/\delta)/(cm)$ dominates the quadratic branch, so the Gaussian-form threshold $\sqrt{2\log(T/\delta)/m}$ used in the calibration experiment is provably too low by the b_z-scaling of the moving-average's heavy tail, and the measured 0.033 exceeds the nominal 0.01 accordingly.

D.4 Ablation and sensitivity

The ablation (`scripts/revision_runs.py -suite ablation`) runs the prediction-unit method and its ablations on 4 representative configurations (abrupt and oscillating drifts, adversarial prior, ill-conditioned spectrum) × 5 seeds. Removing the window estimator (fixed ω) costs +189% overall MSE, removing the two-scale residual detector +190%, and removing the pull-identity refresh +189%; setting $\gamma = 0.5$ costs +560%, which measures the $\gamma = 1$ principle of Appendix A.5. Freezing the strength estimator to its warm-up value *lowers* overall MSE by 2.2% on these configurations, and removing the variance-floor subtraction costs +1.7% under the nominal cap, with the undebiased estimator pinning the window at its cap as Theorem 2 predicts. The three-way dissociation in Table D.4 localizes the prior-immunity mechanism: freezing the spectral anchor refresh costs 2.2× on the warm-prior configurations and 7.4× on the adversarial ones (adversarial early MSE 47.0 vs. 16.4), whereas freezing ρ leaves both within 2.6%. Prior immunity is therefore carried by the refresh, and the strength estimator's role is the drift-onset conflict response measured in Appendix Figure D.1—it matters only when the warm-up prior scale is unknown, where it guards against an oversized anchor. An alternating-trust diagnostic probes the strength estimator's average-case value directly (`diagnostics.py -diag alt_trust`: the target switches every 500 steps between a noisy revisit of the anchor and a drifted target, two spectra, 5 seeds): the adaptive rule attains 6.79 overall MSE against 6.64–7.00 for pinned strengths spanning three orders of magnitude (10^{-3} to 1)—a ±3% spread, because the anchor *refresh*, not the strength estimator, carries the phase-to-phase adaptation.

Table D.4: Component dissociation on the 4-configuration ablation suite (mean ± s.d. over 20 runs per cell; early MSE is the first 500 post-jump steps). Freezing the anchor refresh is the only variant that loses prior immunity; freezing the strength estimator is indistinguishable from the full method.

variant	warm overall	adversarial overall	adversarial early
full	2.26 ± 3.28	1.04 ± 0.21	16.4 ± 3.3
fixed ρ (refresh on)	2.20 ± 3.16	1.05 ± 0.21	16.5 ± 3.3
frozen anchor (no refresh)	4.98 ± 6.49	7.69 ± 1.62	47.0 ± 10.2
no variance floor	2.30 ± 3.33	1.06 ± 0.21	16.7 ± 3.3

The dichotomy is exact: the naive estimator $\hat{\nu}^2_{\text{naive}}(\omega) = (\widehat{\mathbb{E}}[r_t^2] - \hat{\sigma}^2)\omega^2$ yields a cubed estimator $(T_{\text{naive}}(\omega))^3 = (\omega^*)^3 + 4\omega^3/(2-\omega) > 2\omega^3$ for every $\omega \in (0, 2)$, so the cubed trajectory doubles each iteration and reaches the cap in $O(\log)$ steps even at zero drift (`naive_dial_no_fixed_point`), while the variance-floor-subtracted estimator gives the exact affine contraction $u_{m+1} = (1-\eta_\nu)u_m + \eta_\nu(\omega^*)^3 \to (\omega^*)^3$ from any initialization (`debiased_dial_contraction`)—both formalized in `Core.lean`. Replacing $\mathrm{tr}(\bar{A}^{-1})$ with the ridge-damped trace $\mathrm{tr}((\bar{A} + \rho I)^{-1})$ on the rank-deficient configurations (`append_damped_trace.py`, `results/damped_trace.csv`) yields identical error (0.100 vs. 0.100 overall): the residual-detector and floor branches govern the estimator where the trace saturates, and the damped variant is the numerically preferred implementation on ill-conditioned spectra. The sensitivity variants (legacy parameter-unit method, `results/ablation_spar_kf.csv`; the control constants are shared with the prediction-unit method) scale the residual horizons (±), the refresh cadence ($10 \to 50$), the reference-anchor cadence, and the window cap ($0.05 \to 0.02$ and 0.10): six of the seven variants change overall MSE by at most ±10%, and the largest change is the window cap $\omega_{\max} = 0.02$ at +16%.

D.5 The conflict estimator at drift onset

Table D.5 pins the strength at $\rho \in \{0.01, 0.1, 1\}$ with the refresh on and compares it with the adaptive estimator on the abrupt-jump stream with the decay-3 spectrum (5 seeds; `scripts/transient_test.py`, `results/transient_test.csv`). At cadence $k = 10$ the adaptive estimator is 1% below the best fixed strength and the large-anchor policy costs 87% more; at $k = 50$ it is within 5% of the best fixed strength while the large-anchor policy costs 2.8×. The penalty is pre-drift, not post-jump: $\rho = 1$ multiplies the pre-drift

MSE by 4.2 relative to the small fixed anchor (0.049 vs. 0.012 at $k = 10$) because the anchor holds a warm prior that the decaying spectrum no longer supports, whereas the 50-step post-jump window is dominated by the jump magnitude and is flat across the adaptive and small-anchor policies—if anything slightly better for the large anchor (7.7 vs. 9.5). On the isotropic spectrum, where a scalar strength has no structural effect, every policy has the same pre-drift MSE (0.0104) and differs only through the post-jump term. We therefore read ρ_t as a guard against a badly oversized anchor—the failure mode a good warm prior invites—not as a post-jump accelerator.

Table D.5: Transient strength sweep on the abrupt-jump stream, decay-3 spectrum, 5 seeds: pre-drift MSE, MSE over the 50 steps after the jump, and overall MSE. The refresh is on in every row; the adaptive row is the paper's method and the others pin ρ for the whole stream.

k	strength policy	pre-drift MSE	post-jump MSE $[0, 50]$	overall MSE
10	adaptive	0.0136	9.48	**0.166**
10	fixed $\rho = 0.01$	0.0116	9.48	0.168
10	fixed $\rho = 0.1$	0.0271	9.49	0.195
10	fixed $\rho = 1$	0.0489	7.72	0.311
50	adaptive	0.0197	9.52	0.254
50	fixed $\rho = 0.01$	0.0146	9.54	**0.243**
50	fixed $\rho = 0.1$	0.0338	9.56	0.308
50	fixed $\rho = 1$	0.0495	8.52	0.723

D.6 Coupling verification

Theorem 4(i)'s condition is measured directly (`diagnostics.py -diag coupling`, `results/coupling_diag.csv`): on the four ablation configurations × 5 seeds, each estimator is pinned at ±10% of its converged value while the other adapts, and the two sensitivities $L_{\omega\rho}$ (of the cubed window target) and $L_{\rho\omega}$ (of the strength target) are finite-differenced at the empirical coupled fixed point. The product $L_{\omega\rho}L_{\rho\omega} < 1$ holds in 20 of 20 runs, with maximum 5.7×10^{-3}—174 to 1381 times below one on the oscillating configuration, the only one with both estimators interior. On three of the four configurations one estimator saturates at its cap or floor, making the corresponding sensitivity exactly zero—a pinned estimator realizes the triangular specialization of Theorem 4(ii), and the large cubed-coordinate slopes recorded at floored windows are boundary artifacts of perturbing a microscopic $u = \omega^3$, multiplied by an exactly-zero factor: the caps act as the coupling control, and the cross-signs at the interior fixed point match the theory (u^* increasing in ρ, ρ^* decreasing in u).

D.7 Spectral-risk verification

Theorem 3(i)'s closed form was validated at three levels (`diagnostics.py -diag rmt`). First, at 45 fixed operating points (three spectra, including a rank-deficient one, × five nominal windows $1/(1-\lambda) \in \{100, \ldots, 2000\}$ × three anchor strengths $\rho \in \{10^{-3}, 10^{-2}, 10^{-1}\}$), the formula matches direct Monte Carlo simulation of the anchored solve (40 stationary streams per point, oracle anchor) within 4.4% relative error across the entire range, including aspect ratios $\zeta = D\omega$ above one and the rank-deficient spectrum, whose active directions the formula restricts to exactly (`results/rmt_grid.csv` records the grid, `results/rmt_rankdef.csv` the active-subspace invariance, and `results/rmt_verify.csv` the method-level checks). Second, at the method's own operating points—the converged estimator pairs (ω, ρ) measured on stationary streams per spectrum—the formula predicts the method's stationary pre-drift MSE within 1.9% (rank-deficient), 12.7% (ill-conditioned), and 9.8% (isotropic, where the strength estimator sits at its cap), with no fitted parameter. Third, the closed-form sensitivities of part (ii) agree with numerical differentiation of the fixed point to 10^{-6} relative error. The pre-drift MSE reported in Section 5 (1.01–$1.14 \times \sigma^2$ across spectra) is the segment average including the post-warm-up transient; the formula describes the stationary value at the converged operating pair.

D.8 Per-step cost

Table D.6 reports the per-step wall-clock cost of the closed-form methods at $D = 2048$ (fp64 CPU, `scripts/bench_ops.py`; run alone for clean timing). The cached-basis per-direction head is $O(D^2)$ per step with $O(D^3)$ at refreshes; the scalar method solves the anchored system exactly at $O(D^3)$ per step, and its spectral refresh reuses the solve's factorization in $O(D^2)$ (Section 3). The information form deliberately trades the covariance recursion's $O(D^2)$ rank-one update for a factorization that does not wind up on rank-deficient spectra (Proposition 5); where per-step cost dominates, the cached-basis head is the intended path.

Table D.6: Per-step cost at $D = 2048$ (fp64 CPU). The cached-basis per-direction head projects each feature into the last refresh's eigenbasis and solves the diagonal system ($O(D^2)$/step, $O(D^3)$ every k refreshes); the scalar method performs a Cholesky factorization plus triangular solves per step ($O(D^3)$), which motivates the per-direction path at large D.

method	ms/step at $D = 2048$
leaky RLS	32.0
AROW	31.3
Kalman	53.0
directional forgetting	56.3
SPAR-Pull-D (cached basis, $O(D^2)$/step)	52.6
SPAR-Pull-S (exact scalar solve, $O(D^3)$/step)	241

SPAR-EG and SPAR-EG$^+$ run four batch solves per step (the frozen anchor is the remaining expert in the pool), so their per-step cost is about four times the scalar SPAR-Pull solve timed here; the cached-basis path applies to them as well.

D.9 Mismatched generators

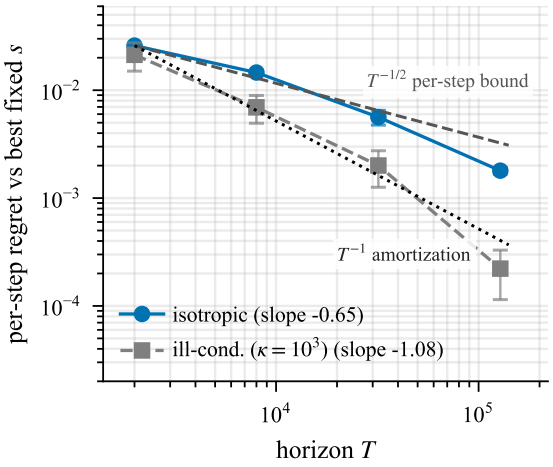
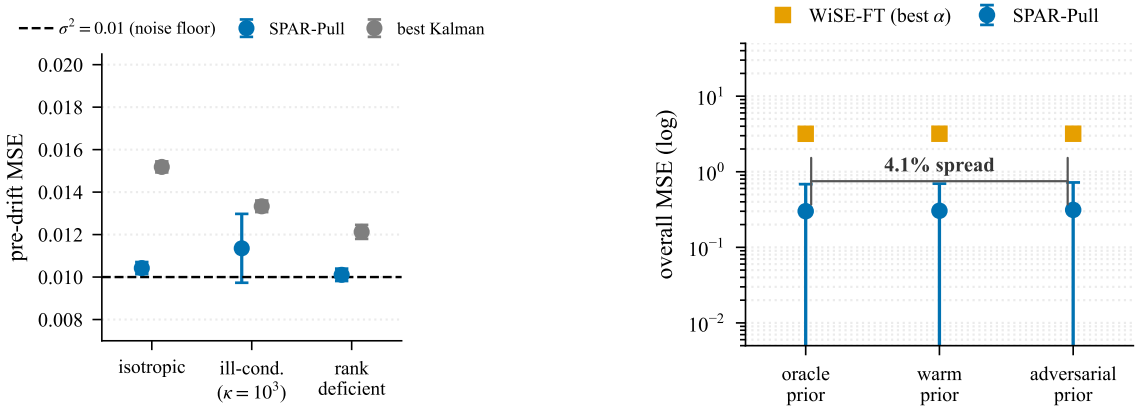


Figure D.3: Average per-step regret of the segment (scalar-pull) OGD method versus horizon T (log–log; whiskers: ± 1 s.d. over 5 seeds; the drift phase lasts a fixed 1,000 steps at every horizon). Fitted slopes -0.65 (isotropic) and -1.08 (ill-conditioned, $\kappa = 10^3$) lie inside the $T^{-1/2}$ per-step rate (dashed) implied by the scalar $O(\sqrt{T})$ Zinkevich bound (the $D = 1$ member of the Proposition 4(i) family, formalized as `ogd_regret_segment` in `Ogd.lean`), with the T^{-1} amortization limit (dotted) as the mechanical floor.

At a fixed horizon, the dimension sweep separates the two gradient variants. With $T = 9{,}000$, the drift phase held at 1,000 steps, $D \in \{8, 32, 128\}$, and 5 seeds (`scripts/regret_dscaling.py`, `results/regret_dscaling.csv`), both methods' per-step regret is measured against the best fixed scalar pull (an 11-point segment grid, each comparator method carrying its own refresh trajectory). The profile method's regret exceeds the segment method's at every operating point, and the gap widens monotonically

with D—from 0.001 to 0.078 on isotropic spectra and from 3×10^{-6} to 0.011 on ill-conditioned spectra—the empirical signature of the $\sqrt{D}$ dimension factor in the Proposition 4(i) bound; on the full grid the same premium appears as the profile method's higher mean MSE (0.707 vs. 0.585, Table 1).



(a) (b)

Figure D.4: (a) Pre-drift MSE remains at the irreducible floor $\sigma^2 = 0.01$ across all three spectra (dots show 5-seed means ± 1 s.d. against the best tuned-q Kalman with true σ^2), illustrating Proposition 1 at the optimum. (b) Overall MSE (log scale) across oracle, warm, and adversarial priors (means over all runs). While the WiSE-FT best-α envelope sits near 3.2 under every prior, SPAR-Pull varies by only 4.1% across all three priors (bracket): Proposition 2.

Three generators violate the linear-Gaussian, constant-rate model class: nonlinear targets ($y = \tanh(x^\top w^*) + \varepsilon$; the linear head is misspecified), covariate shift (the feature scale doubles mid-stream), and a noise-regime change (σ^2 rising from 0.01 to 0.05 at t_1, with no target drift). On nonlinear targets (four configurations: decay3 and isotropic spectra under abrupt, gradual, and stationary drifts; 5 seeds each) SPAR-EG$^+$ attains 0.221 overall MSE, SPAR-EG 0.222, the best single leaky window ($\lambda = 0.997$) 0.232, and SPAR-Pull 0.273: the mixture shifts weight toward the expert that tolerates the nonlinearity. Under covariate shift the Kalman family leads (0.268 with the true noise, 0.284 with the slow estimated R), while SPAR-EG$^+$ and SPAR-Pull lead the anchored and leaky methods (0.490 and 0.492), ahead of the batch-solved leaky method (0.550), SPAR-EG (0.566), and the best single leaky window (0.627), the two-scale residual detector catching the scale change. Under the noise-regime change (two configurations, decay3 and isotropic spectra, 5 seeds each) the fast-horizon estimated-R Kalman attains 0.069–0.098, its 20,000-step-horizon sibling 0.084–0.110, and the true-noise filter 0.096–0.114 (its fixed R is wrong after the change), while the SPAR-EG mixture attains 0.047–0.087 and SPAR-EG$^+$ 0.047–0.053; SPAR-Pull holds 0.052 on the decay3 spectrum but lags at 0.080 on the isotropic spectrum, where its slow noise baseline tracks the pre-change level. No SPAR method diverged under any mismatch, while the $\lambda = 0.995$ leaky method diverged on nonlinear targets (two of five seeds on the decay3 spectrum under abrupt drift) and degraded under the noise change (9.1 and 9.4×10^3 overall MSE).

D.10 A real high-dimensional stream: hourly electricity load

The factorial grid is synthetic, so we also run the grid-default configuration, unchanged, on a real stream: the public hourly electricity-load panel (321 clients, 26,304 hours) (Trindade, 2015; Lai et al., 2018). Features at hour t are the standardized log-loads of all 321 clients plus an intercept and hour-of-day sine/cosine ($D = 324$); the target is the log-load of one held-out client (the 25th-percentile client by mean load) one hour ahead, centered on its warm-up mean. The warm-up is 2,000 hours and the evaluation horizon 6,000 hours, so $T > 5{,}000$ and $D > 128$ as required. The stream is not stationary: load carries daily and weekly cycles and a seasonal trend, and the target series moves with them.

Both SPAR methods use the grid defaults of Table F.1 unchanged: window range $[5 \times 10^{-4}, 0.05]$, refresh cadence 10, residual horizon 30, and the drift-active floor 0.02. The baselines are swept on the same stream and

reported at their best setting, which is oracle tuning in their favour: leaky RLS over $\lambda \in \{0.99, 0.995, 0.999\}$ and the slow estimated-R Kalman over $q \in \{10^{-4}, 10^{-5}\}$.

Table D.7: Real stream (hourly electricity load, $D = 324$, $T = 6{,}000$ evaluation hours, 25th-percentile held-out client): MSE of the next-hour log-load over the full horizon and over the final 2,000 hours. SPAR methods use the factorial-grid defaults with no stream-specific tuning; baseline hyperparameters are swept on the same stream and reported at their best.

method	full horizon	final 2,000 h
leaky RLS, information form ($\lambda = 0.999$)	**0.0209**	**0.0134**
leaky RLS ($\lambda = 0.995$)	0.0219	0.0160
leaky RLS ($\lambda = 0.99$)	0.0228	0.0174
Kalman, slow estimated R ($q = 10^{-5}$)	0.0660	0.0170
Kalman, slow estimated R ($q = 10^{-4}$)	0.3054	0.0183
SPAR-EG$^+$ (grid defaults)	0.0212	0.0143
SPAR-Pull (grid defaults)	0.0232	0.0154

Table D.8: Real stream, held-out client sweep: MSE of the next-hour log-load over the full 6,000-hour horizon for the 25th-, 50th-, and 75th-percentile clients by mean load. The oracle column is the best of the three leaky windows ($\lambda \in \{0.99, 0.995, 0.999\}$) chosen *per client* ($\lambda = 0.999, 0.995, 0.995$ at the 25th, 50th, 75th percentiles)—tuning in the baseline's favour—while the SPAR methods use the grid defaults with no stream-specific tuning. Bold marks the best entry in each row.

held-out client	best oracle leaky	SPAR-EG$^+$	SPAR-Pull
25th percentile	**0.0209**	0.0212	0.0232
50th percentile	0.0086	**0.0083**	0.0090
75th percentile	**0.0112**	0.0113	0.0130
three-client mean	**0.0136**	**0.0136**	0.0151

Across the sweep, SPAR-EG$^+$ lands within 1.2% of the per-client oracle window and 4.3% below it at the 50th percentile, and it matches the oracle on the three-client mean (0.0136 both); SPAR-Pull is 4.2–16.5% above the oracle. The ordering of the closed-form methods is stable across the three clients, so the untuned transfer is not specific to a single reported percentile.

Three observations. First, the untuned SPAR-EG$^+$ lands 1.1% above the best oracle-tuned leaky window over the full horizon and 6.3% above it on the final 2,000 hours, while SPAR-Pull is 10.7% and 14.7% above; a smooth seasonal stream has a long and nearly constant optimal window, which is exactly the regime in which a fixed long window is close to optimal and adaptation has little to recover. Second, the slow estimated-R Kalman is far worse over the full horizon (0.051 and 0.257) and still 26% above the leaky baseline on the final 2,000 hours, i.e. its early noise estimates are poor and it only recovers late. Third, no method diverges and the ordering of the closed-form methods is stable across the horizon, so the control law transfers to a real $D = 324$ stream without retuning.

E BCI IV details

Datasets and protocol. BCI Competition IV 2a (Brunner et al., 2008) and 2b (Leeb et al., 2008), nine subjects each, loaded from the npz cache described in Appendix G. Cross-session protocol: each method fits on the training session and streams the evaluation session (288 trials on 2a, 320 on 2b) predict-then-update. All online methods use supervised adaptation: the label of trial t enters after its prediction, so every prediction is prequential. Labels are one-hot $Y_t \in \{0, 1\}^K$; the streaming head maintains a $D \times K$ coefficient matrix and the scalar residual driving both estimators is $r_t^2 = \|Y_t - \widehat{Y}_t\|_2^2/K$. The metric is balanced accuracy; with $n = 9$ subjects per dataset we report paired tests and effect sizes, not significance alone.

Compute. The complete closed-form suite (feature extraction plus all closed-form methods) for all nine 2b subjects runs in 17 s on the CPU used here (1.9 s per subject; `results/tier1_rerun_2b.log`); one

Table E.1: BCI IV cross-session, balanced accuracy (mean ± s.d. over 9 subjects per dataset). Supervised online protocol for adaptive methods; classical methods are static. ^{T}Session-level Euclidean alignment from the unlabeled evaluation session (transductive); the online-EA row is strictly causal. Highest value per column in bold.

method	BCI IV 2a	BCI IV 2b
classical (canonical implementations)		
MDM (pyriemann)	0.566 ± 0.160	0.728 ± 0.148
FgMDM (pyriemann)	0.641 ± 0.171	0.755 ± 0.149
LDA (raw features)	0.480 ± 0.069	0.651 ± 0.107
LDA (raw) + session EA^{T}	0.563 ± 0.076	0.677 ± 0.133
CSP-LDA (mne)	0.607 ± 0.151	0.749 ± 0.141
tangent cov + LDA	0.637 ± 0.165	0.752 ± 0.147
cov-LDA, offline session-EA^{T}	0.664 ± 0.173	0.728 ± 0.159
ours (closed-form streaming, predict-then-update)		
CSP + SPAR-Pull	0.627 ± 0.159	0.770 ± 0.142
tangent cov + SPAR-Pull	0.527 ± 0.151	0.772 ± 0.129
hybrid + SPAR-Pull^{T}	0.597 ± 0.137	0.808 ± 0.104
hybrid + SPAR-Pull, grid defaults^{T}	0.579 ± 0.137	0.801 ± 0.101
online-EA + hybrid + SPAR-Pull	0.596 ± 0.134	0.801 ± 0.105
hybrid + AAL head (γ=1, fixed anchoring)^{T}	0.608 ± 0.139	0.799 ± 0.107
hybrid + leaky head^{T}	0.554 ± 0.123	0.812 ± 0.108
hybrid + SPAR-EG^{T}	0.580 ± 0.130	$\mathbf{0.816 \pm 0.102}$
hybrid + SPAR-EG$^{+\,\mathrm{T}}$	0.586 ± 0.141	0.813 ± 0.107
backprop references		
EEGNet-lite (train-split tuned, zero-shot)	0.556 ± 0.080	0.776 ± 0.129
WiSE-FT on the net (causal, α-envelope)	0.591 ± 0.079	0.811 ± 0.115
post-hoc ensemble (cov-LDA + hybrid warm fit)^{T}	$\mathbf{0.668 \pm 0.175}$	0.771 ± 0.104

EEGNet-lite training configuration (120 epochs) trains on the WSL GPU environment, and the full backprop protocol trains three configurations per subject plus the online fine-tuning pass.

Feature families. Three feature families feed the readouts of Table E.1. CSP: 6-component CSP (Ledoit–Wolf shrinkage, log output) computed with MNE (Gramfort et al., 2013) on bandpassed epochs (8–30 Hz, 4th-order Butterworth), the classical motor-imagery standard (Ramoser et al., 2000). Tangent covariance: per-trial log-Euclidean covariance tangent features (lower triangle) on bandpassed epochs, the feature family of pyriemann's tangent-space methods. Hybrid: the statistical pipeline (summary statistics, lags $(1, 2, 4, 8)$, FFT bins) concatenated with the tangent-covariance features. Classical readouts are MDM and FgMDM as implemented in pyriemann (Barachant et al., 2020), raw-feature LDA with and without session-level Euclidean alignment (He & Wu, 2020), CSP-LDA, tangent-covariance LDA (shrinkage LDA in all cases), and a covariance-LDA method with the same offline alignment. Streaming readouts are the two-estimator SPAR-Pull head (anchor refresh every 5 trials), the AAL head (fixed $\gamma = 1$ anchoring), and a forgetting-only leaky head, each applied to CSP, tangent-covariance, or hybrid features.

Alignment protocol. The hybrid methods use session-level Euclidean alignment: per-trial covariance whiteners are computed from each session separately, so the alignment is transductive at the session level (the offline classical methods share this convention). The strictly online method recomputes its whitener causally from a trailing window of the evaluation stream (40 trials, refreshed every 10) for *both* feature halves: the statistical features and the tangent-covariance features are each whitened with the trailing-window estimate, so no stream feature depends on trials beyond the current one. All SPAR methods in Table E.1 run the same variance-floor-subtracted window inversion as the factorial grid (Theorem 2), with the estimator ranges rescaled to the 288-trial stream (window range $[5 \times 10^{-3}, 0.02]$, refresh cadence 5, residual horizon 40, calibration blocks 2). The *grid defaults* row is the same hybrid head with the factorial-grid window range, refresh cadence, residual horizon, and calibration blocks ($[5 \times 10^{-4}, 0.05]$, 10, 30, 10); the gap to the EEG-range row prices those four settings (0.018 balanced accuracy on 2a and 0.007 on 2b). The reduced range therefore helps, but the ranking of the readouts and the qualitative conclusion (closed-form streaming matches the causal backprop reference) do not depend on it.

WiSE-FT protocol measurement. Transplanting the original WiSE-FT protocol (Wortsman et al., 2022) (fine-tune on the target data, then evaluate on it) into cross-session evaluation is train-on-test. Measured directly, the offline fine-tuned endpoint reaches accuracy 1.000 by memorizing the evaluation stream. The

comparison we report is therefore the causal reformulation: the network is fine-tuned online with one SGD step per trial *after* predicting it, trial t is predicted with the interpolation $(1-\alpha)\,W_{\text{zero}} + \alpha\,W_t$ where W_t is the online model before trial t's label, and the α-envelope is swept post-hoc over 11 values with all trajectories causal (the prequential evaluation of Table E.1).

Backprop references. EEGNet-lite is a small EEGNet variant (Lawhern et al., 2018) ($F_1 = 8$, $D = 2$, $F_2 = 16$) trained with Adam. Its hyperparameters (epochs, learning rate) are selected on an 80/20 split of the training session from the grid $(120, 10^{-3})$, $(300, 10^{-3})$, or $(300, 3 \times 10^{-4})$ with no evaluation-session access, then retrained on the full training session and evaluated zero-shot on the evaluation session. The WiSE-FT-on-network reference uses this zero-shot model as its frozen endpoint and the causal fine-tuning protocol above.

F Hyperparameters and provenance

Fixed constants and the default control parametrization. The complete fixed control parametrization is enumerated in Table F.1: the three headline constants (anchor-refresh cadence 10 steps, fast-EMA residual horizon 30 steps, drift-active anchor-strength floor $\rho \geq 0.02$), the clip ranges of both estimators, the slow-noise-EMA baseline horizon (20,000, matched to the estimated-noise Kalman), the reference-anchor EMA horizon (200 steps), the conflict-floor multiplier ($\frac{1}{2}$), the estimator smoothing gains ($\eta_C = 0.15$, $\eta_\nu = 0.2$), and the 10^{-8} numerical ridge of the trace and active-dimension computations. These were fixed once, before the factorial grid, and are shared by every SPAR method on it; all remaining behavior is self-calibrated at runtime by the two online estimators. The sensitivity analysis of Appendix D perturbs the headline constants directly.

Table F.1: Fixed default control parametrization (grid: $T = 10^4$). The EEG head uses a reduced estimator range and shorter horizons for the short evaluation streams, hand-set once and disclosed here; no per-configuration search was performed for any SPAR method.

parameter	grid default	EEG head
refresh cadence k	10	5 trials
fast-EMA horizon h_f	30	40
warm-up anchor strength ρ_0	0.1	0.1
stationary ρ floor	10^{-5}	10^{-5}
drift-active ρ floor	0.02	—
anchor-strength cap $\rho_{\max}$	1.0	1.0
window clip $[\omega_{\min}, \omega_{\max}]$	$[5 \times 10^{-4},\ 0.05]$	$[5 \times 10^{-3},\ 0.02]$
slow-noise-EMA horizon	20,000	2,000
reference-anchor EMA horizon	200	30
conflict-floor multiplier	$\frac{1}{2}$	$\frac{1}{2}$
smoothing gains (η_C, η_ν)	$(0.15,\ 0.2)$	$(0.2,\ 0.1)$
active-dimension / trace ridge	10^{-8}	10^{-8}

Estimator floors and caps. The window estimator is clipped to $\omega \in [5 \times 10^{-4},\ 0.05]$ on the grid (the floor is no forgetting at the grid's horizon; the cap bounds the forgetting rate), and the anchor-strength estimator enforces the drift-active floor above together with a hard cap. The EEG head uses a reduced range suited to the short evaluation streams ($\omega \in [5 \times 10^{-3},\ 0.02]$, fast horizon 40, refresh every 5 trials, reference-anchor horizon 30, smoothing gains $(\eta_C, \eta_\nu) = (0.2, 0.1)$) and omits the drift-active floor; these are the only stream-dependent settings in the paper.

Conflict estimator branches (prediction units). Algorithm 1 step (iv) takes the maximum of three branches, all in prediction-risk units $[y^2]$ (the residual EMAs are σ^2_{fast} = fast EMA of r_t^2 at horizon h_f, σ^2_{slow} = slow EMA at horizon 20,000; d_{act} is the number of eigenvalues of $\bar{A}$ above the 10^{-8} numerical ridge): (i) the reference branch $C^{\Sigma}_{\text{ref}} = (G_{\text{ref}} - \widehat{w})^\top \bar{A}\,(G_{\text{ref}} - \widehat{w}) - \widehat{V}^{\Sigma}$, where $\widehat{w}$ is the minimum-ridge data solve, $\widehat{V}^{\Sigma} = \hat{\sigma}^2 d_{\text{act}}/n$, and G_{ref} is the reference anchor (a slow-moving copy of G at the reference cadence, updated every $5k$ steps); (ii) the residual branch $C^{\Sigma}_{\text{resid}} = E_t = \max(0, \sigma^2_{\text{fast}} - \sigma^2_{\text{slow}})$; (iii) the floor $C^{\Sigma}_{\text{floor}} = \frac{1}{2}\,\hat{\sigma}^2 d_{\text{act}}/T_w$

(the warm-up prediction variance is an irreducible lower bound on the conflict of a warm-up anchor). The per-clip strength is $\rho = \mathrm{clip}(\bar{\mu}\,\widehat{V}^{\Sigma}/\widehat{C}_t^{\Sigma},\ \rho_{\min}^{\mathrm{drift}},\ \rho_{\max})$ with the drift-active floor 0.02 engaged while the window is at least half open ($\omega_t \geq \frac{1}{2}\omega_{\max}$) and the stationary floor 10^{-5} otherwise. Every branch has units $[y^2]$ and ρ has units $[x^2]$; no inverse of $\bar{A}$ is formed, so the estimator is invariant to feature rescaling and immune to the rank-deficient trace blow-up. The legacy parameter-unit form (with $\mathrm{tr}(\bar{A}^{-1})$ and $C_{\mathrm{resid}} = E_t\mathrm{tr}(\bar{A}^{-1})/(n\bar{\mu})$) is reported only as the ablation of Appendix D.

Method identifiers. The paper's display names map to frozen artifact identifiers (internal Python/CSV keys) as follows: SPAR-Pull is `spar_pull` (its legacy parameter-unit form is `spar_kf2`), SPAR-Pull-S is `spar_scalar`, SPAR-Pull-D is `spar_diag`, SPAR-EG is `spar_eg`, SPAR-EG$^+$ is `spar_eg_plus`, the causally-tuned Kalman filters are `kf_causal` and `kf_est_causal`/`kf_est_slow_causal`, the AAL head is `hybrid-pam_g1`, leaky RLS is `leaky[`λ`]` in covariance form and `leaky_batch` in information form, directional forgetting is `df[`λ_f`]`, AROW is `arow[`r`]`, the Kalman filters are `kf[`q`]` (true noise), `kf_est` (100-step estimated R), and `kf_est_slow` (20,000-step estimated R), the gradient-descent baselines are `sgd` and `adam`, the OGD variants are `ogd_profile` (scalar segment) and `ogd_profile_d` (full D-dimensional profile), the variable forgetting factor baseline is `vff[fortescue]` in covariance form and `vff_info[fortescue]` in information form, and the WiSE-FT envelope is `wiseft`. All renaming in the paper is display-layer only; the CSVs, scripts, and tests key on the artifact identifiers.

Seeds. All seeds are pinned in the drivers: the factorial grid uses seeds 0–4 per configuration; the ablation, regret scaling, coupling diagnostic, and mismatched-generator studies use seeds 0–4; the false-escalation calibration uses 300 streams (seeds 1000–1299); the estimator-trajectory figure uses seed 0; the EEG evaluation uses subjects 1–9 per dataset (all available subjects); the BP references train with per-subject seeds equal to the subject index. Seeds are shared across methods at each configuration.

Data manifests. Synthetic streams are generated from the pinned seeds by the committed generator (`rls_baselines.make_stream_fac`), so no synthetic data files are needed. The EEG evaluation reads the committed npz cache (`data/eeg_cache/{2a,2b}/subject*.npz`), exported from the MNE data directory by `scripts/export_eeg_npz.py`. Every result table in the paper is machine-printed from a committed CSV under `results/` (Appendix G).

G Reproducibility

One driver. A single driver regenerates every figure and every wired result table from the committed caches:

```
py scripts/run_all.py -list  list the pipeline steps
py scripts/run_all.py -steps quick  verify+aggregate+figures
py scripts/run_all.py  everything (hours; resumable)
```

The quick preset takes about 2 minutes; the full run takes hours and is resumable.

Table G.1: One driver, `scripts/run_all.py`, runs these stages in order; each is resumable.

stage	driver	output
`verify`	`verify_spar_theorems.py`	symbolic identities
`factorial`	`synthetic_factorial.py`	`factorial_d_*.csv`
`sparkf`	SPAR-Pull append	`factorial_extras.csv`
`append`	`append_arms.py`, `regen_spar_eg_plus.py`, `revision_runs.py -suite grid`	`factorial_pu_refresh.csv`
`aggregate`	`aggregate_factorial.py`	Table 1
`ablation`	`revision_runs.py -suite ablation`	`ablation_spar_pull.csv`
`falseesc`	300-stream calibration	`false_esc.csv`
`regret`	horizon and dimension scaling	`regret_*.csv`
`ops`	wall-clock timing at $D = 2048$	`ops_table.csv`
`eeg`	`bench_eeg_tier1.py`	`eeg_tier1_2*.csv`
`bp`	backprop baselines	`bp_tta_2*.csv`
`figures`	`make_figures.py`	`results/figures/`

Environment. Python 3.12 with the committed `requirements.txt` (jax, numpy, scipy, pandas, scikit-learn, matplotlib, sympy). Float64 is required for the streaming solves and is enabled by setting `JAX_ENABLE_X64=1` (the drivers set it); a CPU runtime is recommended for the fp64 streams. The backprop methods run on the WSL GPU virtual environment: `scripts/run_bp_tta_2a.sh` and `scripts/run_bp_tta_2b.sh`, with `BP_PYTHON` pointing at that environment's interpreter and GPU extras in `requirements-wsl.txt`.

EEG data. Raw BCI IV files live under the MNE data directory (`MNE_DATA` environment variable or the mne configuration; default `~/mne_data`) at the BNCI database path. The loader prefers the npz cache: run `scripts/export_eeg_npz.py` once to produce `data/eeg_cache/{2a,2b}/subject*.npz` for all nine subjects per dataset.

Electricity data. The real stream of Appendix D.10 reads the public hourly electricity-load panel (321 clients, 26,304 rows, headerless comma-separated) from `data/ts/electricity.csv` (the loader also accepts the `.txt` variant); the file is a public download and is not shipped with the artifact. It regenerates with `scripts/bench_stream_real.py -clients 11,82,171` (`results/stream_electricity.csv`); the script accepts any client indices, and the tables report the 25th-, 50th-, and 75th-percentile clients. The transient strength sweep of Appendix D.5 regenerates with `scripts/transient_test.py` (`results/transient_test.csv`). The grid-default BCI row of Table E.1 is the `hybrid-spar_kf_grid` column of `scripts/bench_eeg_tier1.py`; on 2a it is also reproduced by the cheaper `scripts/eeg_grid_default_probe.py`, which saves `results/eeg_griddefault_2a.csv`.

Lean. The proof artifact builds with the elan-managed Lean 4 toolchain and Mathlib via lake; the full recipe, including the Mathlib cache fetch, is `theory/lean/build_ogd.sh` (Appendix C).

Artifact layout. Every number in the paper is machine-printed into `results/*.csv` by the drivers above, and every figure is redrawn from those CSVs: the factorial tables (`factorial_*`), the ablation (`ablation_spar_pull.csv`), the false-escalation calibration (`false_esc.csv`), the regret scaling (`regret_scaling.csv`), the cost table (`ops_table.csv`), the EEG tables (`eeg_tier1_2*.csv`, `spar_eeg_2*.csv`, `bp_tta_2*.csv`), the estimator trajectory (`estimator_trajectory.csv`), the real electricity stream (`stream_electricity.csv`), and the transient strength sweep (`transient_test.csv`). The offline session-EA cov-LDA method and the post-hoc ensemble (members: the cov-LDA method and the static hybrid warm fit, identifier `ens-covlda+hybrid`) regenerate from the `eeg` step's baseline pass (`scripts/bench_eeg.py`, recorded in `eeg_2*.csv`); all other rows regenerate from the tier-1 pass. The Lean material (six files, the build script, and the axiom-listing script) is under `theory/lean/`.